BIF401 MID TERM HIGHLIGHTED & SHORT HANDOUTS 2022

COMPOSED BY M ZAMAN

(I just highlight the main and important topics for midterm.

If you found any mistake please correct it and tell me also.)

Just take Concept Because Concept is Power

Chapter 1-Introduction to Bioinformatics

Module 001: INTRODUCTION TO BIOINFORMATICS

 BACKGROUND

Bioinformatics is an interdisciplinary science at the cross-roads of biology, mathematics, computer science, chemistry and physics. With the digitalization of the biological information, doors have

been wide opened towards the analysis of this information using computer algorithms and software.
Now we know well that the human genome has over 25,000 genes and these genes code for thousands of different proteins which perform day-to-day functions in the living cell

EXPERIMENTS IN BIOLOGY

We have several next generation instruments and techniques available for obtaining digitalized biological information on genes and proteins etc. These instruments include:

1. Next Generation Sequencers (NGS) for whole genome sequencing
2. High Resolution Mass Spectrometry for whole proteome profiling
3. Nuclear Magnetic Resonance Spectroscopy for structural studies

DIGITALIZATION OF BIOLOGY

In today's world, when a biologist performs an experiment in the wet-lab, he or she in fact produces digital data which is continuously being stored on computer disks. The data may include text, numbers, symbols or images.

SPEED OF DATA GROWTH

Due to advancement in instrumentation used in biological experiments, data is being accumulated at exponentially increasing rates. For example; genome sequences in genome databases are doubling every few years.

Module 002: INTRODUCTION TO BIOINFORMATICS

MOTIVATION

- It is an **interdisciplinary field** as it covers the information of biological digital information including human, plants, animals and microorganisms.
- Although it is a new field but it is rapidly **developing field**.
- It demands a very **low cost** infrastructure and hardly any lab equipment.
- It presents us with **plenty of opportunities** in scientific discovery.

SCOPE OF BIOINFORMATICS

- Storage of data
- Organization data
- Analysis of many experiments
- For representation of biological information

ACTIVITIES

In modern biological sciences, bioinformatics is used for activities such as:

- Developing algorithms for organizing data collected from experiments
- Writing software and tools for data analysis
- Data processing to determine the role of underlying biomolecules
- Statistical evaluation of data using methods such as t-test and ANOVA
- Data visualization for meaningful presentation of biological information

Module 003: INTRODUCTION TO BIOINFORMATICS

NEED FOR BIOINFORMATICS –I

Biomedical Engineering, Electrical and Computer Engineering, Computer Science, Applied Mathematics, Genetics, Biology, Anatomy and Cell Biology, Micro Biology, and Biostatistics are the principal allied disciplines.

The need for bioinformatics is on a rapid rise as biological data is rapidly increasing and becoming available online, free of any cost.

Module 004: NEED FOR BIOINFORMATICS –II

If we observe the growth of gene bank than from 1982 it comprised of 2 billion base pairs but by year 2002 it had risen to 56 billion base pairs.

The phylogenetic “tree of life” which consist of:

- Bacteria
- Archaea
- Eucarya

WHAT IS IT THAT BIOINFORMATICS CAN DELEIVER?

- Provide us better understanding of life, evolution, molecular mechanisms as well as disease.
- Moreover, we can make better drugs with the availability of an enhanced molecular understanding of disease.

POSSIBLE CONTRIBUTIONS

- It can help us to organize the large datasets from new experiments instruments
- Bioinformatics can help store and process this data as well.
- It can provide insights into the meanings of our research results and findings.
- Overall, it can help us to better understand paradoxes defining the life forms.

Module 005: APPLICATIONS OF BIOINFORMATICS – I

JUST MAIN POINTS YAD KARAIN

GENOMICS

- Bioinformatics can help in assembling DNA sequencing data.
- It can help in gene finding (markers).
- Gene assembly can be performed using bioinformatics tools (nucleotide alignments)
- It can help transcribe the gene data to RNA data
- Also, databases can be generated from such data.

EVOLUTIONARY STUDIES

- Evolutionary relationships between different organisms can be derived from data.
- Evolutionary distance among species can be computed by using bioinformatics tools.
- Phylogenetic trees can be constructed to find relationships between species.
- Ancestry can be better understood between several species and organisms.

PROTEOMICS

- Bioinformatics can help us in decoding protein sequences.
- It can also help us in understanding protein structure.
- We can also understand post translational changes in proteins with the help of bioinformatics.
- We can better understand the protein-protein interaction in different biological reactions.
- It can also help us in generating databases of these sequences and structures.

SYSTEMS BIOLOGY

- Bioinformatics can assist us in modelling regulatory mechanisms in gene and protein networks.
- Such models can be analyzed to identify the key regulators in these networks.
- Moreover, the models can help evaluate drugs to treat these key regulators.

CONCLUSION

Recent trends in bioinformatics involve development of personalized therapeutics for cancer and diabetes.

Module 006: APPLICATIONS OF BIOINFORMATICS - II

Bioinformatics is being applied in routine life in many ways like in Genomics, transcriptomics, Proteomics, Metabolomics, Structural Proteomics, Designing Drugs, System Biology and in personalization of medicines for cure.

Except these applications Bioinformatics introduced us the techniques which enabled us to generate the large data regarding biology and also its use. And step by step the applications of bioinformatics increased from genomic level to entire system level.

▯ SMALL TO BIG

- Bioinformatics helps us to understand the systems from small to big like from gene findings to entire system prediction
- In structure findings and modeling of many biological system to understand them in better ways.
- Bioinformatics helped the human to understand the protein, protein interaction in many biological systems.
- And provide us the concept how these biological process are interconnected with each other and how they affect each other.
- Now we are able to understand the modeling of molecules and genome at cell level.
- Signaling pathways are easy just because of bioinformatics.
- Now morphology of tissue can be understand by creating the models with help of bioinformatics tools.

Module 007: FRONTIERS IN BIOINFORMATICS - I

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FRONTIER IN GENOMICS

Now we are able to sequence the whole genome with the bioinformatics tool of Next generation sequencing (NGS)
We are able to save, store and analyze the massive amount of biological data which is in (Terabyte files)
We can handle the large number of data easily and can process it as well in easy way.
Whole genome can be assemble in sequence and can flaws can be identified easily.

 FRONTIER IN TRANSCRIPTOMICS

Now in genomics we are able to identify those matters which are unknown yet or under discussion.
Role of RNA in making proteins and its dynamics can be understood easily now.
Interactions of RNA molecule can be easily understood by simple model.

 FRONTIER IN PROTEOMICS

Deficiency of low proteins in any patient tissue sample can be identify.
Expression and manufacture of protein in large molecular level in any organism can be identified.
Pathways before and after any biological reaction are easy to design.

Module 008: FRONTIER IN BIOINFORMATICS-II

Frontier in Bioinformatics includes

- Next generation genomics
- Transcriptomics
- Proteomics

 FRONTIER IN PROTEIN STURUCTURE

Bioinformatics helps us to understand the layer folding of proteins that how they are processed, and helps to know that how protein interact with each other and how a drug can affect or stimulate a protein.

 FRONTIER IN SYSTEM BIOLOGY

It helps us to understand the whole system of a single cell, in that cell how organelles, gene, proteins and metabolites are interconnected in a single unified system (cell). And bioinformatics also give us the idea how these models can be applied to real-time.

 FRONTIER IN PERSONALIZED MEDICINE

This is the important thing for this century and upcoming generation that personalize the medicine for exact cure of a disease. Because all the medicine cannot work exact some effect patient badly Bioinformatics helps us to evaluate the medicine.

CONCLUSION: If we talk about the 21st century than it's the century of bioinformatics it will enable the human to cure many disease with one drug by personalizing it.

Chapter 2 -Sequence Analysis

Module 001: Gene, mRNA and Protein Sequences

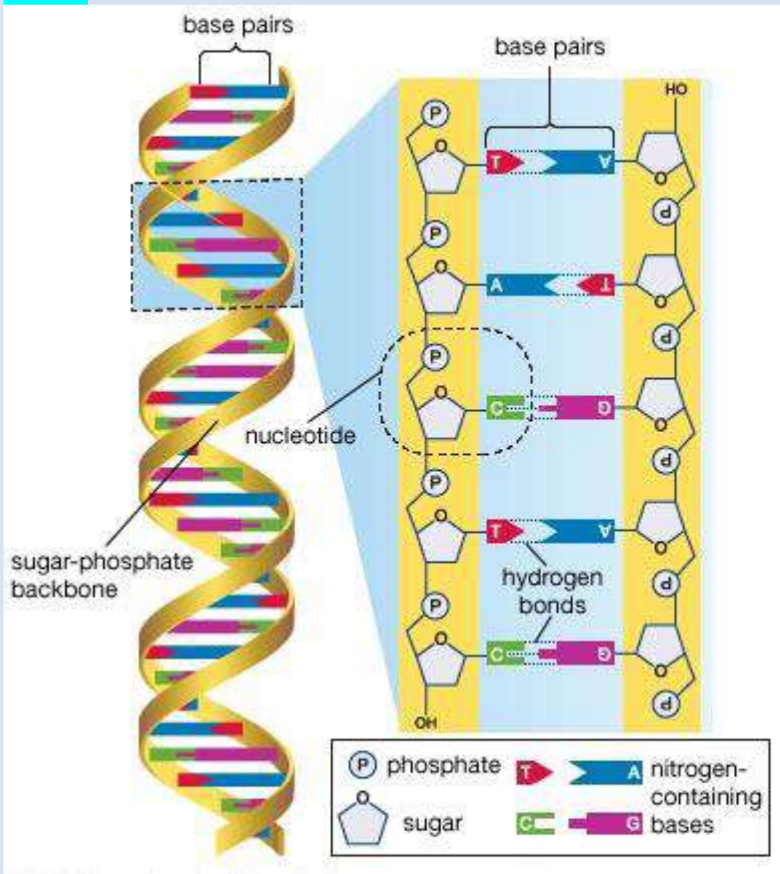
"Central Dogma"

"Flow of information from DNA to RNA and then to Protein"

Proteins are than use in constructing the cell.

 DNA

DNA molecule is double helix structure contain base pairs composed of nucleotides and these nucleotides are composed of sugar phosphate group and are bind with each other with hydrogen bonds.



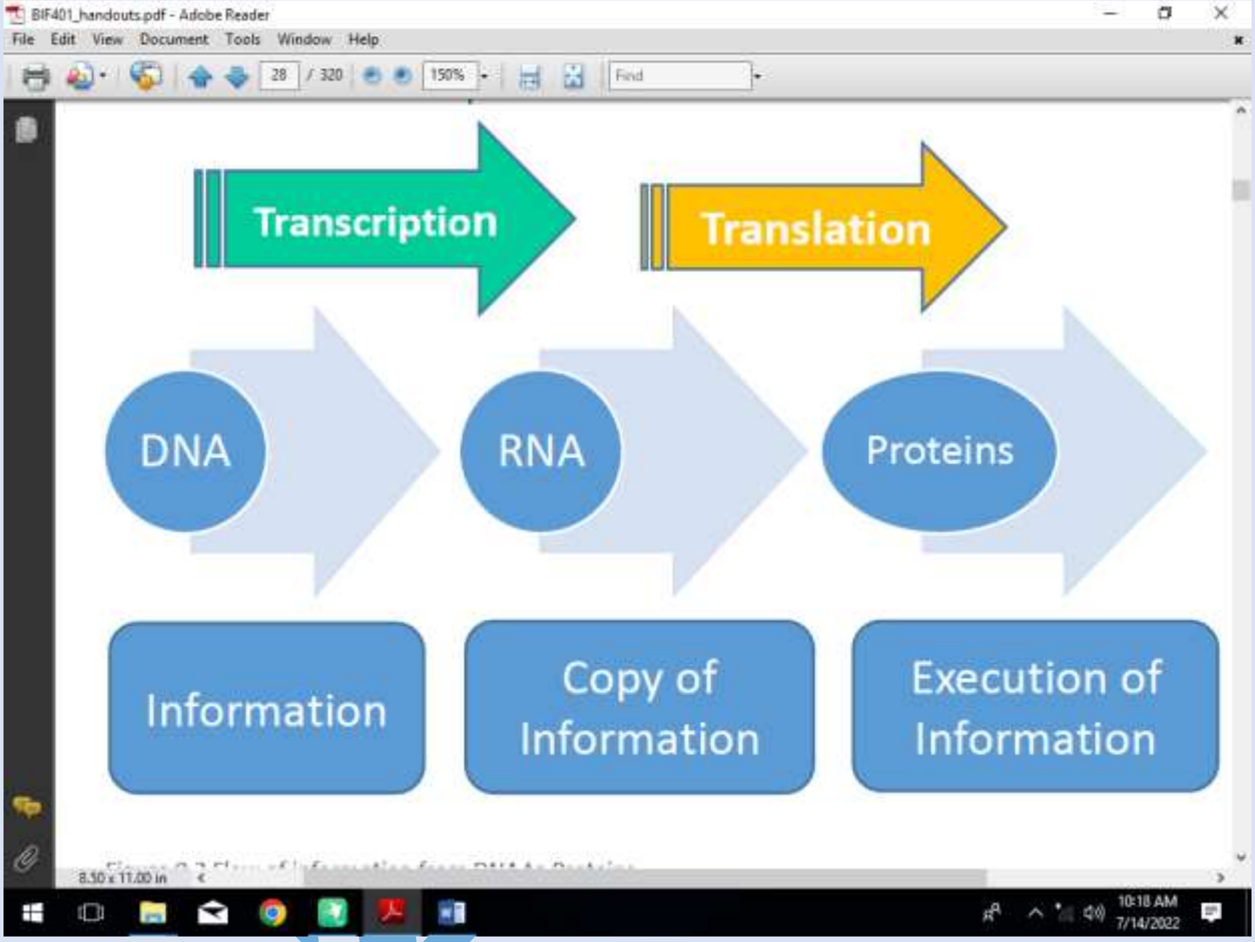
Normally all the nucleotides are same in both DNA and RNA except one position in RNA which is U (Uracil) and in DNA it is T (Thiamin)

DNA sends the information to cell via mRNA and that sequence the amino acids according to coded information and protein structure is formed and that protein form a cell.

Module 002: TRANSCRIPTION

All cells are made of carbohydrates and proteins and for these cells DNA codes the information which makes the RNA and protein both.

Figure 0.2 Flow of information from DNA to Proteins



Transcription: DNA codes the information and converted into RNA where mRNA copies the information and it execute the information in cell and amino acids combine with each other according to coded information of DNA and protein formation takes place. Which is known as **Translation.**

Molecule of DNA contains only four base pairs (A, T, C, and G) which are repeated thousands of time and Adenine "A" pairs with Cytosine "C", While Thymine "T" binds with Guanine "G" and all pairings are with the help of Hydrogen bonding.

Same like DNA, the RNA contains four base pairs but Thymine is replaced with Uracil "U" and RNA is single stranded.

DNA just codes the information for protein but RNA helps in making protein.

Module 003: NUCLEOTIDES

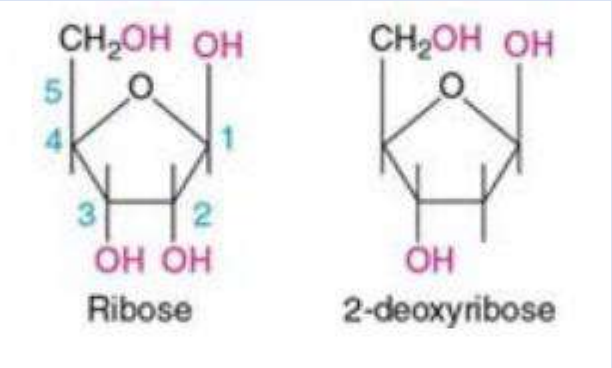
If we talk about the composition of DNA and RNA molecule than these are composed of four other molecules which are named as Nucleotides.

These molecules are Adenine (A), Cytosine (C), Thymine (T), Uracil (U), and Guanine (G).

DNA molecule although is double stranded and RNA is single stranded but there is difference in sugar composition.

RNA has Ribose sugar and DNA has de-ox ribose sugar:

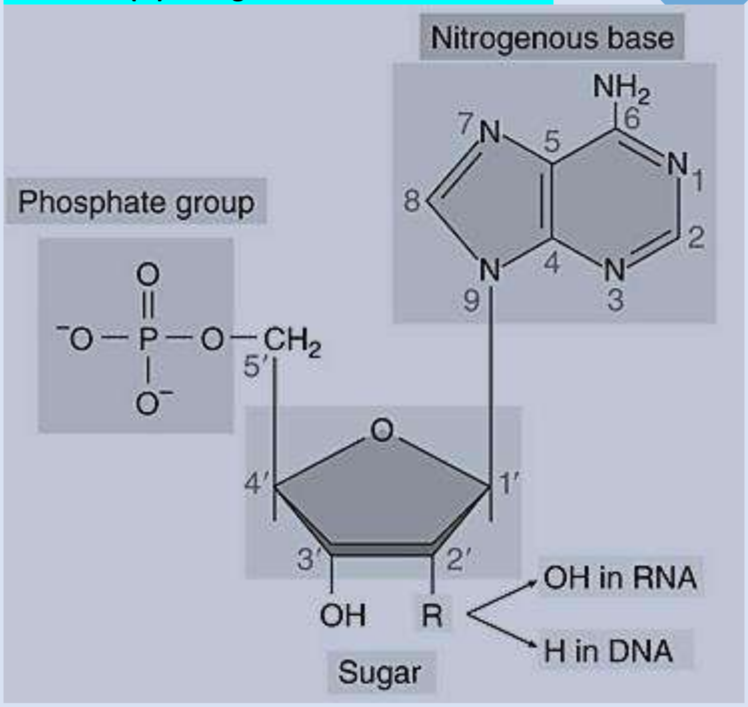
Figure 0.3 Difference between RNA and DNA sugar



RNA **DNA**

Adenine and Guanine collectively called **Purines** while Cytosine, Uracil, and Thymine are called as **Pyrimidine.**

When phosphate, nitrogen base and sugar come together if there is **(OH)** than molecule is RNA and if there is **(H)** in sugar than molecule is DNA.



Module 004: TRANSLATION

Cells are built of proteins and carbohydrates and these proteins are made in results of transformation of RNA molecule and this transformation is called as translation.
Translation takes place in ribosome of cell and ribosomes after reading the information of mRNA collects the amino acids from cell cytosol which is the part of the cytoplasm that is not held by any of the organelles in the cell.

□ **MECHANISM**

At ribosome three nucleotides are read at a time from mRNA, this set of three nucleotide is called as codon and each codon correspond to a specific amino acid.

Figure 0.4 sixty four codons combinations

Amino Acid	3-Letters	1-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic Acid	Asp	D
Cysteine	Cys	C
Glutamic Acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Module 005: AMINO ACIDS

RNA decodes the information at ribosomes in form of Codons each codon select a specific amino acid. Because there are 20 different amino acids in nature therefore they fold together and make a protein structure by polymerizing themselves.
If we observe the structure of amino acid it contains nitrogen, hydrogen, oxygen and two carbon atoms. And a variable group R.

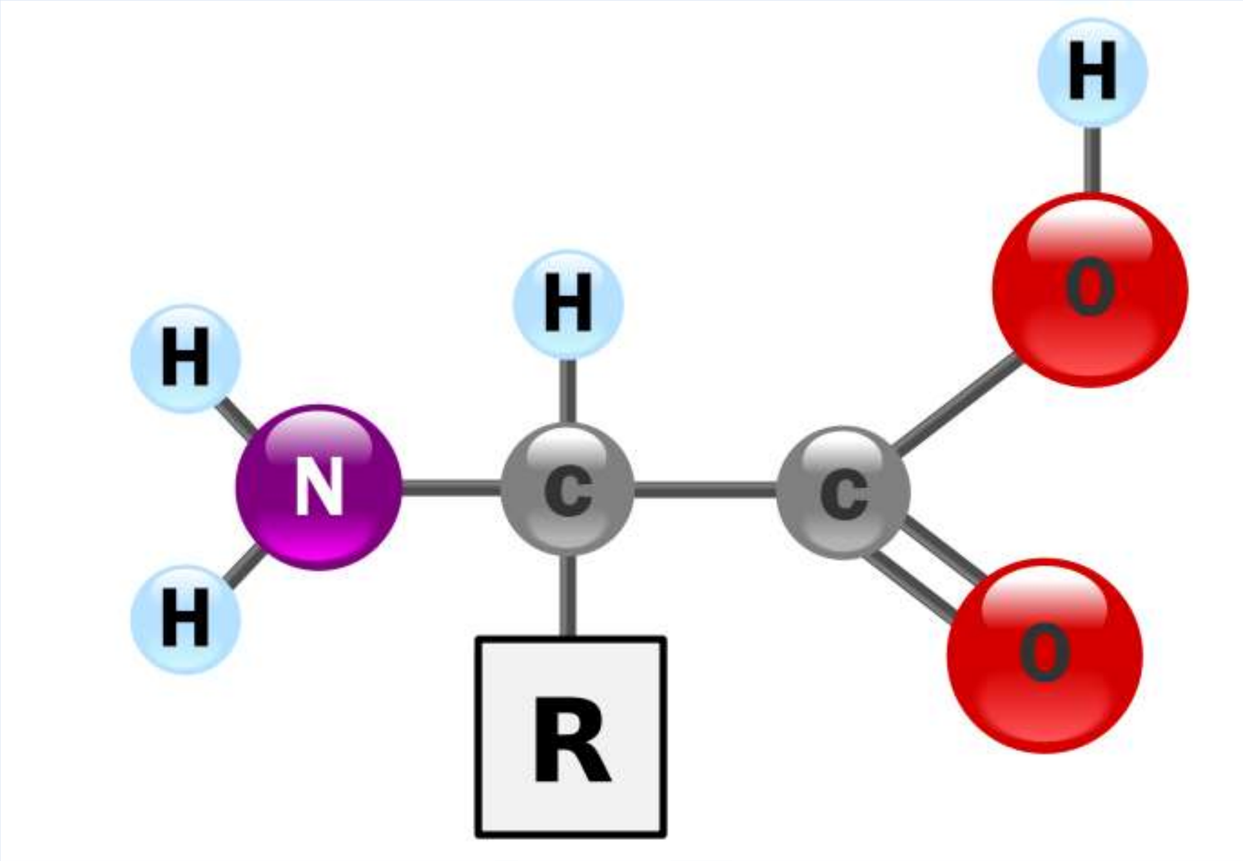


Figure 0.5 structure of amino acid

When polymerizations takes place water is formed and if any compound attached with R group than structure of protein is changed.

These amino acids are joined with each other with peptide bonds and fold with each other in 3D form they make protein structure.

Module 006: STORAGE OF BIOLOGICAL SEQUENCE INFORMATION

We know that sequence of DNA contain A,C,T&G nucleotides and sequence of RNA contains A,C,U&G while sequence of protein contain A,R,N,D,C,E,Q,G,H,I,L,K,M,F,P,S,T,W,Y&P these are actually 20 different amino acids in nature which compose a protein.

▯ SOLUTIONS DATABASES

Public sequence data bases for DNA & RNA the public database is **GenBank (by NIH).**
For proteins the public database is **UniProt (by Uniprot Consortium)**
Both **GenBank** and **UniProt** are online database and the DNA, RNA and Protein sequences are available here online for public and researchers.

Module 007: USING GENBANK

GenBank is online database where researcher can get access to the sequences of DNA, RNA and proteins.

To find any sequence we go online to NCBI GenBank website which is Public database site. Which is;
www.ncbi.nlm.nih.gov/genbank

Sequences can be searched from GenBank by typing;

- Sequence name
- ID
- Name
- Species
- Locus
- Accession Number
- Author
- Journal

Module 008: USING UNIPROT

UniProt is public database which is being used to search the sequence of proteins.
www.Uniprot.org
For example we want to search a sequence of a protein which is Ubiquitin which plays an important role in cytosol for recycling the proteins. We have to go online to the website www.Uniprot.org and above page will appear.
We have to write the name of protein in search box and press enter.

By clicking on any result you can **download** or **Blast** the sequence.
In home page there is a box named **"Swiss Prot"** which contains human curated protein information, molecular mass, observed and predicted modifications etc.
Uniprot can be searched by typing amino acid, Name, ID or sequence.

Module 009: COMPARING SEQUENCES

By comparing sequences of DNA, RNA and Proteins we can get

- Similarity among sequences
- There might be some specific difference due to some disease or mutation
- There may be some evolutionary relationship.

Module 011: SIMILARITIES & DIFFERENCES IN SEQUENCES

If some sequence are exactly similar to each other it means;

- They might have some regular expression in cell or system.
- Or they indicate some specific presence like signature of any protein or gene.
- Or they might have similar nucleotide just one or two between them are different from rest.

Module 012: PAIR WISE ALIGNMENT –II

In pair wise alignment of nucleotides the nucleotides comes in pairs and matching are colored while missing amino acids are indicated with “” and this empty space is called as gap.
And these Gaps are inserted for deletion or insertion of any nucleotide. Increase in Gaps can increase the chance of plenty in sequencing and less number of Gaps can increase the similarity rate of sequences.
There are two types of pair alignments.

1. Global

2. Local

In Global ways of sequence pair alignment we introduce the Gaps in all sequence to know over all matching. While in Local type of sequence pair alignment we find those regions where nucleotides are maximum matching with each other it is used to find the similarity or some nutation.

Pairwise Sequence Alignment is used to identify regions of similarity that may indicate functional, structural and/or evolutionary relationships between two biological sequences (protein or nucleic acid).

Module 013: PAIR WISE SEQUENCE ALIGNMENT –III

Three ways:

- Substitutions ACGA AGGA
- Insertions ACGA ACCGA
- Deletions ACGA AGA

But among above Substitution increases mismatch of sequence.

A C A C G

A c C G G

Module 014: DOT PLOTS

To visualize the sequence alignment we have a method called Dot Plots in this method the sequence is written top and left side of dot matrix grid.

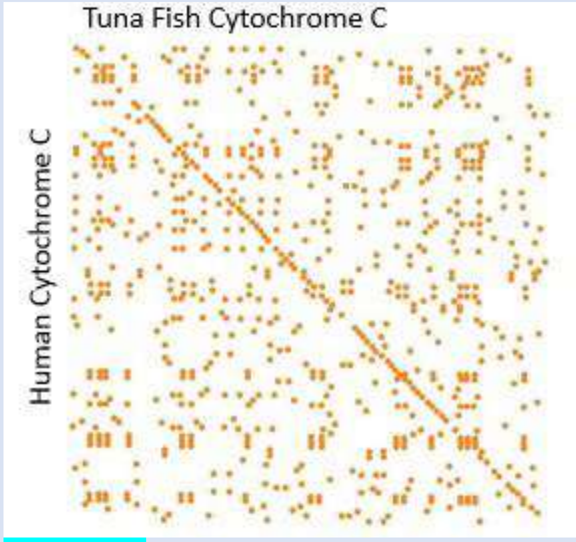
Where one nucleotide or amino acid match with each other the dot is placed in grid position in each row for one time.

Similar dots are match with diagonal pattern and which remain separate differ from similar sequence.

Module 015: EXEMPLE OF DOT PLOTS

In dot plot the matching nucleotides are connected in diagonal way and represent the sequence alignments.

When we compare the **human Cytochrome** and **Tuna Fish Cytochrome** than the diagonal alignment of sequence we find is in this below diagram.



? BENEFITS

Dot plots provides us the Global similarity between the two sequences and helps us to visualize the alignments of sequences and sequence repeats appear as diagonal stacks in plot.

? CONCLUSION

Dot plot help us to find the threshold difference among two sequences.

Module 016: IDENTY VS SIMILARITY

There are two concepts for sequence analysis

1. Identity

2. Similarity

Identity means the counting number of nucleotides or amino acids which exactly match when two biological sequences are matched.

For example:

Number of match = 5

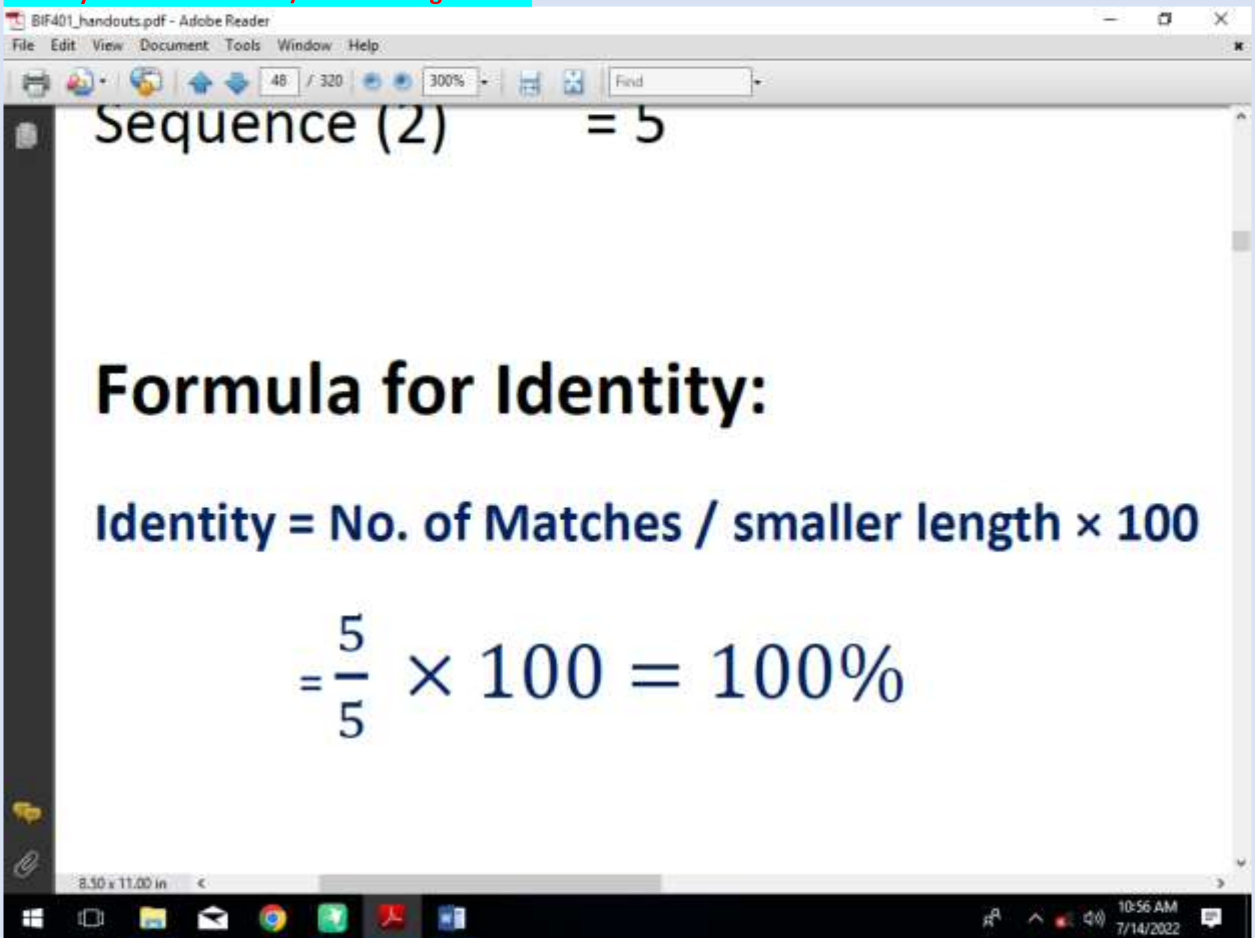
Smaller length = 5

Sequence (1) = 7

Sequence (2) = 5

Formula for Identity:

Identity = No. of Matches / smaller length × 100



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And **Similarity** means the comparison between two different sequences calculated by alignment approach.

In both identity and similarity the dots are not counted.

Module 017: INTRODUCTION TO ALIGNMENT APPROACHES

Two different approaches to align the sequence.

1. Global Alignment

2. Local Alignment

In **Local alignment** we compare one whole sequence with the one portion of other like this.

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
              |||| |||||| |||||||||||||||
          5' TACTCACGGATGAGGTACTTTAGAGGC 3'
```

While in **Global alignment** we compare both sequence from end to end completely.

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
   |||||||||||    |||||||  |||||||||||||| |||||||
5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'
```

Local alignment just focus on highly matching portions of sequence while in Global one whole sequence is compared with other one.

Module 018: WHY LOCAL ALIGNMENT

Because Local alignment have power to detect the smaller regions with high similarity and such matches are motifs or domains which remain hidden in case of protein function.

DOMAIN SHUFFLING

Aligned portions of sequence can be considered in varying orders and this process is called as domain shuffling.

** ADVENTAGES**

- We can compare the different length sequences
- Conserved domains can be determined from proteins
- Common function features can be identified.

** CONCLUSION**

Local alignment is used to compare the segments for high matching sequencing.

Module 019: ALIGNING, INSERTION & DELETION

Insertion means addition of amino acids in protein sequence and addition of nucleotides in DNA sequences.

And deletion means removal of amino acids from protein sequence and removal of nucleotides from DNA or RNA sequences.

- **ALIGNING INSERTION**
  1- Add gap 1st
  2- Add gap where we delete the nucleotide from sequence such insertion of gap is called as –ve or plenty.

Module 020: ALIGNING MUTATION IN SEQUENCES

Removal and addition of amino acids in proteins and nucleotides in DNA, RNA by using Gaps named as Indels.

Mutation is totally different from Indels, because in Mutation we replace the amino acid with other amino acids and replace the nucleotides with other and we don't use Gaps is inserted in template or target for mutation.
CONCLUSION

In identity alignment we use Gaps and in mutation we use substitution penalties and penalties depend upon the substitution.

Module 021: INTRODUCTION TO DYNAMIC PROGRAMMING

To find matching in nucleotides and amino acids of two sequences we use dot plot method. But dot plot cannot capture the insertions, deletions and gaps in the sequences.

We represent the matching nucleotides with +1 while gaps, substitutions, insertions and mutations can be represented as -1 in dot plot.

Dynamic programming is an algorithmic technique used commonly in sequence analysis.
Dynamic programming is used when recursion could be used but would be inefficient because it would repeatedly solve the same sub problems.

Handout Chapter 02: Sequence Analysis
Module 022: DYNAMIC PROGRAMMING ESSENTIALS

In algorithm we calculate the step involve in sequence compression for example if we compare two sequences of length "n" than it would be "n square "
And its order is O (n 2)

Handout Chapter 02: Sequence Analysis
Module 023: DYNAMIC PROGRAMMING METHODOLOGY

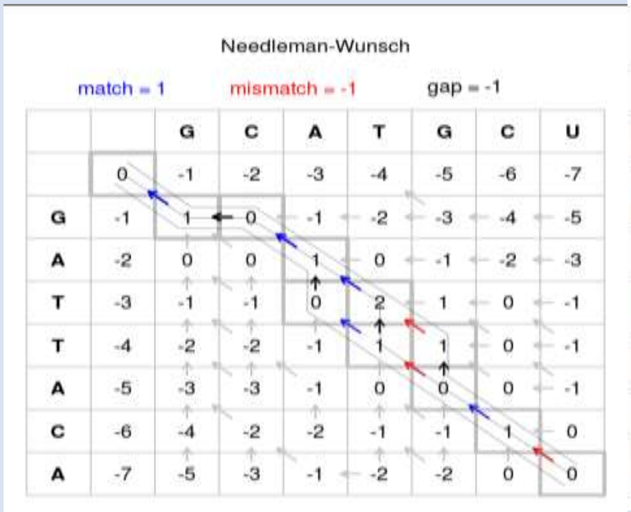
Dynamic programming helps us to reduce the computational cost in sequence comparisons and it works on the method of "scoring function".
For example
Match = +a
Mismatch = -b
Gap = -c
Score = #match + #Mismatch +#Gaps
Total =11
All the alignments are done in diagonal way in dot plot matrix. For total score we make calculations in diagonal way and after calculation best one is selected.

Handout Chapter 02: Sequence Analysis
Module 024: NEEDLEMAN WUNSCH ALGORITHM-I

In two different sequences alignments are arranged in a diagonal pattern of dot matrix. Needleman Wunsch Algorithm is the way for alignment. The method is same like dot plot but it computes the scores in different way. We Start with a zero in the second row, second column. Move through the cells row by row, calculating the score for each cell.

(Just take concept from below lines)

The score is calculated as the best possible score (i.e. highest) from existing scores to the left, top or top-left (diagonal). When a score is calculated from the top, or from the left this represents an indel in our alignment. When we calculate scores from the diagonal this represents the alignment of the two letters the resulting cell matches to. Given there is no 'top' or 'top-left' cells for the second row we can only add from the existing cell to the left. Hence we add -1 for each shift to the right as this represents an indel from the previous score. This results in the first row being 0, -1, -2, -3, -4, -5, -6, and -7. The same applies to the second column as we only have existing scores above.
Figure 11 Needleman-Wunsch pairwise sequence alignment



		G	C	A	T	G	C	U
	0	-1	-2	-3	-4	-5	-6	-7
G	-1	1	0	-1	-2	-3	-4	-5
A	-2	0	0	1	0	-1	-2	-3
T	-3	-1	-1	0	2	1	0	-1
T	-4	-2	-2	-1	1	1	0	-1
A	-5	-3	-3	-1	0	0	0	-1
C	-6	-4	-2	-2	-1	-1	1	0
A	-7	-5	-3	-1	-2	-2	0	0

		G	C	A	T	G	C	U
	0	-1	-2	-3	-4	-5	-6	-7
G	-1							
A	-2							
T	-3							
T	-4							
A	-5							
C	-6							
A	-7							

Handout Chapter 02: Sequence Analysis
Module 025: NEEDLEMAN WUNSCH ALGORITHM-II

Alignments are represented by unbroken diagonal dot matrix plot. In this way we can create numerous combinations.

If the sequence is too long then there will be many diagonal alignments and at the end we select the best alignment by combinations of all. And for this we use Needleman Algorithm
In Needleman Algorithm we use 0, 0 in first row and first column.

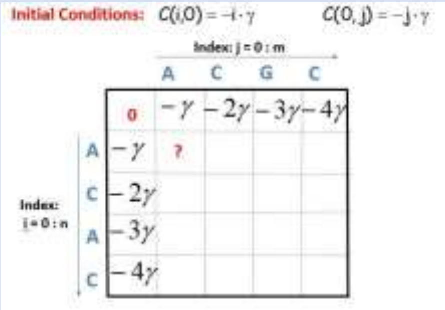


Figure 14 initial column and row are kept zero (0)

Left to right and top to bottom the best element (having high score) is selected.

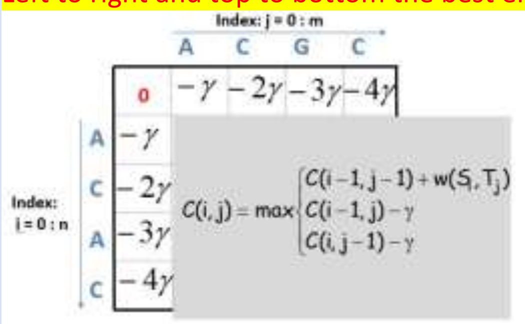


Figure 15 maximum score element is selected from all three sides comparison

The terms for match, mismatch are:
The matrix is computed progressively until the bottom right element
Alpha = Match reward
Beta = Mismatch penalty
Gamma = Gap penalty

Module 026: NEEDLEMAN WUNSCH ALGORITHM-III

We follow the diagonal route for scoring in Needleman Wuncsh Algorithm. Left to right and top to bottom in diagonal way we select the highest score.

Module 027: NEEDLEMAN WUNSCH ALGORITHM EXAMPLE

Top left and diagonal element are considered to calculate an element in the matrix. Match, mismatch and gap penalty is computed from all there sides (Left to right) (Top to bottom) and (Diagonal).
For example:

+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5						
A	-10								
T	-15								
T	-20								
C	-25								
A	-30								
C	-35								

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Figure 17 score computation from all sides) Figure 18 the best score is computed in diagonal way

DNA, RNA and Protein sequences can be computed by using Needleman algorithm.

Module 028: BACKTRACKING ALIGNMENTS

To find an optimal alignment in Needleman Wunsch Algorithm we use traceback method.
Trace back method starts from end maximum score.

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Handout Chapter 02: Sequence Analysis
Module 029: REVISITING LOCAL AND GLOBAL ALIGNMENTS

We use Needleman Algorithm to align the sequences in scoring way and traceback method to find the optimal alignment.

In Needleman Algorithm we start traceback from bottom right towards top left progressively and this provides us the global alignment.

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

We can start traceback from any point in matrix and smith waterman algorithm helps elicit the local alignments.

Handout Chapter 02: Sequence Analysis
Module 030: OVERLAP MATCHES

Dot plot and Needleman wucsch are algorithm method with little difference. Dot plot help us in finding matching residues of two sequences while Needleman wunsch helps us to find the global alignments.
If some sequences have different regions of nucleotides which does not match to any other for that alignment we prefer Global alignment not local, but that does not penalize leading or trailing end.

And "Traceback" is the technique by which we can check the sequences from any end of the matrix box. And such "Tracebacks" helps us to find the overlaps in aligned sequences.

Module 031: EXAMPLE

A slight variation in traceback can helps us to find the overlaps in sequences and can apply some interesting strategies in sequences alignments.
In following example of amino acids alignment we can understand the ways of tracback.

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	−2	−1	−1	**−2**	−1	−4	−2	−2	−1	−1
A	0	−2	−2	4	−1	**3**	−4	−4	−4	−3	−2
W	0	−3	−5	−4	1	−4	**18**	**10**	2	6	−6
H	0	10	2	6	−6	−1	10	16	**20**	12	4
E	0	2	16	8	0	7	2	8	16	**26**	18
A	0	−2	8	21	13	5	3	2	8	18	**25**
E	0	0	4	13	18	12	4	4	2	14	24

Figure 23 Traceback in amino acid sequence alignment

Scoring stagey is:
Match = +2. Mismatch = -1, Gap = -2

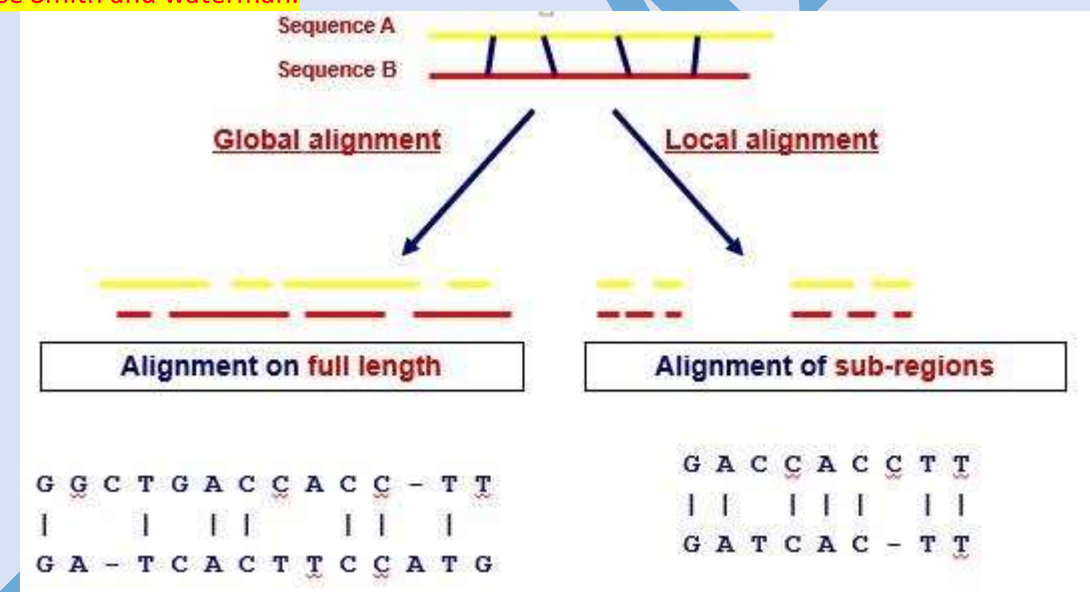
```
GAWGHEE
PAW-HEA
```

Handout Chapter 02: Sequence Analysis
Module 032: MOVING FROM GLOBAL TO LOCAL ALIGNMENT

DNA has coding and noncoding regions. Coding regions are called "EXON" expressed as protein and they remain more conserved due to their role in making functional proteins.
And noncoding regions of DNA are called as "INTRONS" which are more likely involved in mutations than coding ones. It means high degree of alignment can be find among two exons.
In local alignment we use small segments of sequences and through which we can find exons.
Through this we can find "functional subunits".
However, the term exon is often misused to refer only to coding sequences for the final protein.
This is incorrect, since many noncoding exons are known in human genes.

Handout Chapter 02: Sequence Analysis
Module 033: SMITH WATERMAN ALGORITHM

In global alignment we compare the sequence from end to end but in local alignment we compare the sequences in segments.
For Global sequences we use Needleman and Wunsch algorithm while for local pairwise alignment we use Smith and waterman.



The Smith Freshman algorithm is different from Needle man.
- Top row and Colum are set to zero.
- Alignment can end anywhere.
- Traceback starts from highest score.

Handout Chapter 02: Sequence Analysis
Module 034: EXAMPLE OF SMITH WATERMAN ALGORITHM

The only difference between Needleman and Smith Waterman is that zero "0" is placed in the relationship.

And in the matrix we place top line of zero and first Colum of zero.

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0								
A	0								
T	0								
T	0								
C	0								
A	0								
C	0								

+1 for a match, -1 for a mismatch, -5 for a space

Figure 25 top line and first Colum are filled with zero in Smith Freshman Algorithm

Local alignments can be extracted by starting from a high score till reaching '0'

Handout Chapter 02: Sequence Analysis
Module 035: REPEATED ALIGNMENTS

We can find the best local alignments by using Smith Waterman algorithm.

By making some change in strategy of traceback we can find the repeated sequences. We use threshold "T" score for matching and it avoids low scoring local alignment. And traceback can help us to find multiple aligned regions in multiple ways.

Standard Smith Waterman Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0	1	0	1	0	1	0	0	1
A	0	0	0	0	0	0	2	0	0
T	0	0	1	0	0	0	0	1	0
T	0	0	1	0	0	0	0	0	0
C	0	1	0	2	0	1	0	0	1
A	0	0	0	0	1	0	2	0	0
C	0	1	0	1	0	2	0	1	1

+1 for a match, -1 for a mismatch, -5 for a space

Figure 27 (-5) is threshold score in table

This threshold scoring method with some modifications in waterman algorithm can help us to find many matching sequence of amino acids or DNA.

Handout Chapter 02: Sequence Analysis

Module 036: EXAMPLES OF REPEATED ALIGNMENTS

Slight modification in waterman model can help us to find the Exons as well as the functional units in any sequence. Matches should be end at the threshold score or we should keep track of maximum score in sequence.

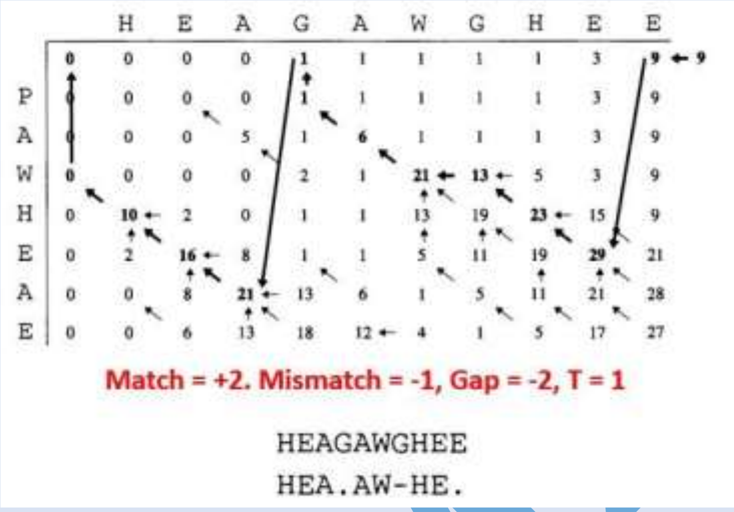


Figure 28 Trackback from different sides to find maximum or Threshold (T) Score.

Traceback should start from last element of the row and should reach at the top of row element and then move to the highest score of the Column. And this traceback is done twice and end at the point where score become “0, 0”

Module 037: INTRODUCTION TO SCORING ALIGNMENTS

There are two types of alignments;
- Optimal Alignments
- Best Alignment

Scoring scheme used in sequences matches play crucial role in producing optimal alignment. An optimal alignment should be:
- appropriately rewarded for matches and mismatches.

- INTRODUCTION

We identify the pairs of symbol which most frequently appear in a sequences it helps us to find the substitution of specific pair of amino acid or nucleotide with other on in a sequence.
For example AA nucleotide have a specific pattern of substitution. And same pairs of amino acids does in protein sequences because it can help to preserve the function of protein.

Module 038: MEASURING ALIGNMENTS SCORES

Score of match and mismatch both are equally observed while sequence alignments.

Module 039: SCORING MATRICES

Alignments are used to align the biological sequences. Amino acids and nucleotides are more easily substituted because they have similar chemical nature.
Scoring Matrices have both values +ve and –ve. Positive value for matches and negative value for mismatches.

Module 040: DERIVING SCORING MATRICES

Each amino acid have different property.

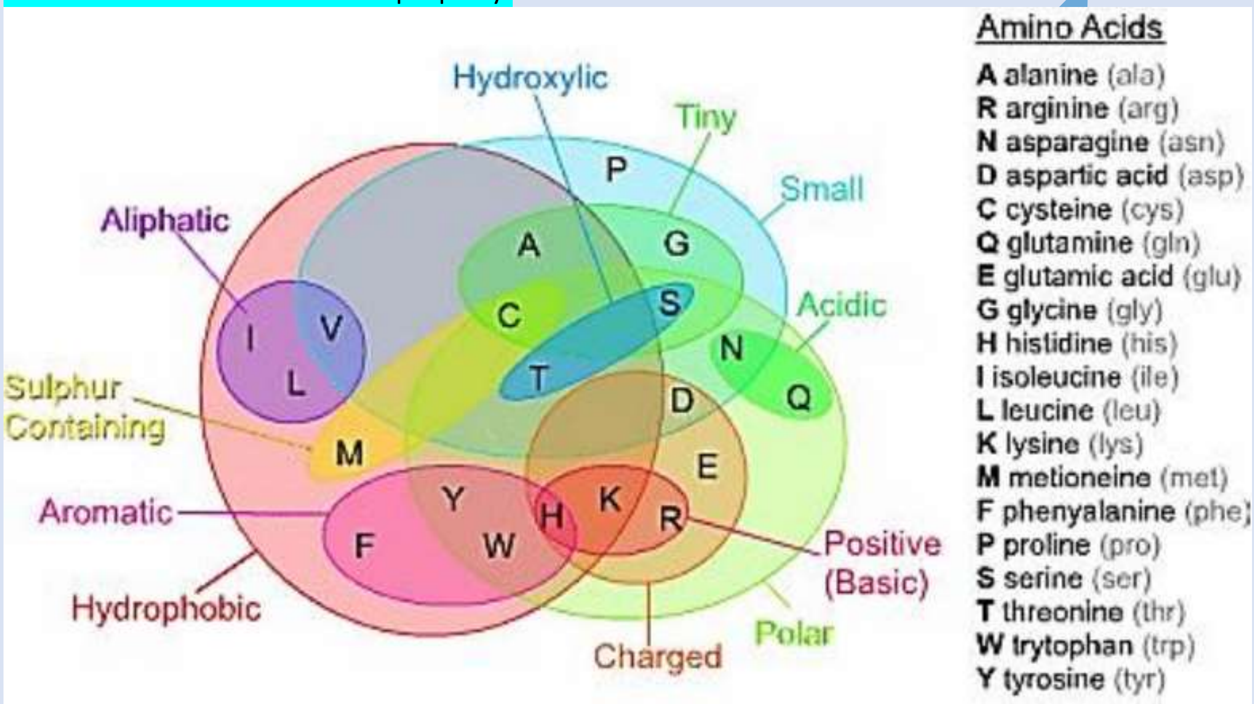


Figure 0.13 properties of amino acids (Image Esquivel et al. (2013)

And each amino acid have different frequency.

Table 7.1 The average amino acid frequencies reported by Robinson and Robinson in [173].

Amino acid	Freqency	Amino acid	Freqency	Amino acid	Freqency	Amino acid	Freqency
Ala	0.078	Gln	0.043	Leu	0.090	Ser	0.071
Arg	0.051	Glu	0.063	Lys	0.057	Thr	0.058
Asn	0.045	Gly	0.074	Met	0.022	Trp	0.013
Asp	0.054	His	0.022	Phe	0.039	Tyr	0.032
Cys	0.019	Ile	0.051	Pro	0.052	Val	0.064

Module 041: PAM MATRICES

Alignment matrices scoring is very useful method to score the sequences alignment for match and mismatch.
There are two types of scoring matrices.
- PAM

 BLOSUM

PAM means “Point Accepted Mutation”

Point accepted mutations means the substitution of one amino acid in a sequence with another that protein function remain conserved.

 PAM UNIT

PAM unit is actually that time during which 1% amino acid undergo for acceptable mutation. If two sequences diverge by 100 PAM units, it does not mean that they will be at totally different positions.

STEP TO COMPUTE PAM MATRICES

1. Align the protein sequence which are 1-PAM Unit diverge.
2. Let Ai,i be the number of times Ai is substituted by Ai.
3. Compute the frequency fi of amino acid Ai

Then, PAM1=pii= $\frac{A_{ij}}{\sum_k A_{ik}}$

PAM ‘n’= (PAM1) n

Module 042: BLOSUM MATRICES

BLOSUM matrices can be used to align the protein sequences. BLOSUM matrices was first purposed in 1992 by Henikoff et al.

BLOSUM matrices is also called the Block substitution matrix without any gap although it has mismatches in sequences.

There are three steps to compute the BLOUSM Matrices.

Step 1: Eliminate sequences that are identical in x% positions

Step 2: Compute observed frequency fi, j of aligned pair Ai to Aj. Hence, fi,j becomes the probability of aligning Ai and A j in the selected blocks.

Step 3: Compute fi which is the frequency of observing Ai in the entire block

$$s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}, \quad p_{ij} = \begin{cases} f_i f_i & i = j \\ 2 f_i f_j & i \neq j. \end{cases}$$

Typically used matrices: BLOSUM62 or PAM120 in PAMx, larger x detects more divergent sequences.

Module 043: MULTIPLE SEQUENCE ALIGNMENT

In pair wise sequence alignments we use pairs of sequence to compare them. And scoring matrices were used to score the sequence ranks.

In Multiple sequence Alignments we compare multiple number of protein and DNA sequences to identify the matches and mismatches.

For pair wise alignment we use Dynamic programming but for multiple alignment it would be very expensive computationally. So solution for this is progressive alignment.

M Q V K L F T P L H D K S D H G K Y H
M Q V K I F T P L H D K S - H G K S H
M Q V H L Y - P L H D K S - T G K S H
M Q V H L F - P L H D K S D T G K S H
M Q V K L Y T P L H D K S D H G K Y H

Figure 0.16 multiple sequence alignment

Module 044: MORE ON MULTIPLE SEQUENCE ALIGNMENT (MSA)

MSA helps compare several sequences by aligning them. MSA can extract consensus sequences from several aligned sequences. Characterize protein families based on homologous regions.

APPLICATION OF MSA
 Predict secondary and tertiary structures of new protein sequences
 Evaluate evolutionary order of species or "Phylogeny"
METHODOLOGY
 Pairwise alignment is the alignment of two sequences
 MSA can be performed by repeated application of pairwise alignment

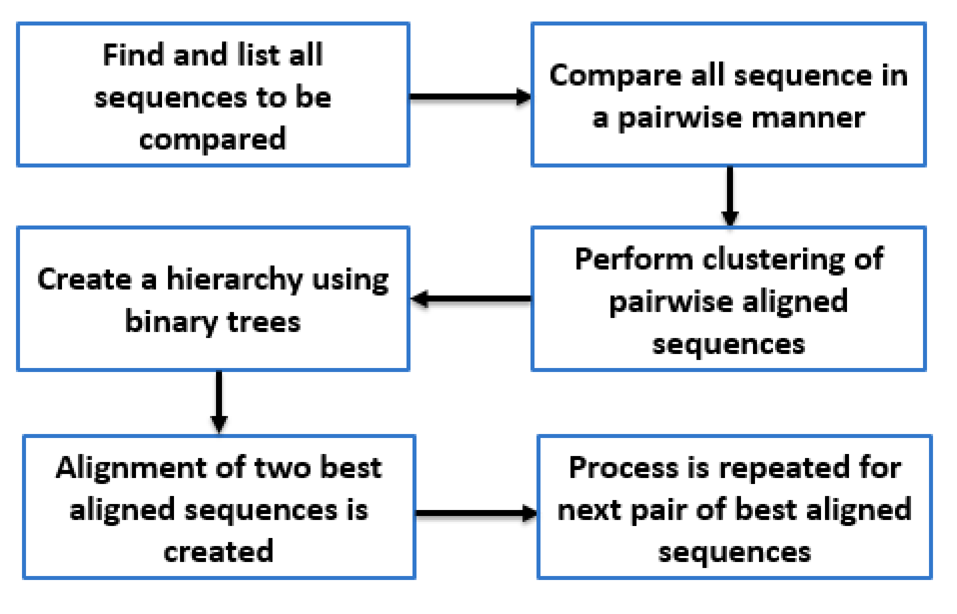


Figure 0.17 Methodology

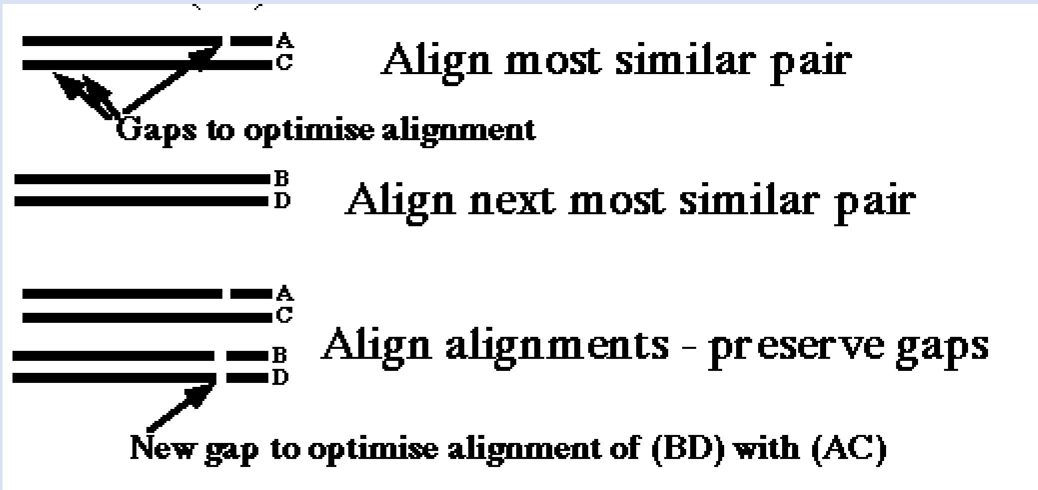


Figure 0.18 sequence alignments

CONCLUSION

MSA can help align multiple sequences. Progressive alignment can help perform MSA. Need to remove sequences with >80% similarity.

Module 045: PROGRESSIVE ALIGNMENT FOR MSA

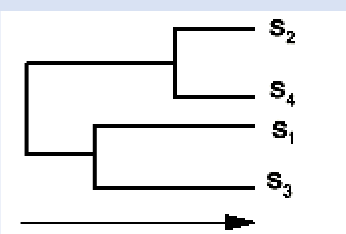
MSA involves progressive alignment of sequences. Doing so many progressive alignments can be slow.

STEPS:

 Step 1 : Pairwise Alignment of all sequences

Example: S_1 , S_2 , S_3 , S_4, so that is 6 pairwise comparisons.

 Step 2: Construct a Guide Tree (Dendogram) using a Distance Matrix.



$$Distance = 1 - \frac{Matches}{Comparisons}$$

$$Similarity = \frac{Matches}{Comparisons}$$

 Step 3: Progressive alignment following branching order in tree.

$$Similarity = \frac{Matches}{Comparisons}$$

	v_1	v_2	v_3	v_4
v_1	-			
v_2	.17	-		
v_3	.87	.28	-	
v_4	.59	.33	.62	-

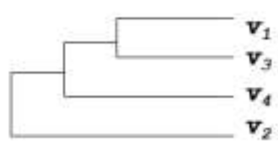


Figure 0.20 Similarity Matrix

SHORTCOMING OF THIS APPROACH

 Dependence upon initial alignments

- If sequences are dissimilar, errors in alignment are propagated
- Solution: Begin by using an initial alignment, and refine it repeatedly

Progressive alignments are used in aligning multiple sequences. Iterative approaches can help refine results from progressive alignments.

Module 046: MSA-EXAMPLE

MSA involves progressive alignment of sequences. Doing so many progressive alignments can be slow.

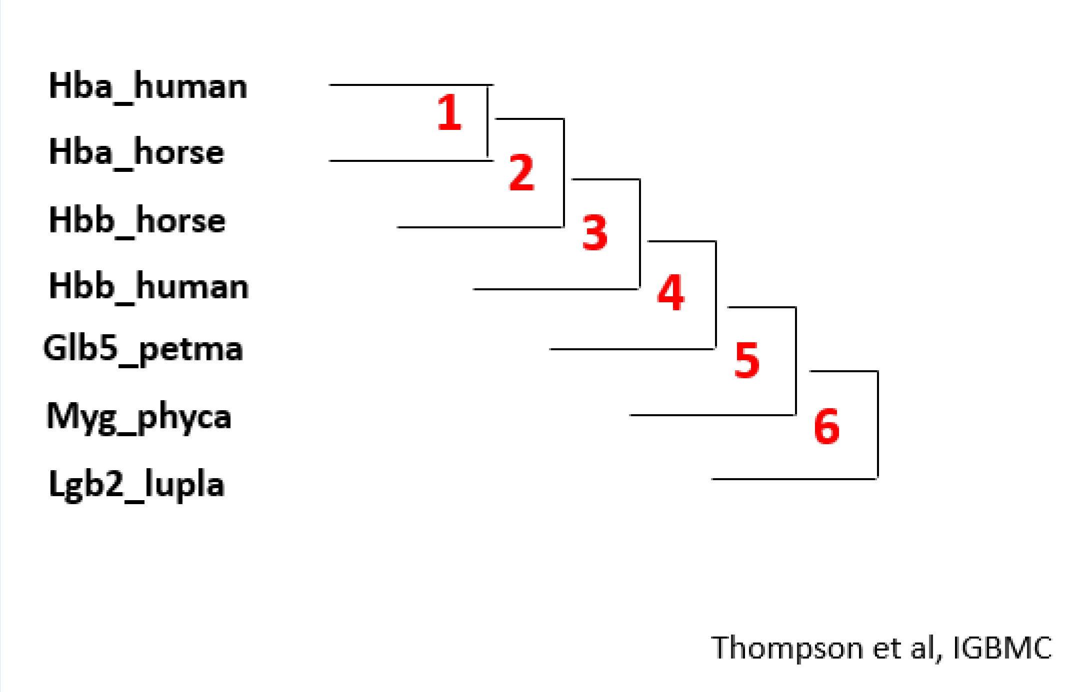
For example:

$$\text{distance between 2 sequences} = 1 - \frac{\text{No. identical residues}}{\text{No. aligned residues}}$$

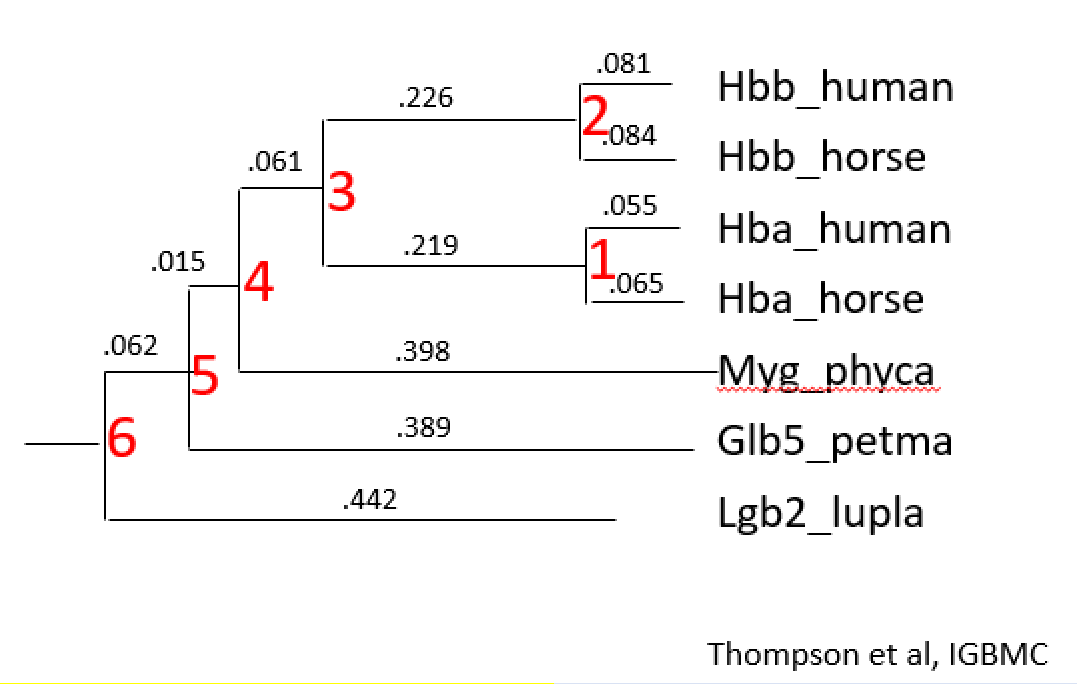
		Hbb_human	Hbb_horse	Hba_human	Hba_horse	Myg_phyca	Glb5_petma	Lgb2_lupla
Hbb_human	1	-						
Hbb_horse	2	.17	-					
Hba_human	3	.59	.60	-				
Hba_horse	4	.59	.59	.13	-			
Myg_phyca	5	.77	.77	.75	.75	-		
Glb5_petma	6	.81	.82	.73	.74	.80	-	
Lgb2_lupla	7	.87	.86	.86	.88	.93	.90	-

Thompson et al, IGBMC

Figure 0.21 MSA on globin sequences



Progressive alignment using sequential branching



Progressive alignment following a guide tree

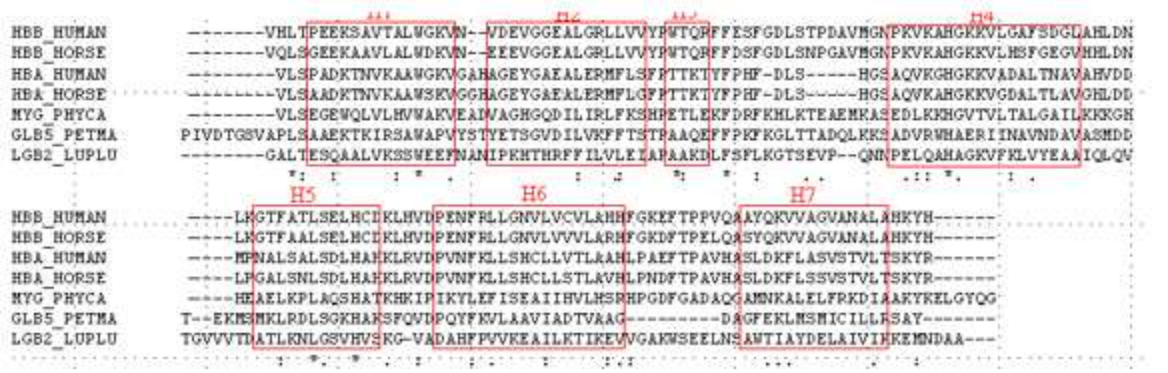


Figure 0.24 Alignment results

MSA can be better performed using clustering strategies followed by alignment of the alignments later. CLUSTAL is a free online tool that does all of this for us!

Module 047: CLUSTALW

MSA involves progressive alignment of sequences. Doing so many progressive alignments can be slow. CLUSTALW is an online tool to perform MSA.
Developed by European Molecular Biology Laboratory & European Bioinformatics Institute.
Performs alignment in:
- slow/accurate
- fast/approximate

SCOPE
- create multiple alignments,
- optimize existing alignments,
- Profile analysis &
- create phylogenetic trees

Module 048: INTRODUCTION TO BLAST-I

National Center for the Biotechnology Information (NCBI) – USA. BLAST developed in 1990. "Basic Local Alignment Search Tool". Searches databases for query protein and nucleotide sequences. Also searches for translational products etc. Online availability

BLAST can be used to search for local alignment of protein and nucleotide sequences. It is available online. Can perform searches across species and organisms

Module 049: INTRODUCTION TO BLAST-II

Smith Waterman can align complete sequences. BLAST does it in an approximate way. Hence, BLAST is faster BUT does not ensure optimal alignment. BLAST provides for approximate sequence matching. Input to BLAST is a FASTA formatted sequence and a set of search parameters
OUTPUT OF BLAST
Results are shown in HTML, plain text, and XML formats. A table lists the sequence hits found along with scores. Users can read this table off and evaluate results

Module 050: BLAST ALGORITHM

BLAST can search sequence databases and identify unknown sequences by comparing them to the known sequences. This can help identify the parent organism, function and evolutionary history.
For example: (Just read)
Query sequence: PQGELV
Make list of all possible worlds (length 3 for proteins)
PQG (score 15)
QGE (score 9)
GEL (score 12)
ELV (score 10)
Assign scores from Blosum62, use those with score> 11: PQG & GEL
Mutate words such that score still > 11
PQG (score 15) similar to PEG (score 13)

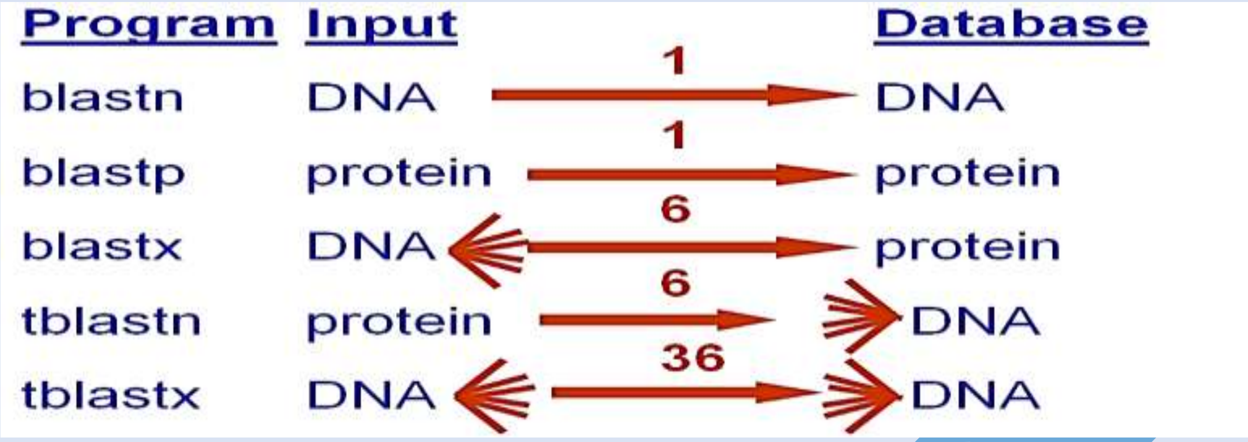
Query:	M	E	T	P	Q	G	I	A	V
Database:	-	-	-	P	Q	G	E	L	V
				8	5	5	2	0	8

At the end, we get: PQG, GEL and PEG
Find all database sequences that have at least 2 matches among our 3 words: PQG, GEL & PEG.
Find database hits and extend alignment (High-scoring Segment Pair):
High Scoring Pair: PQGI (score 8+5+5+2)
If 2 HSP in query sequence are < 40 positions away
Full dynamic alignment on query and hit sequences
BLAST performs quick alignments on sequences

Module 051: TYPES OF BLAST (important topic for quiz and short questions)

There are two main types of BLAST.
Nucleotides
• Blastn: Compares a nucleotide query sequence against a nucleotide database.
Proteins
• Blastp: Compares an amino acid query sequence against a protein database.

There are also many other types of BLAST:
▯ Blastx:
Compares a nucleotide query sequence against a protein sequence database.
Helps find potential translation products of unknown nucleotide sequences
▯ tblastn:
Compares a protein query sequence against a nucleotide sequence database
Nucleotide sequence dynamically translated into all reading frames
▯ tblastx:
Compares the six-frame translated proteins of a nucleotide query sequence against the six- frame translated proteins of a nucleotide sequence database.



Handout Chapter 02: Sequence Analysis
Module 052: SUMMERY OF BLAST

Step1: obtain a query of sequence
Step2: choose a type of BLAST
Step3: search parameter
Step4: tabulated search results
Figure 0.28 tabulated search results

Handout Chapter 02: Sequence Analysis
Module 053: INTRODUCTION TO FASTA

For comparing two sequences we use pair wise sequencing and for the comparison of many sequences we use multiple sequence alignment. To handle the multiple alignments we perform alignment through smith-waterman algorithm for local one. And for global alignment we use Needleman-wunsch algorithm.
Both local and global alignments are the dynamic approaches.
FASTA took a similar approach.
Developed in 1988. It does Fast Alignment. Searches databases for query protein and nucleotide sequences. Was later improved upon in BLAST.

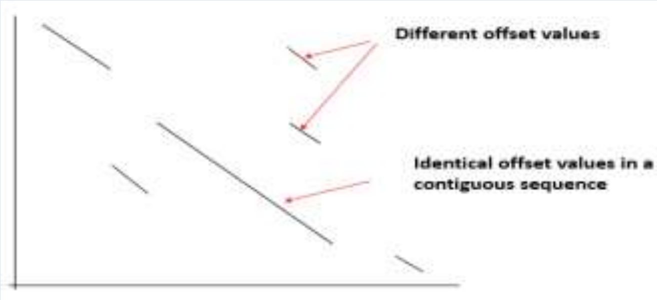
Handout Chapter 02: Sequence Analysis
Module 054: INTRODUCTION TO FASTA-II

FASTA – Fast Alignment Algorithm. Classical global and local alignment algorithms are time consuming. FASTA achieves alignment by using short lengths of exact matches.

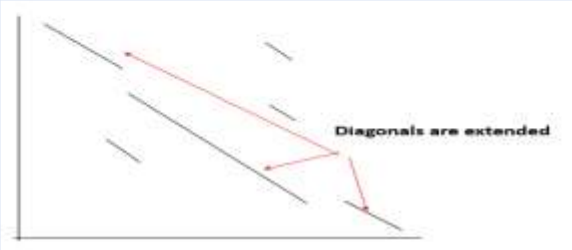
 USES OF FASTA
FASTA relies on aligning subsequences of absolute identity. Input to FASTA search can be in FASTA, EMBL, GenBank, PIR, NBRF, PHYLIP or UniProt formats
 OUTPUT OF BLAST
Results are output in visual format along with functional prediction. Makes table lists the sequence hits found along with scores.

Module 055: FASTA ALGORITHM (remember the points)

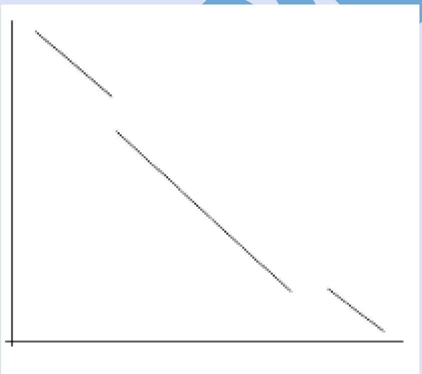
STEP1: Local regions of identity are found



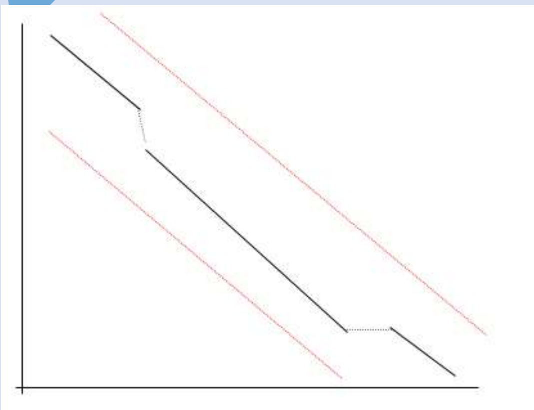
STEP2: Rescore the local regions using PAM or BLOSUM matrix



STEP3: Eliminate short diagonals below a cutoff score



STEP4: Create a gapped alignment in a narrow segment and then perform Smith Waterman alignment.



Module 056: TYPES OF FASTA (important lectures)

There are six types of FASTS:
 fasts35
Compare unordered peptides to a protein sequence database
 fastm35
Compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database
 Fasta35
Scan a protein or DNA sequence library for similar sequences
 Fastx35
Compare a translated DNA sequence (6 ORFs) to a protein sequence database
 tfastx35
Compare a protein sequence to a DNA sequence database (6 ORFs)
 fasty35
Compare a DNA sequence (6ORFs) to a protein sequence database

Module 058: BIOLOGICAL DATABASE AND ONLINE TOOLS

All molecular information of RNA, DNA, Proteins have need to be stored and retrieved.
Sequences are obtained from genome sequencing and mass spectrometry
Structures are obtained from X-Ray Crystallography, Atomic Force Microscopy & Nuclear Magnetic Resonance Spectroscopy.
Online Databases

 OBJECTIVE
 Make biological data available to scientists in computer-readable form
 For handling, sharing and analysis of the data
 The best way to share is to keep this data on the web

Module 059: EXPASY

It is developed by Swiss Bioinformatics Institute (SIB). Website provides access to databases and tools Proteomics, genomics, phylogeny, systems biology, population genetics, transcriptomics etc. can be searched.
http://www.expasy.org/

Module 060: UNIPROT AND SWISSPROT

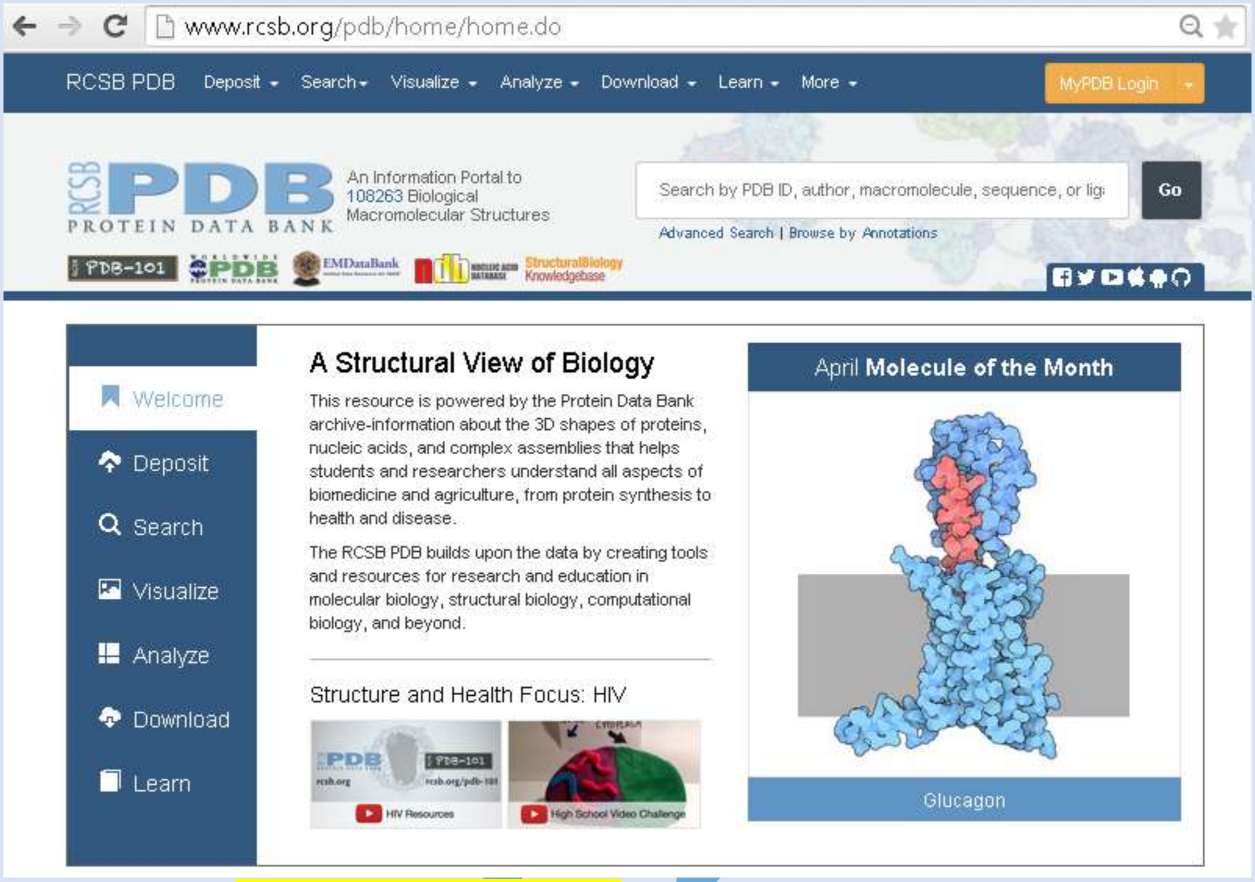
Both UniProt and SwissProt are the online database for proteins.

Swiss-Prot contains human curated protein information
- Accession number, unique identifier
- The sequence
- Molecular mass
- Observed and predicted modifications

Protein sequences from various species and organisms can be found in uniprot. SwissProt is the manually annotated version of the UniProt Database.

Module 061: PROTEIN DATA BANK

Protein Data Bank is the premier resource of protein structures. These structures have been determined using experimental techniques. It's Open & Free



Protein Data Bank provides Cartesian coordinates of each atom in the protein structure. Over 50,000 protein structures are reported and present in this database

Module 062: REVIEW OF SEQUENCE ALIGNMENT

We use next generation sequencing and whole genome sequencing to obtain the genetic information. For protein sequencing we use Mass Spectrometry and Edman Degradation

STORAGE:
 Sequence information is stored digitally
 Databases are designed to store sequence data
 Several databases exist depending on the type of sequence data

SHARING AND ACCESS:
 Sequence databases are shared via online websites
 Access to several such websites is free
 Data can be downloaded or searched on these website

USAGE OF DATA:
Sequence data can be used to obtain:
 Similarity of sequences
 Evolutionary History

 Predict the function of molecules

Handout Chapter 02: Sequence Analysis
Module 063: GENBANK

 Developed by Swiss Bioinformatics Institute (SIB)
 Website provides access to databases and tools
 Proteomics, genomics, phylogeny, systems biology, population genetics, transcriptomics etc.

http://www.ncbi.nlm.nih.gov/genbank/

Module 064: ENSEMBLE
For genome finding.
ESEMBLE is genome search engine which is used to search the genome of every recorded species.
http://asia.ensembl.org/index.html

Chapter 3 - Molecular Evolution
Module 001: Molecular Evolution & Phylogeny

Molecular evolution is the process of change in the sequence composition of cellular molecules such as DNA, RNA, and proteins across generations.

The field of molecular evolution uses principles of evolutionary biology and population genetics to explain patterns in these changes.

Phylogenetic is the science of studying evolutionary relationships. Phylogenetic has led to the creation of relationship trees between various species of Bacteria, Achaea, and Eukaryote.

Types of Phylogenetic Trees
Scaled Trees
• Branch lengths are equal to the magnitude of change in the nodes
Unscaled Trees
• Only representing the relationship between sequences

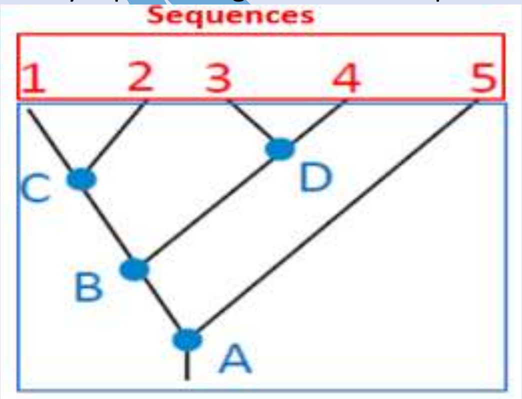


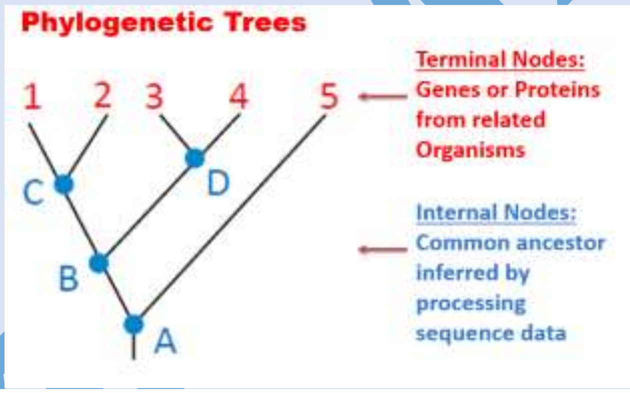
Figure 0.2 phylogenetic tree interference

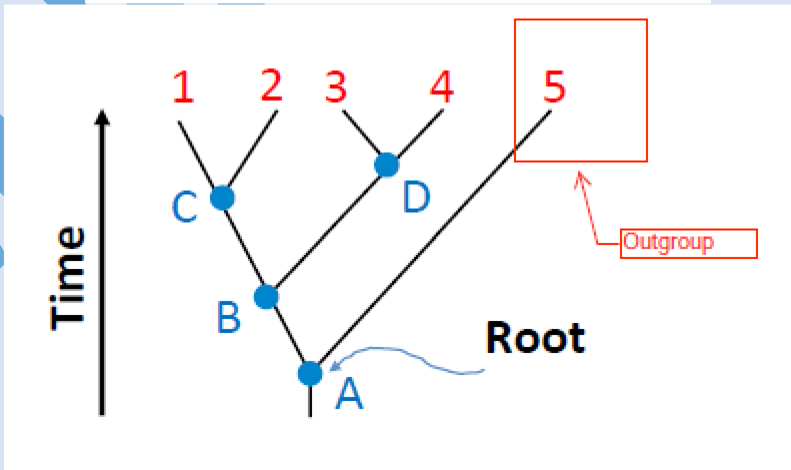
Handout Chapter 03: Molecular Evolution
Module 002: Evolution of Sequences

DNA acts as cellular memory unit and protein are the translated product of DNA coded information.
And evaluation is very important to survive in different type of environments.
There are some methods which brings change or evolution in any organism. (Kluger 2015)
Method of Change
DNA gets modified by:
- Mutation & Substitution
- Insertion
- Deletion

Handout Chapter 03: Molecular Evolution
Module 003: Concepts and Terminologies – I

Phylogenetic involves processing sequence information from different species to find evolutionary relationships.
Root node is the ancestor of all other nodes. The direction of evolution is from ancestor to the terminal nodes.

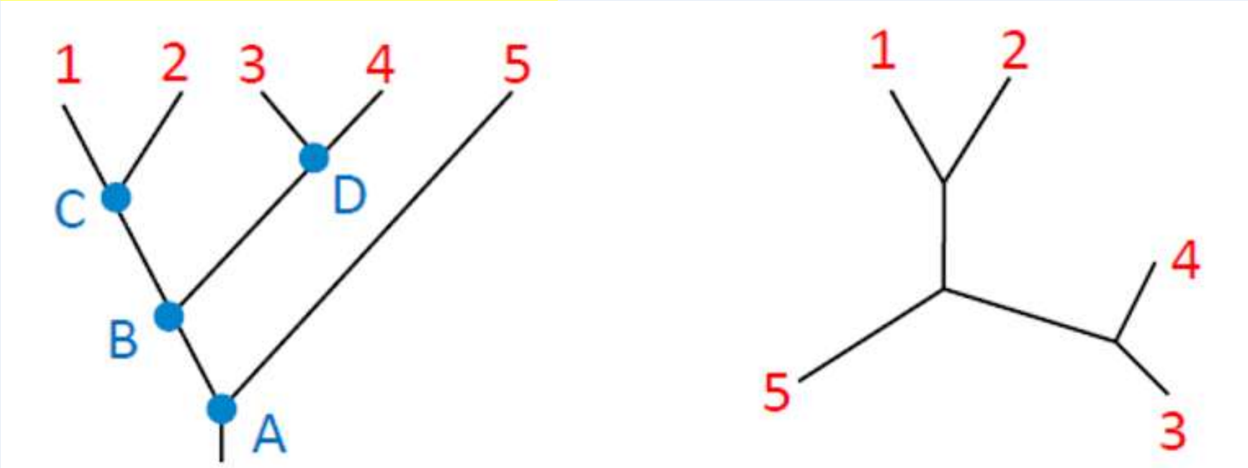




Conclusion
Phylogenetic specifies evolutionary relationship with the help of trees. Trees can be rooted or unrooted. Rooted trees can show temporal evolutionary direction.

Handout Chapter 03: Molecular Evolution
Module 004: Concepts and Terminologies – II

Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences. Rooted trees are computationally expensive.



Rooted and Unrooted trees have their own advantages and disadvantages. Depending on our requirement, we can choose between them.

Handout Chapter 03: Molecular Evolution
Module 005: Algorithms and Techniques

UPGMA (Unweighted Pair Group Method with Arithmetic Mean) is a simple agglomerative (bottom-up) hierarchical clustering method. The method is generally attributed to Sokal and Michener.

Least Squares Distance Method. Branch lengths, represent the "observed" distances between sequences (i & j).

Module 006: Introduction to UPGMA

Phylogenetic trees can be used to show phylogenetic relationships between sequences. To construct these trees, several types of algorithms exist which are divided into two classes.

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

- **Calculating distance between two clusters:**

Cluster X + Cluster Y = Cluster Z

Calculate the distance of a cluster (e.g. W) to the new cluster Z

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

N_x is the number of sequences in cluster x

- **Calculating distance between two trees:**

Assume we have N sequences

Cluster X has N_X sequences, cluster Y has N_Y sequences

d_{XY} : the evlotionary distance between X and Y

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

Methods for constructing trees

Introduction to UPGMA

Methods for constructing trees

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1 * 6 + 1 * 6}{1 + 1} = 6$$

Module 007: UPGMA-I

(Just take concept of all Upgma topics)

UPGMA has two components to it. These include distance calculations between two clusters and between two trees.

□ Building trees using UPGMA

Combining Clusters: Cluster X + Cluster Y = Cluster Z

Calculate the distance of each cluster (e.g. W) to the new cluster Z

UPGMA - I

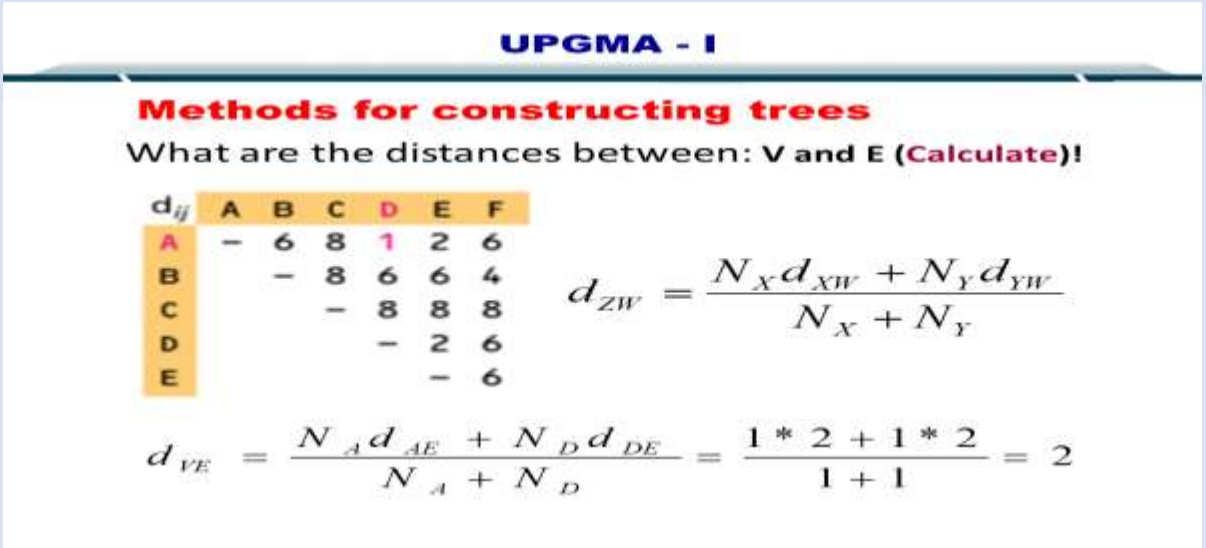
Methods for constructing trees

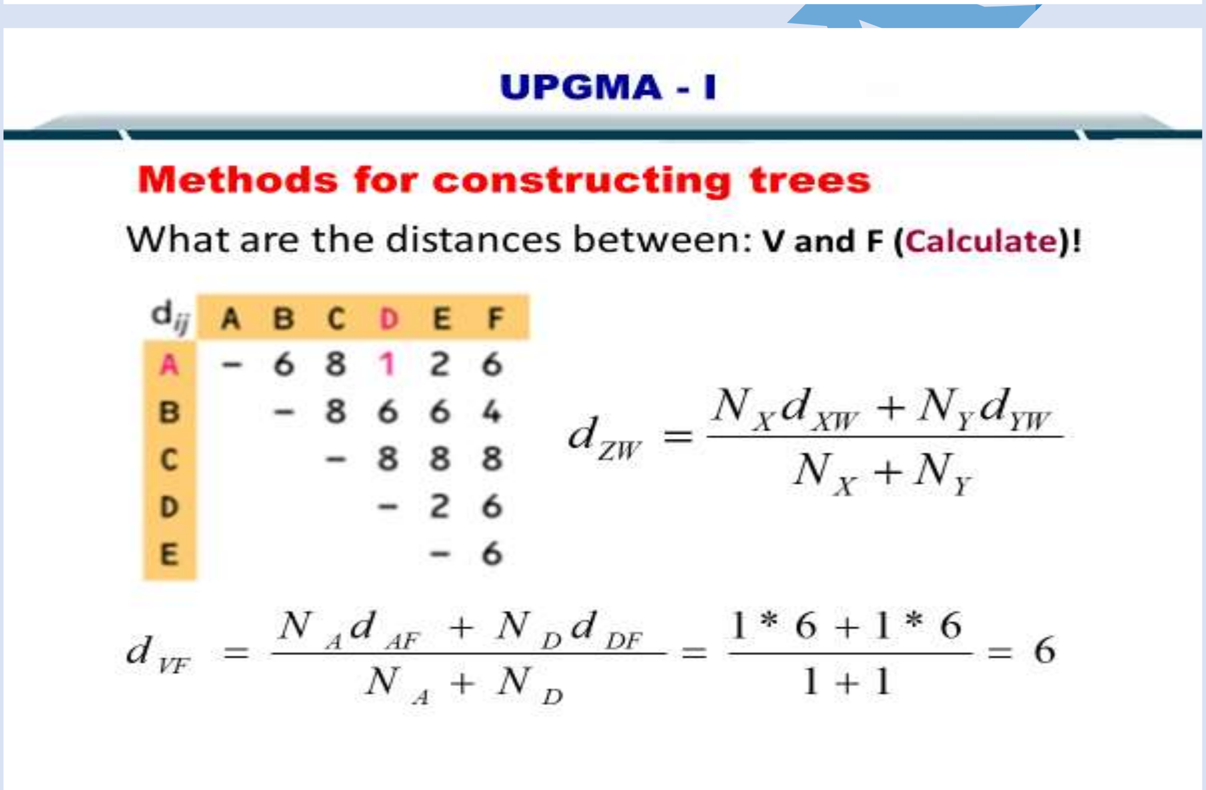
What are the distances between: **V and C (Calculate)!**

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1*8 + 1*8}{1+1} = 8$$





UPGMA starts with creating clusters of sequences which are the closest.

Module 008: UPGMA-II

UPGMA steps include distance calculations between two clusters and between two trees. We formed clusters from sequences which had the shortest distance.

Building trees using UPGMA

The distance matrix is obtained using pairwise sequence alignment.

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

→

d_{ij}	B	C	E	F	V
B	–	8	6	4	6
C		–	8	8	8
E			–	6	2
F				–	6

UPGMA - II

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1*6+1*6}{1+1} = 6$$

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1*8+1*8}{1+1} = 8$$

$$d_{VE} = \frac{N_A d_{AE} + N_D d_{DE}}{N_A + N_D} = \frac{1*2+1*2}{1+1} = 2$$

$$d_{VF} = \frac{N_A d_{AF} + N_D d_{DF}}{N_A + N_D} = \frac{1*6+1*6}{1+1} = 6$$

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

V – E becomes a new cluster let's say W

Handout Chapter 03: Molecular Evolution

Module 009: UPGMA-III

Building trees using UPGMA

Combining Clusters: Cluster X + Cluster Y = Cluster Z. Calculate the distance of each cluster (e.g. W) to

Calculating the distance between two trees

 Assume we have N sequences

- Cluster X has N X sequences, cluster Y has N Y sequences
- D XY: the evolutionary distance between X and Y

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

V – E becomes a new cluster lets say W. Now we have to modify the distance matrix again.

What are the distances between:

W and B, W and C,

W and F.

New matrix

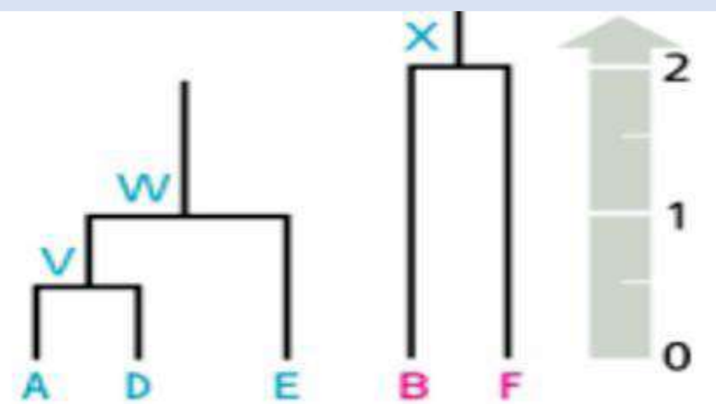
$$d_{WB} = \frac{N_V d_{VB} + N_E d_{EB}}{N_V + N_E} = \frac{2*6+1*6}{2+1} = 6$$

$$d_{WC} = \frac{N_V d_{VC} + N_E d_{EC}}{N_V + N_E} = \frac{2*8+1*8}{2+1} = 8$$

$$d_{WF} = \frac{N_V d_{VF} + N_E d_{EF}}{N_V + N_E} = \frac{2*6+1*6}{2+1} = 6$$

Cluster according to min distance

d_{ij}	B	C	F	W
B	–	8	4	6
C		–	8	8
F			–	6



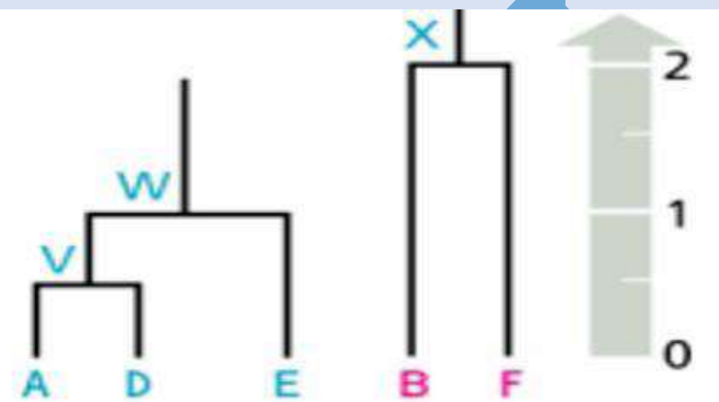
Now we have formed three clusters. Also, two separate trees have been formed. Next, we need to join these trees to create a complete tree.

Handout Chapter 03: Molecular Evolution

Module 010: UPGMA-IV

Application of UPGMA resulted in formation of two sub-trees. The need now was to join them into a single tree. Let's see how that is done.

d_{ij}	B	C	F	W
B	–	8	4	6
C		–	8	8
F			–	6



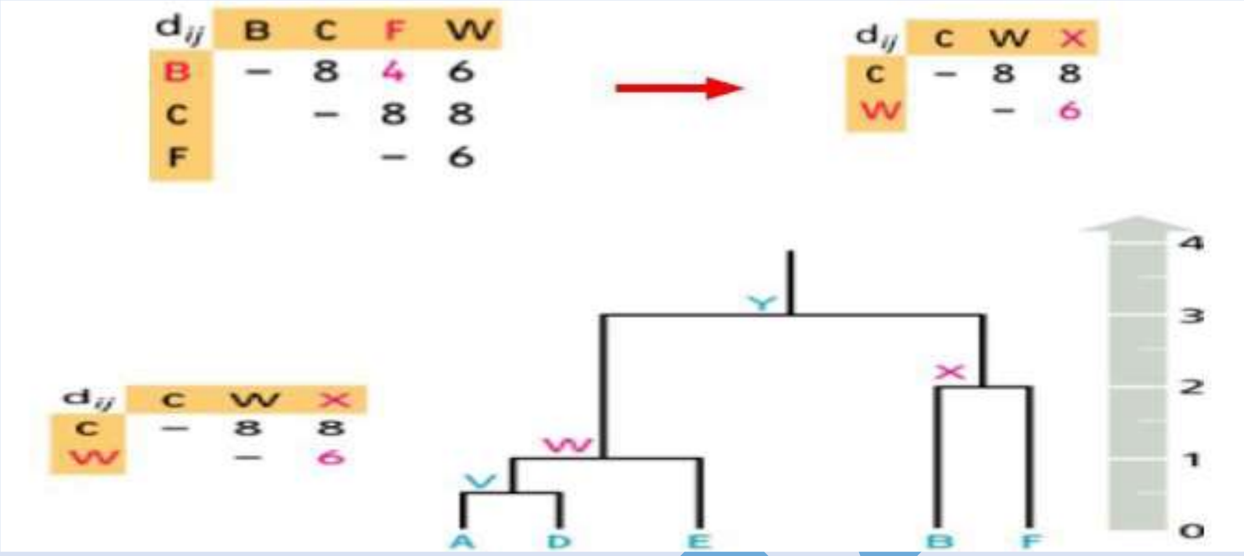
F – B becomes a new cluster lets say X. We have to modify the distance matrix yet again. What is th
distance between trees: W and X.

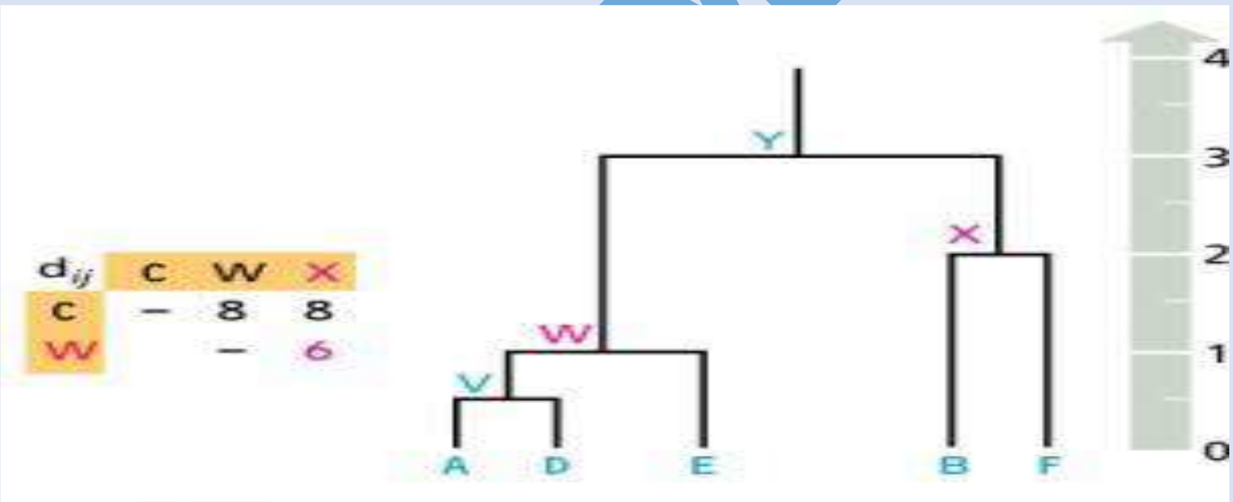
d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

$$d_{WX} = \frac{1}{N_W N_X} \sum_{i \in W, j \in X} d_{ij} =$$

$$\frac{1}{N_W N_X} (d_{AB} + d_{AF} + d_{DB} + d_{DF} + d_{EB} + d_{EF}) =$$

$$\frac{1}{3*2} * (6+6+6+6+6+6) = 6$$

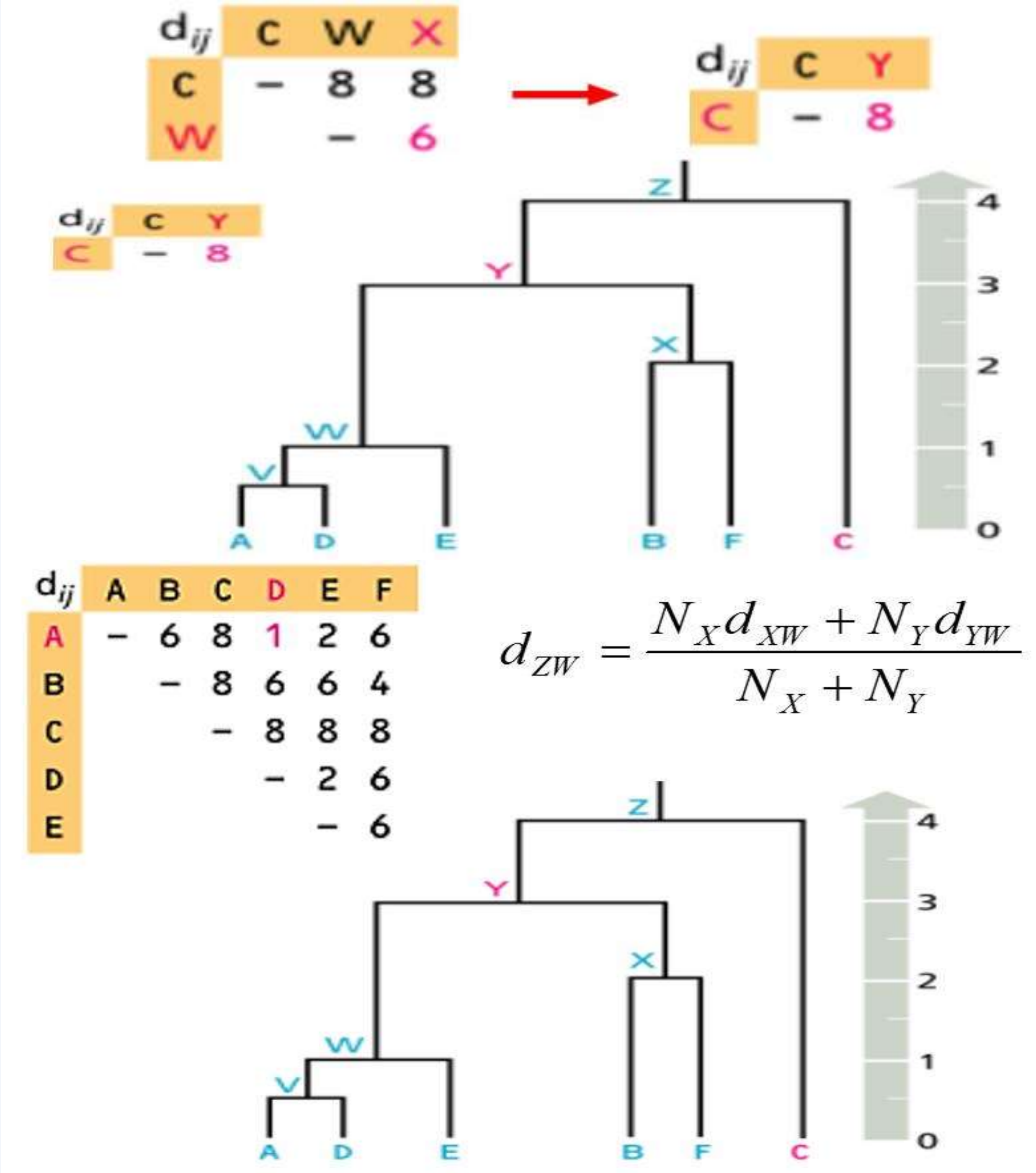




Module 011: UPGMA-V

Application of UPGMA resulted in formation of two sub-trees. The need now was to join them into a single tree. Let's see how that is done.

X – W becomes a new cluster let's say Y. We have to modify the distance matrix. What is the distance between: Y and C



Un-weighted Pair Group Method using Arithmetic Averages is a clustering method to construct phylogenetic trees. Non-clustering methods such as Maximum Parsimony may be used for making trees as well.

Chapter 4 - RNA Secondary Structure Prediction
Module 001: DNA TO RNA SEQUENCES
(Important lecture)

? MOTIVATION
In early days RNA was a considered as a structure which was involve between DNA and protein, means takes information from DNA and converts that information into protein synthesis.

Types like mRNA, tRNA, rRNA, miRNA and siRNA. (5 types of RNA)

Not All RNA molecules are same, they differ in nucleotide sequences and functions also.

Many viruses assemble their genomes from RNAs. They are therefore called RNA viruses.
Examples include Human Immunodeficiency Virus and Hepatitis C Virus.

There is little difference between RNA and DNA:
? Thymine is replaced by Uracil in RNA molecule.
? RNA molecule is single stand.
? RNA contain ribose sugar.
Because RNA has two (OH) groups that's why it has shot life spam because of both (OH) repulsion.

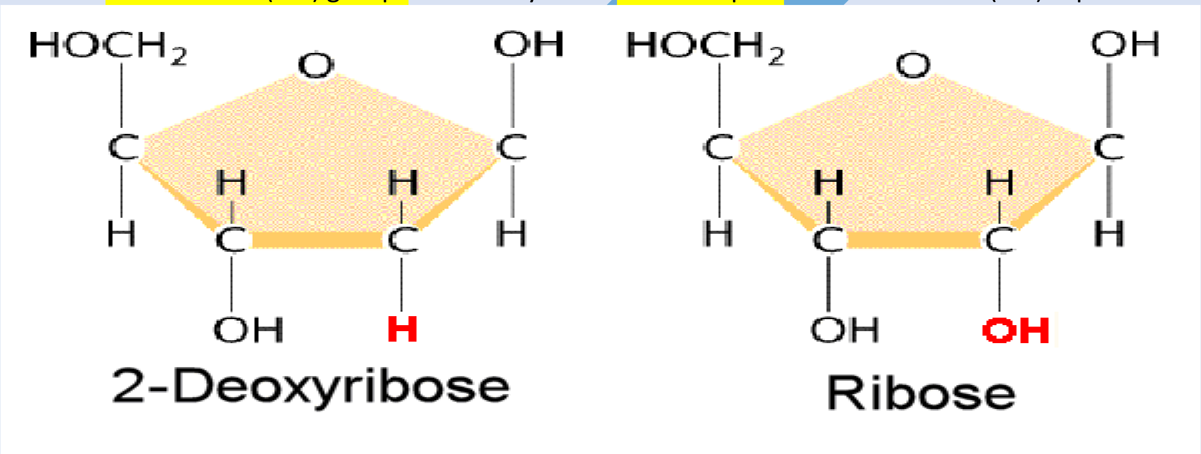
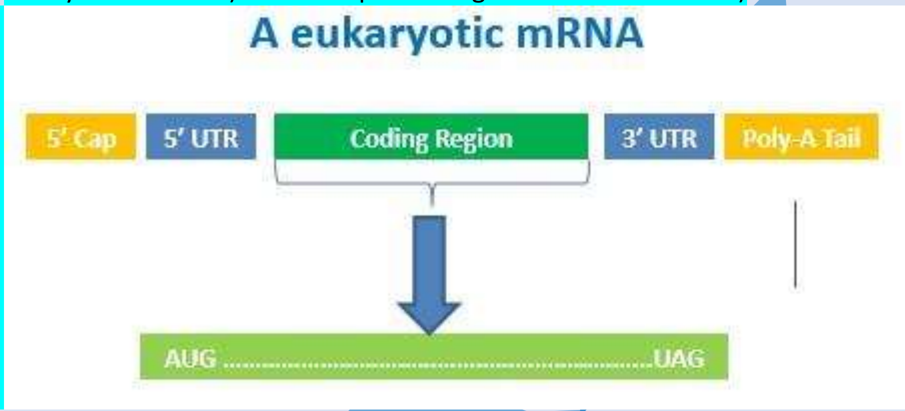


Figure 0.1 Ribose sugar has (OH) and Deoxyribose (H)

Handout Chapter 04: RNA Secondary Structure Prediction
Module 002: TYPES OF RNA &THEIR FUNCTIONS
(Important lecture)

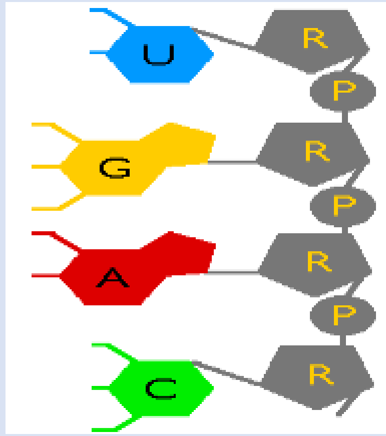
There are two categories of RNA:
? Coding RNA
? Non-Coding RNA
Coding RNA perform their coded function in protein synthesis. And Non-coding RNA helps in translation process.
? TYPES OF RNA

- Messenger RNA (mRNA)
- Transfer RNA (tRNA)
- Ribosomal RNA (rRNA)
- Micro RNAs (miRNA)
- Small Interfering RNA (siRNA)
-
- **MESSENGER RNA**
  - Only 5-10% of this RNA type is present in cell.
  - Which has variable sequence, variable size and it carries the genetic information form DNA to Ribosomes where proteins to be assembled.
  - (Important quiz)
  - Messenger RNA 5’ end is capped with (7-Methyl Guanosine Triphosphate) which helps the
  - Ribosomes to identify the mRNA. And 3’ end of the mRNA is poly A tail (around 30-200 adenylate residues) which help shield against 3’ exonucleases)
  - 

Handout Chapter 04: RNA Secondary Structure Prediction
Module 003: SIGNIFICANCES OF RNA STRUCTURE

RNA can form 3D structures {Sarver, 2008 #5}, such structural properties helps the RNA molecule to perform different functions. As RNA is composed of sugars, phosphate and nucleotides and these nucleotides have ability to form hydrogen bonds.

- A’ can make hydrogen bonds with ‘U’
- ‘G’ makes hydrogen bonds with ‘C’
- ‘G’ can also make hydrogen bonds with ‘U’ (Wobble Pair)



RNA functions:

- DNA information transfer
- Regulatory roles
- Catalytic roles
- Defense & immune response
- Structure-based special roles

Handout Chapter 04: RNA Secondary Structure Prediction
Module 004: RNA FOLDING AND ENERGY FOLDING

RNA molecules form many structures for stability and different functions.
"Gibs Free Energy" (LANGRIDGE and KOLLMAN 1987) is the free energy available for RNA molecule for reactions and RNA structure formation takes place at this lower energy. Incase if RNA has two structure we can select the one with lowest energy state.
We can calculate the overall energy of RNA structures by summing up energies given out during the process of folding. For knowing the positive and negative values of calculations of stabilizing and destabilizing energies we may factor in ways in which RNA can be destabilized.

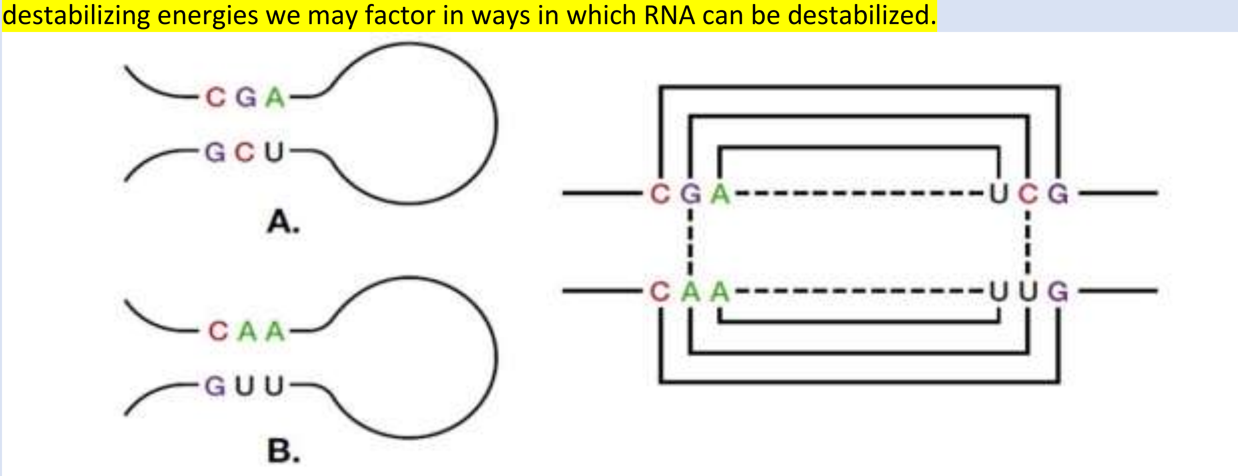
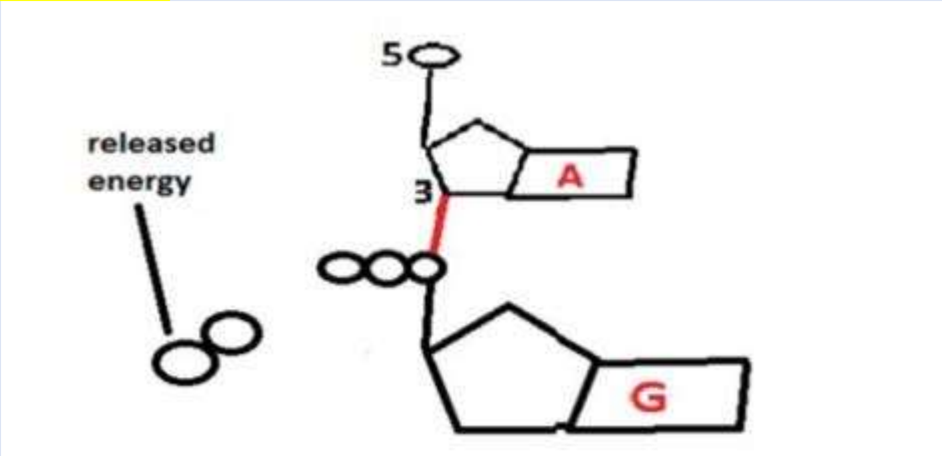


Figure 4 Energy is continuously given out as the RNA molecule folds by pairing complementary bases

We can calculate the overall energy of RNA structures by summing up energies given out during the process of folding.

Handout Chapter 04: RNA Secondary Structure Prediction
Module 005: CALCULATING ENERGIES OF FOLDING-AN EXAMPLE

RNA is composed of four nucleotides (A, U, C and G) and these nucleotides are attached with ribose sugar in backbone. And these nucleotides have hydrogen bonding between them. G always bond with C and A always bonds with U through hydrogen bonding and energy is released.



That's why RNA molecule become more stable.

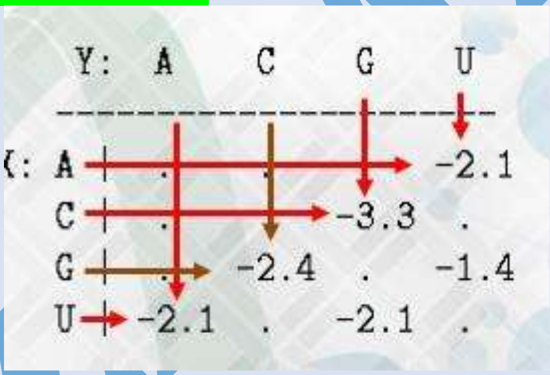


Figure 0.2nucleotides are held together by strong bonds which are created by the release of energy

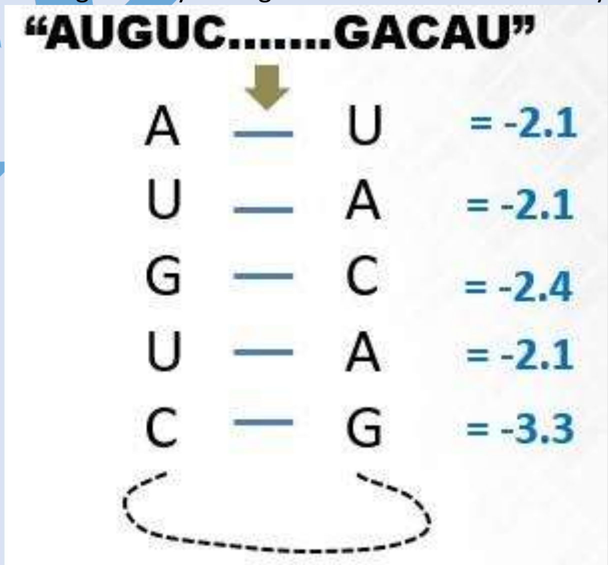


Figure 7 RNA Sequence energy

5 nucleotides formed H-Bonds. This bond formation released energy (-12.0 kcal/mol) RNA molecule took up a 2’ structure. Hence became more stable.

Module 006: TYPES OF RNA SECONDARY STRUCTURES-I

All the complimentary bases of RNA combine together to form RNA secondary structures. A simple nucleotide sequences of RNA is called as Primary structure and denoted by 1’ while when these nucleotides fold together and form a complex structure that is called secondary structure and denoted by 2’.
The preferred structure of RNA is 2’ which has many structural patterns like Helices, Loops, Bulges and Junctions

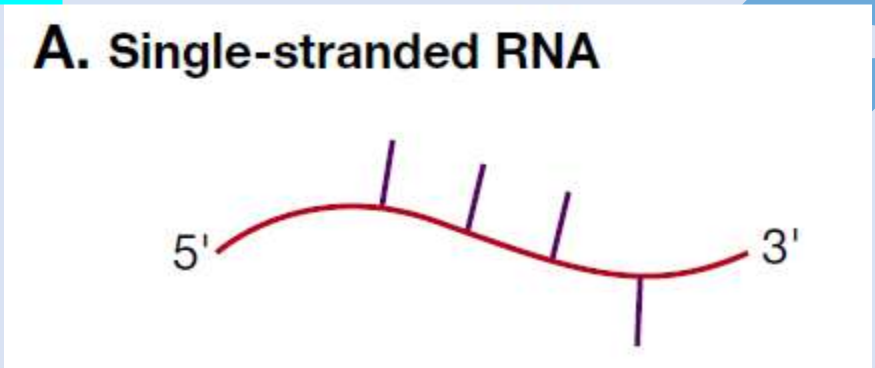
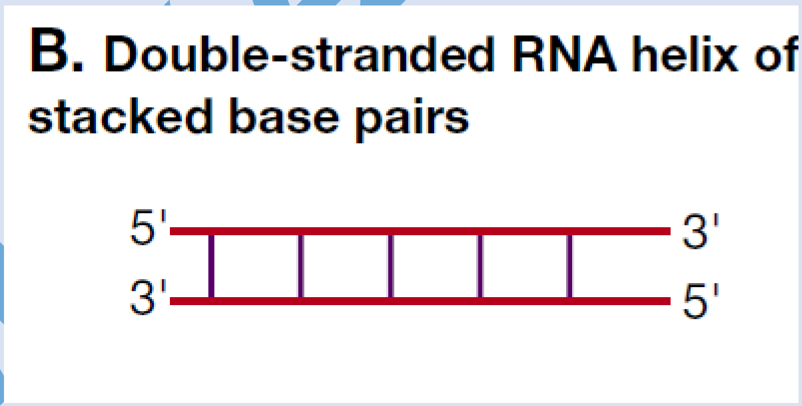
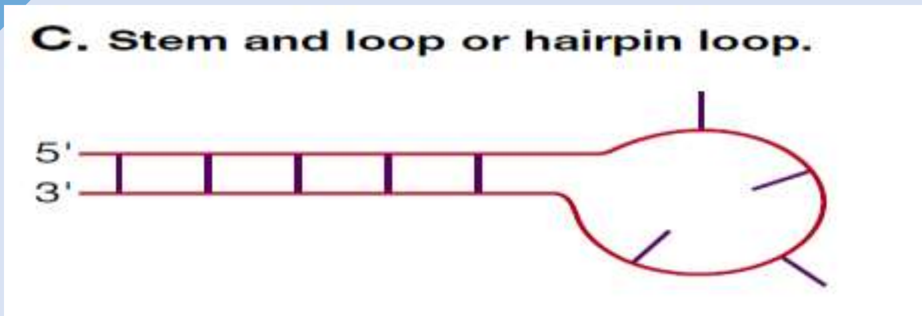


Figure 8 RNA sequence extends from its 5’ end to 3’ end. Upon folding, 3’ end may fold on to the 5’ end

The first 2’ RNA structure is called helix. Unlike the DNA helix, the RNA helix is formed when the RNA folds onto itself.





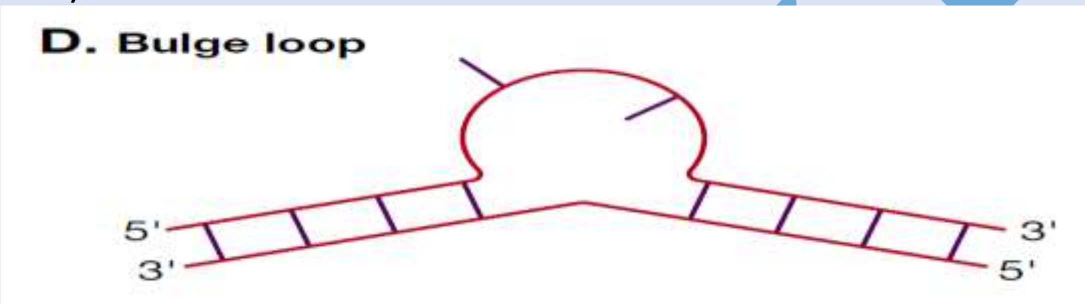
The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure.
Note that hairpins reverses the chemical direction of the RNA molecule.

Module 007: TYPES OF RNA SECONDARY STRUCTURES-II

RNA 1’ structure fold the (5’-3’) ends and make RNA 2’ structure just like helix and hairpin structure. The third type of 2’ structure is bulge loop.

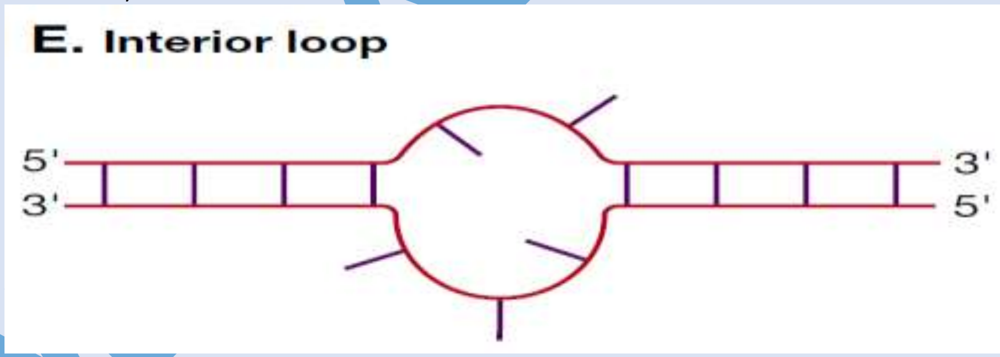
What is bulges???

Bulges, are formed when a double-stranded region cannot form base pairs perfectly. Bulges can be asymmetric with varying number of base pairs on one side of the loop. Bulge loops are commonly found in helical segments of cellular RNAs and used to measure the helical twist of RNA in solution. **(Tang and Draper 1990)**



Interior loop??

The forth type of 2’ RNA structure is interior loop.
Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.(Turner, Sugimoto et al. 1988)



Handout Chapter 04: RNA Secondary Structure Prediction
Module 008: TYPES OF RNA SECONDARY STRUCTURES-III

Another 2’ RNA structure is the Junction or Intersection.

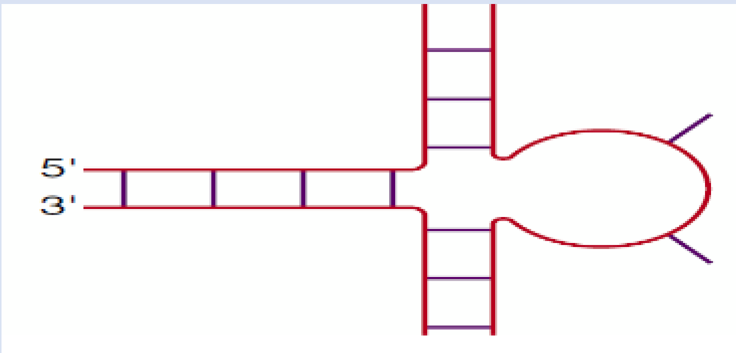


Figure 9 2' RNA structure called junction

Junctions include two or more double-stranded regions converging to form a closed structure. The unpaired bases appear as a bulge.(Zuker and Sankoff 1984)

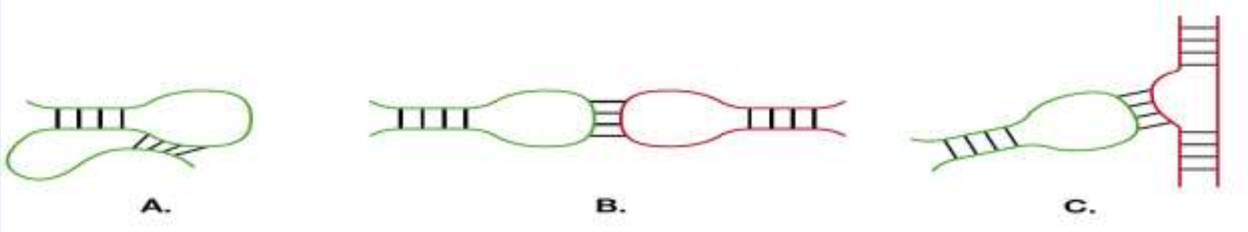


Figure 10 Unpaired bases in two 2’ structures form hydrogen bonds with each other

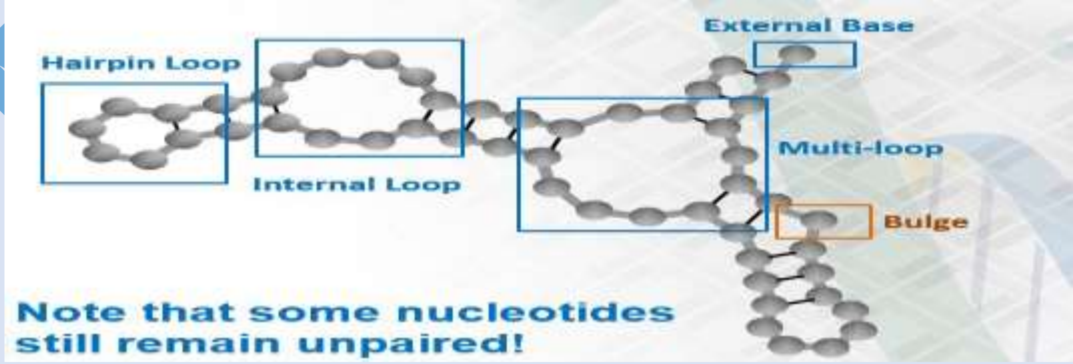
RNA tertiary structures are formed when RNA unpaired base bond in 2’ region bond.

Handout Chapter 04: RNA Secondary Structure Prediction

Module 009: RNA TERTIARY STRUCTURES

2’ RNA structures is formed due to folding of nucleotide with in RNA molecule but after folding some nucleotides remain open for interaction. And they form hydrogen bonds together.

These unpaired nucleotides of 2’ structure interact with other unpaired nucleotides and form a third structure called tertiary 3’ structure. For example 4 nucleotides in hairpin loop structure does.



The unpaired bases in 3’ structure remain paired by abnormal folding called (pseudoknots) but instead of pairing they remain available or pairing.

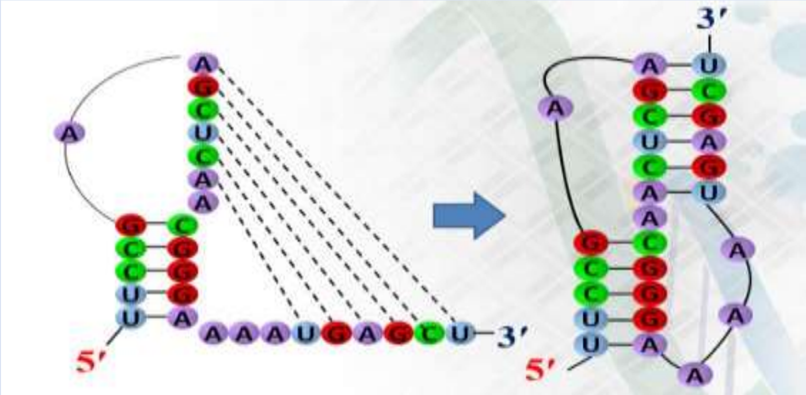
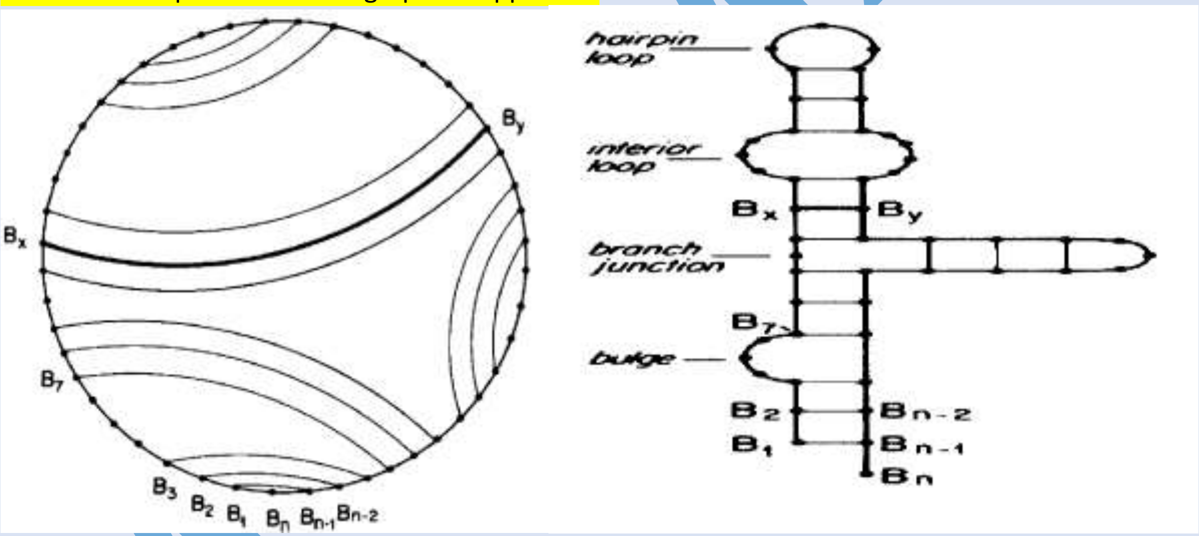


Figure 12 pseudoknots

Handout Chapter 04: RNA Secondary Structure Prediction Module 010: CIRCULAR REPRESENTATION OF STRUCTURES

Tertiary or 3’ structure of RNA may form pseudoknots to detect the pseudoknots in RNA structure we need “circular plot” which is a graphical approach.



Intersecting arcs in circular plot are the pseudo knot.

Handout Chapter 04: RNA Secondary Structure Prediction Module 011: EXPERIMENTAL METHODS TO DETERMINE RNA STRUCTURES

RNA has 1’, 2’ and 3’ structures. 1’ has simple nucleotide sequence and 2’ has nucleotides folding and 3’ has knots.

For measuring the RNA structure we use X-ray crystallography (Smyth and Martin 2000), which works according to the principle of diffraction. Crystallized RNA diffracts X-rays which helps estimate atomic positions

All isotopes that contain an odd number of protons and/or of neutrons (see Isotope) have an intrinsic magnetic moment and angular momentum, in other words a nonzero spin, while all nuclides with even numbers of both have a total spin of zero. The most commonly studied nuclei are 1H and 13C, al

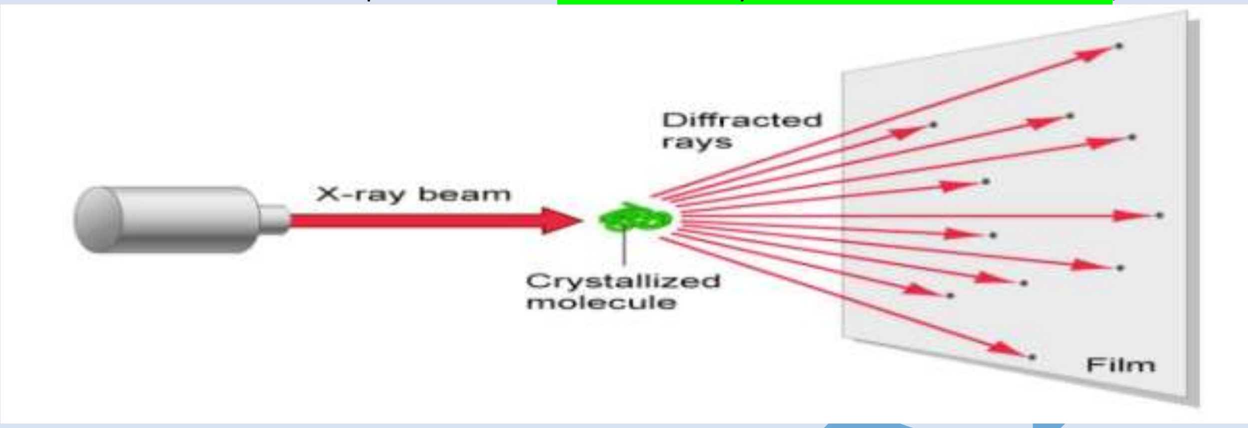


Figure 13 X-ray Crystallography

Another method to measure the RNA structure is called as Atomic Force microscopy in this technique a laser connected to a Si 3 N 4 piezoelectric probe scans an RNA sample. It works well in air and liquid environment.

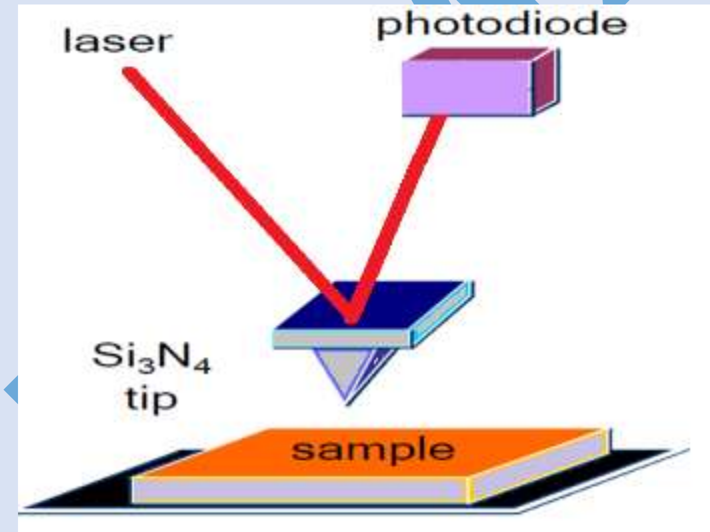
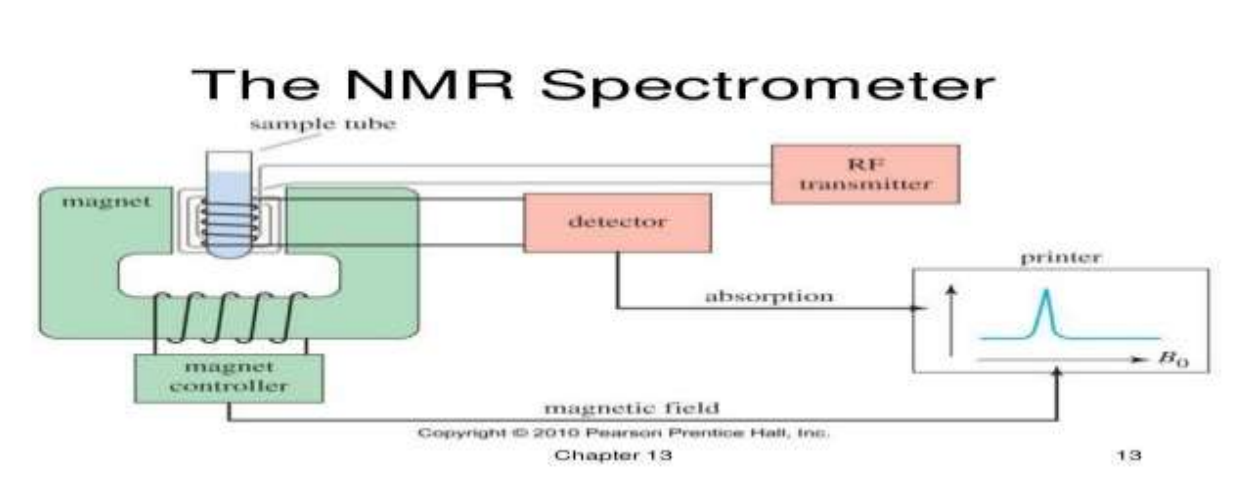


Figure 14 Atomic microscopy

The third method for measuring the RNA structure is Nuclear Magnetic Resonance Imaging in this method Hydrogen atoms in RNA resonate upon placement in a high magnetic field. It Works well without crystallizing RNA.



 STORAGE OF STRUCTURES
Reported structures are stored in online databases. Example includes RNA Bricks and RMDB etc. Bioinformaticians can refer to these databases for RNA structure studies

RNA Bricks is a database of RNA 3D structure motifs and their contacts, both with themselves and with proteins
Stanford University’s RNA Mapping Database is an archive that contains results of diverse structural mapping experiments performed on ribonucleic acids.

Module 012: Strategies for RNA Structure Prediction

RNA structure 2’ and 3’ can be measured experimentally, but RNA molecule readily degrade due to their short shelf life.

Give 1’ RNA structure creates the 2’ structure because the simple nucleotides folds and form 2’ structure. And on the base of folding we can predict the stability of the RNA molecule.
For example.

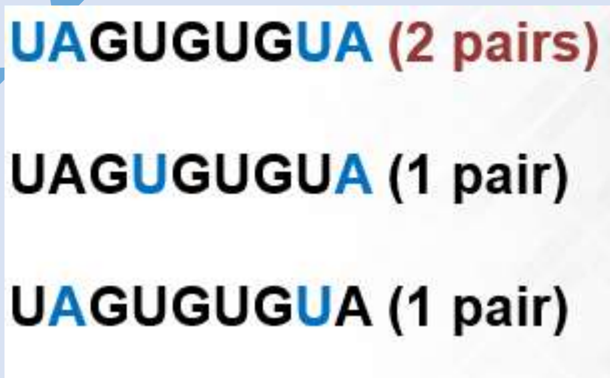


Figure 15 pairs represent the stability of the RNA molecule

Maximizing the number of nucleotides can increase the structure and we have to select the structure according to the stability.

Module 013: Dot Plots for RNA 2' Structure Prediction

The dot plot method for RNA structure prediction is easy. Draw a square and partition by drawing gridlines. Put RNA sequence on top and left sides of the square. Put a “DOT” on complementary nucleotides

For example:

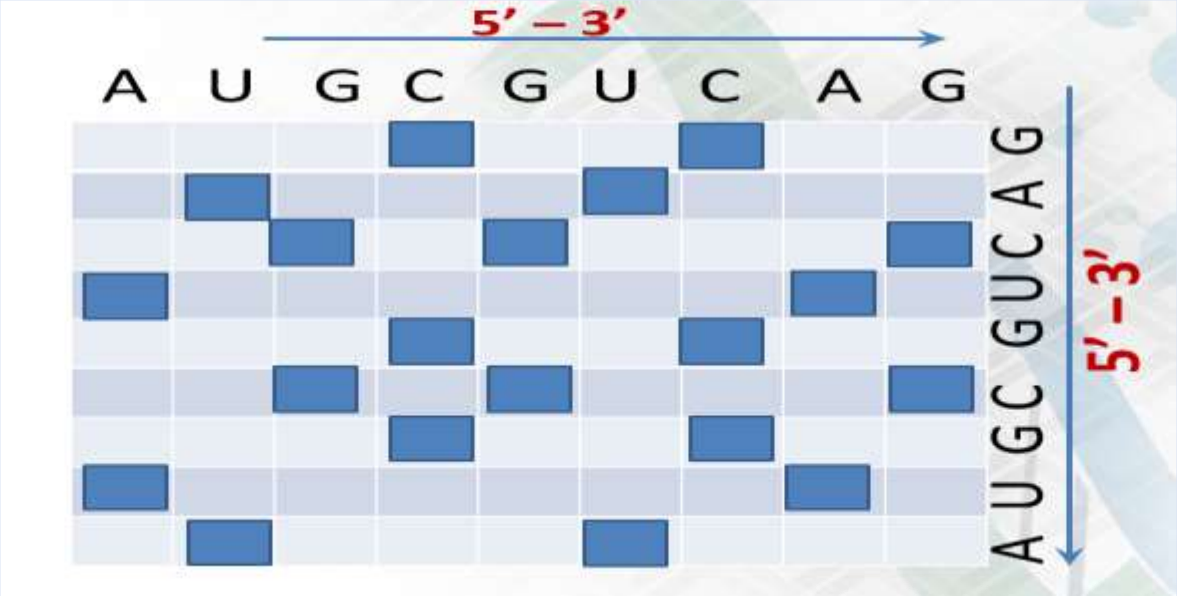


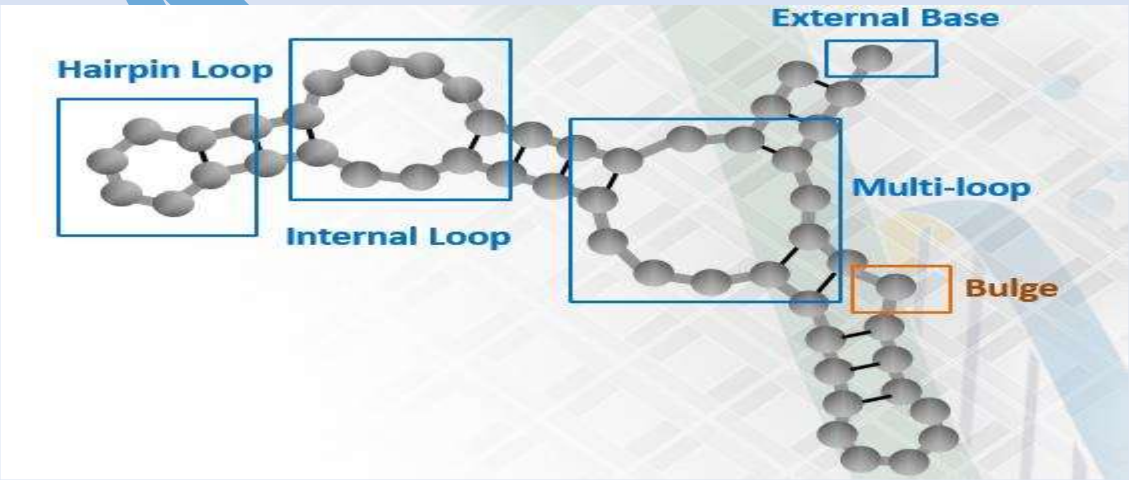
Figure 16 dot are placed at complimentary base pair place.

Module 014: ENERGY BASES METHODS

Experimental prediction of RNA structure is slow and costly that’s why a few 2’ RNA structures are reported experimentally.

 STABILIZING ENERGY

Energy table helps us to find the optimal prediction of structure because energy is released when complementary nucleotides make bonds.



DESTABILIZING ENERGY
Remaining unpaired nucleotides destabilized the RNA structure in form of hairpin or bulge structure.

SUM OF ENERGIES
Sum of stabilizing and destabilizing energies can help determine the quality of a 2’ RNA structure. 2’ structure with longest coupled sequences vs. one with lowest energy

Module 015: Zuker’s Algorithm

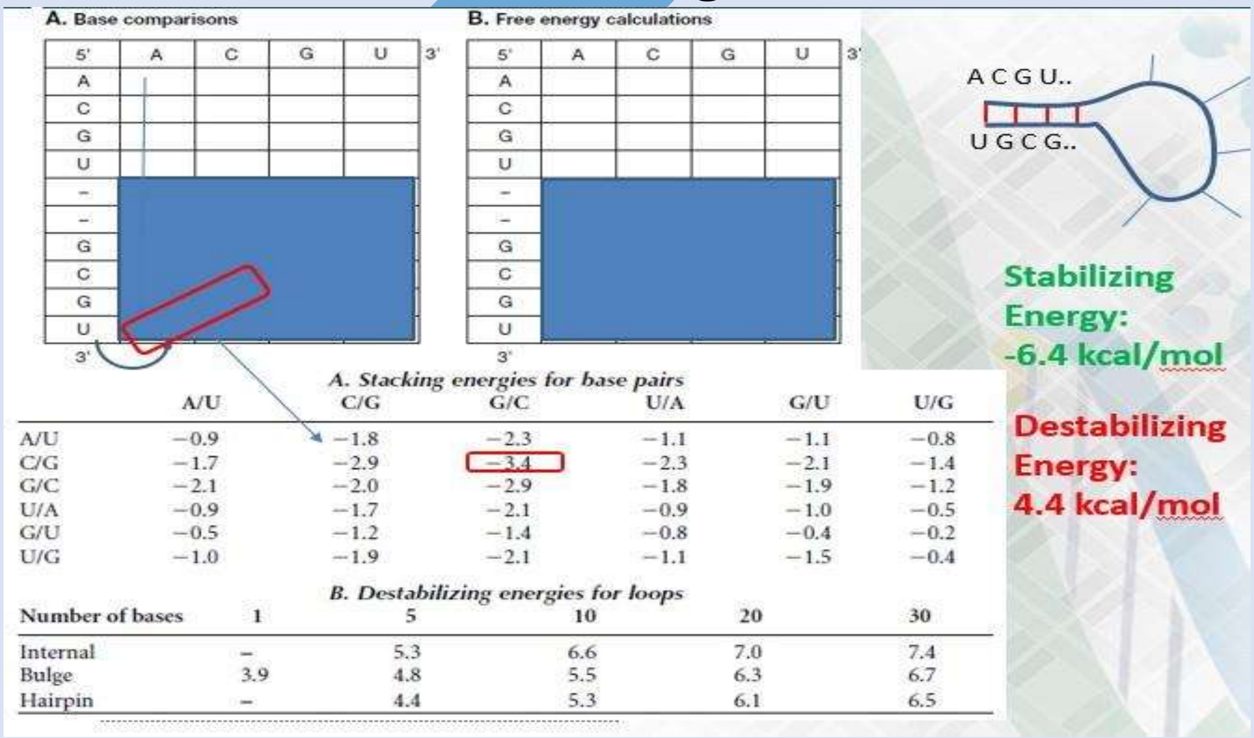
Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1’ or 2’ optimal structure prediction we use Zuker’s Algorithm.

Zuker’s Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and –ve energies.

Working principle of Zuker’s Algorithm (2003)

It Compute energies of all possible 2’ structures. Generate combinations of all computed 2’ structures. Select the one with lowest energy.

Module 016: Zuker’s Algorithm EXAMPLE



A. Stacking energies for base pairs

	A/U	C/G	G/C	U/A	G/U	U/G
A/U	−0.9	−1.8	−2.3	−1.1	−1.1	−0.8
C/G	−1.7	−2.9	−3.4	−2.3	−2.1	−1.4
G/C	−2.1	−2.0	−2.9	−1.8	−1.9	−1.2
U/A	−0.9	−1.7	−2.1	−0.9	−1.0	−0.5
G/U	−0.5	−1.2	−1.4	−0.8	−0.4	−0.2
U/G	−1.0	−1.9	−2.1	−1.1	−1.5	−0.4

B. Destabilizing energies for loops

Number of bases	1	5	10	20	30
Internal	–	5.3	6.6	7.0	7.4
Bulge	3.9	4.8	5.5	6.3	6.7
Hairpin	–	4.4	5.3	6.1	6.5

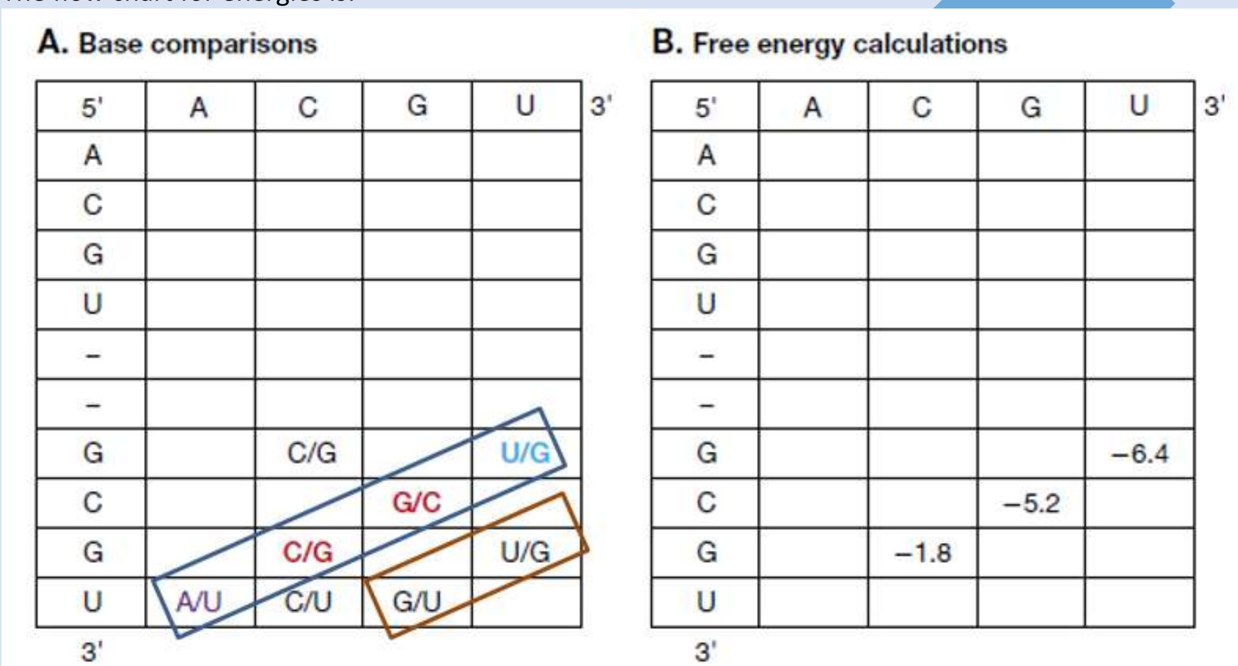
Figure 23 Calculation of all possible structure combinations

We need to construct all the possible combinations of nucleotides for selection of optimal 2’ RNA structure.

Handout Chapter 04: RNA Secondary Structure Prediction

Module 017: Zuker’s Algorithm – A Flow Chart

Zuker’s Algorithm involves computing stabilizing and destabilizing energies of 2’ structure. And it also computes the overall energy by summing up the positive and negative energies.
The flow chart for energies is:



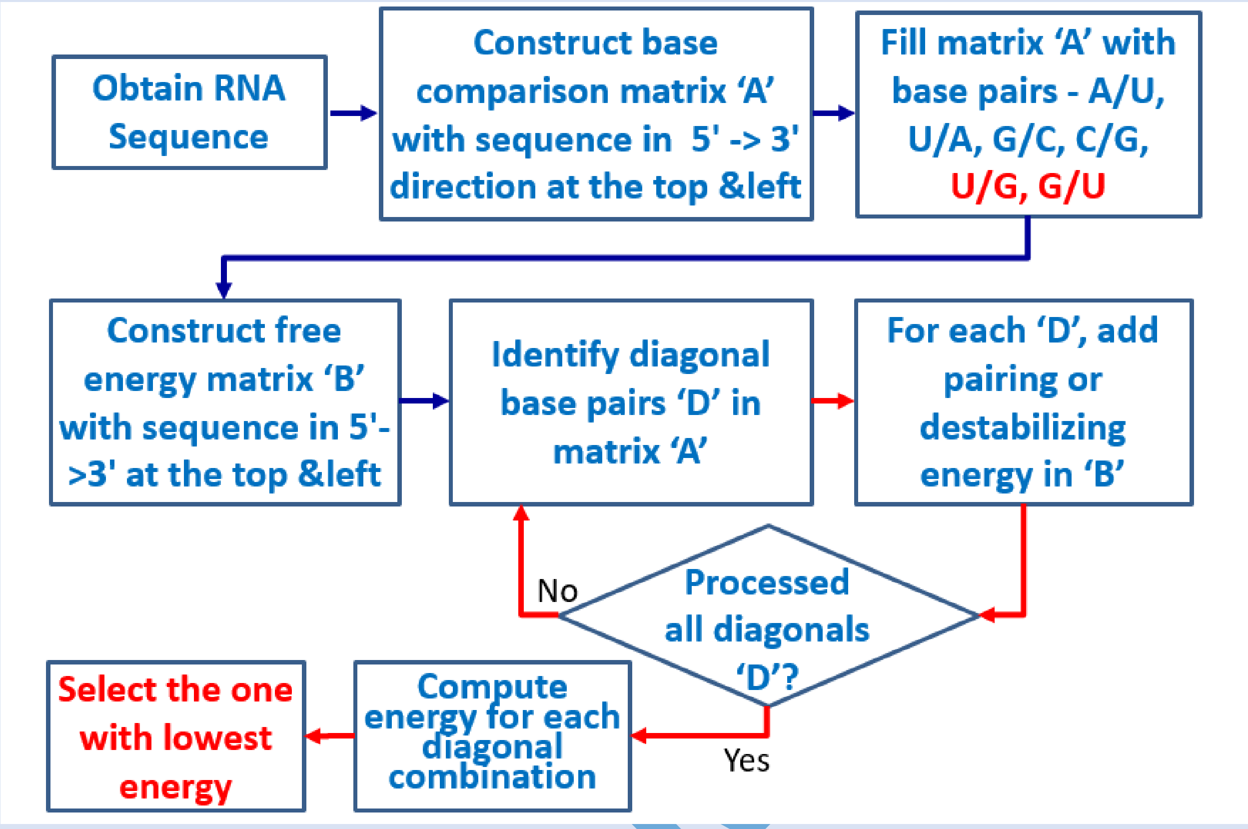
A. Base comparisons

5'	A	C	G	U	3'
A					
C					
G					
U					
–					
–					
G		C/G		U/G	
C			G/C		
G		C/G		U/G	
U	A/U	C/U	G/U		
3'					

B. Free energy calculations

5'	A	C	G	U	3'
A					
C					
G					
U					
–					
–					
G				−6.4	
C			−5.2		
G		−1.8			
U					
3'					

Flow chart

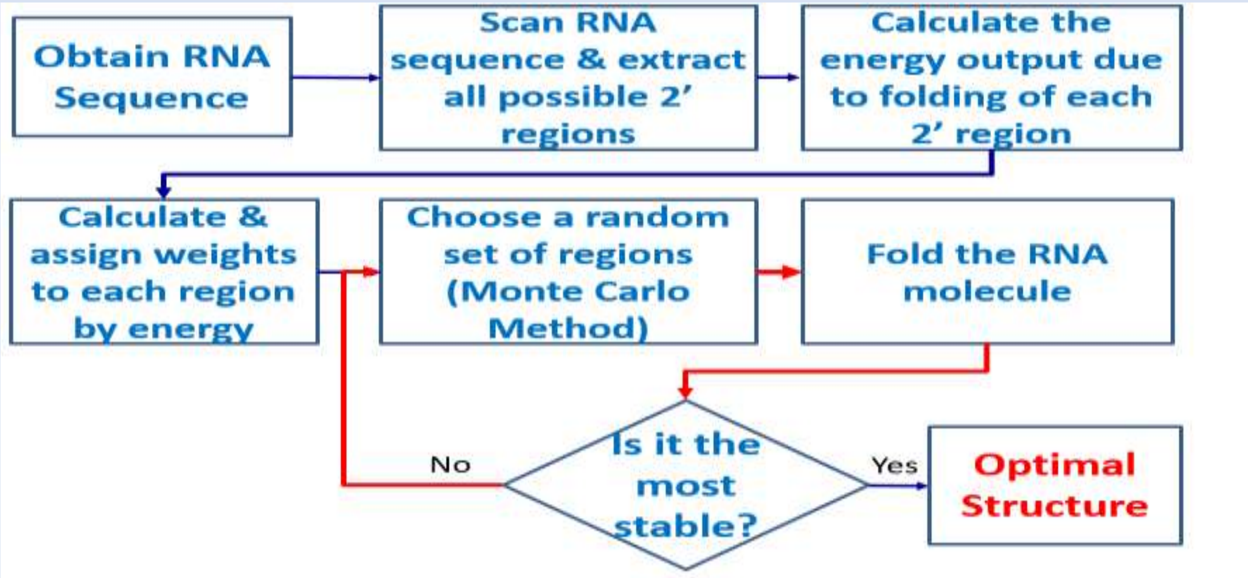


The diagonal combination from all possible is selected with overall lowest energy.
The two diagonals ('D') given above include:
1. A/U, C/G, G/C, U/G
2. G/U, U/G

Handout Chapter 04: RNA Secondary Structure Prediction
Module 018: Martinez Algorithm

Zuker's Algorithm involves computing stabilizing and destabilizing energies of 2' structure. And it also computes the overall energy by summing up the positive and negative energies. Martinez Algorithm is improvement on it.
Making combination of all possible structures is time consuming, Martinez Algorithm favors those 2' structure which are energetically more feasible.

Martinez Algorithm flow chart



In Martinez algorithm all the 2' structures are weighed by its stability and optimal one is sorted out.

Monte Carlo methods (or Monte Carlo experiments) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other mathematical methods. And Monto Carlo method do not provide a definitive solution.

Module 019: Dynamic Programming Approaches

In 2' RNA structure there may be large number of nucleotide sequence with large number of combinations hence it is hard to find the optimal one and for this prediction we us Dynamic Programming (DP) which breaks the larger problems into smaller one.

PRINCIPLE OF DYNAMIC PROGRAMMING

For optimal structure combination selection we use the Dynamic Programming (DP) and we select the sequence of RNA nucleotides and list all the possible complementary positions for nucleotides in the given complete sequence.

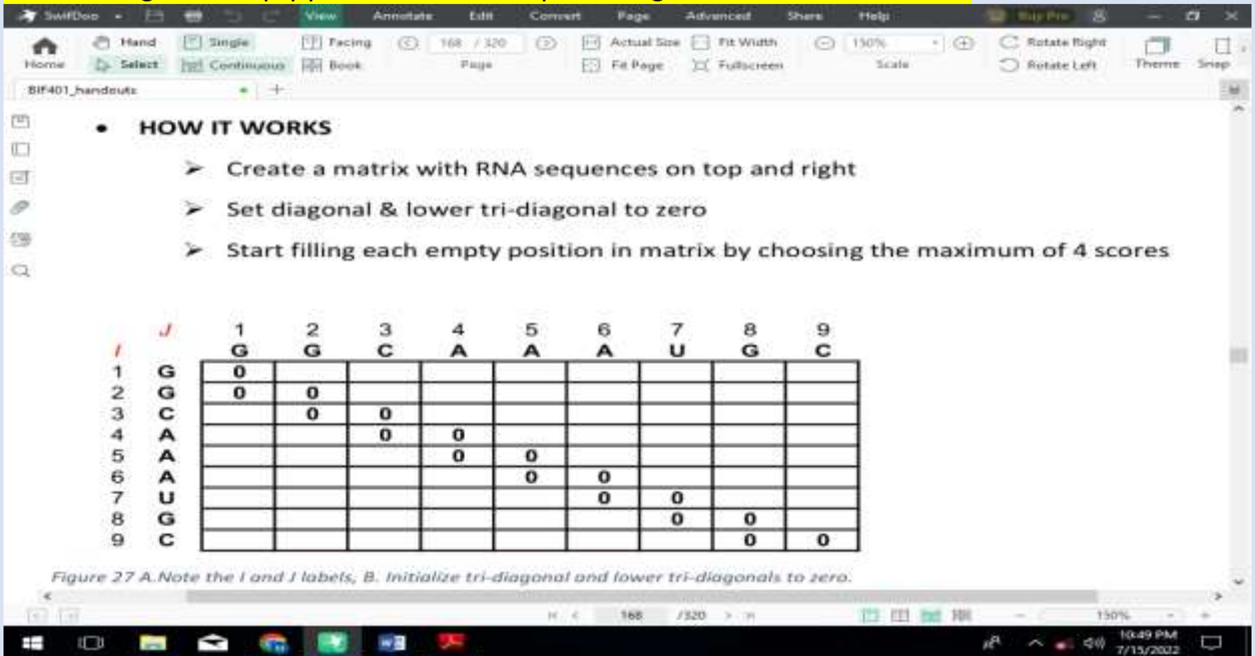
Dynamic Programming then recombines such combinations in a process called "Traceback" to ensure that the highest coupled 2' structure is reported

Handout Chapter 04: RNA Secondary Structure Prediction

Module 020: Nussinov-Jacobson Algorithm – An Overview

Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal RNA 2' structures, Proposed in 1980. Computes 2' structures with most nucleotide coupling.
http://ultrastudio.org/en/Nussinov_algorithm

HOW IT WORKS
Create a matrix with RNA sequences on top and right
Set diagonal & lower tri-diagonal to zero
Start filling each empty position in matrix by choosing the maximum of 4 scores



Module 021: Nussinov-Jacobson Algorithm – The Flowchart

NJ algorithm is actually a dynamic programming (DP) approach to predict the 2’ RNA structure.
A scoring matrix is initialized to record scores in NJ Algorithm .**For filling scoring matrix, the maximum score from 4 matrix positions is chosen.**

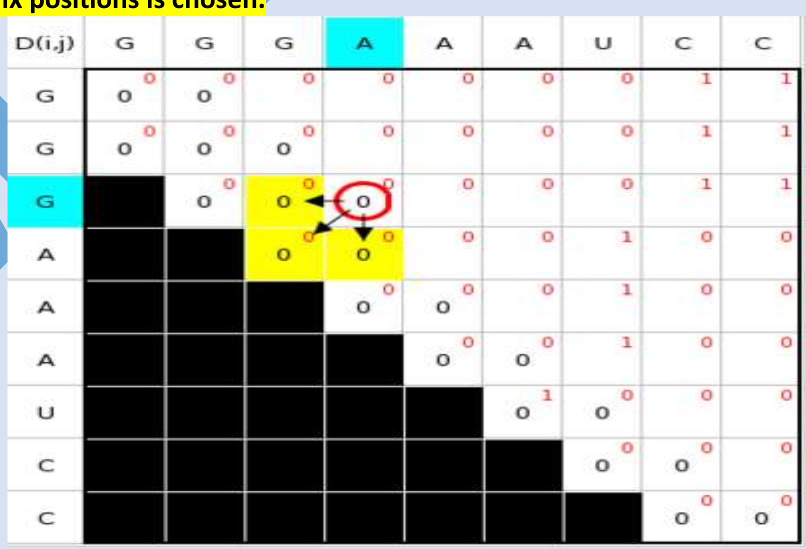
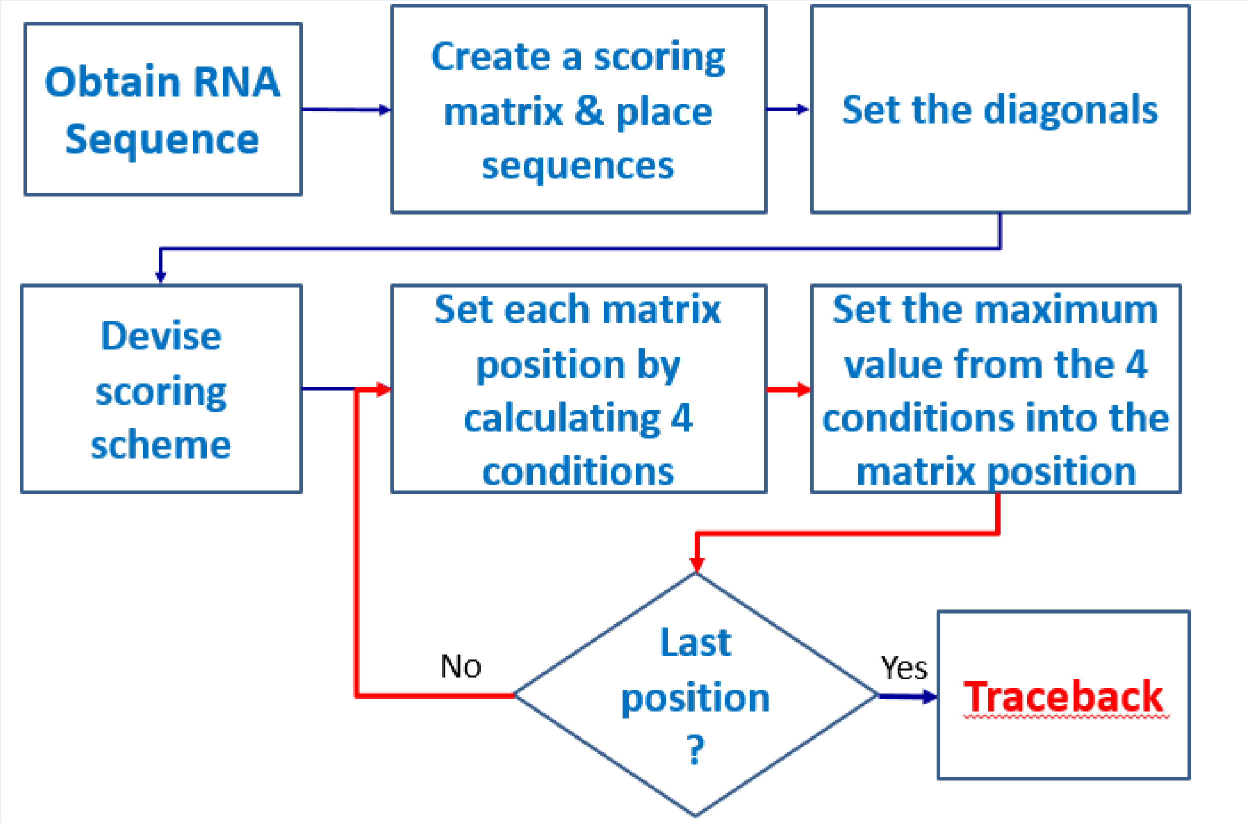


Figure 0.3 for maximum score 4 positions are used in scoring

Flow chart of NJ Algorithm



Traceback is used to report the coupling of structures in sequences.

Handout Chapter 04: RNA Secondary Structure Prediction

Module 022: Nussinov-Jacobson Algorithm – EXAMPLE

The main points to be focused in N-J Algorithm are:
- Scoring Matrix
- Matrix Initialization
- Scoring method
- The 4 different positions to be considered for calculating matrix

The matrix is filled by four different positions. Left, Bottom, Diagonal, and Left/Bottom elements. In this way all complementary nucleotides coupling is catered.

Module 023: Score Calculations & Traceback

From four positions the score is calculated and from each position we calculate the score contribution. And maximum score is sorted out.

		1	2	3	4	5	6	7	8	9	10	11	12	13
		A	A	U	C	U	G	U	U	A	C	G	C	A
1	A	0	0	2	2	4	5	7	7	9	9	12	12	12
2	A	0	0	2	2	2	5	5	5	7	7	10	10	12
3	U		0	0	0	0	3	3	3	5	5	8	8	10
4	C			0	0	0	3	3	3	5	5	8	8	10
5	U				0	0	1	1	1	3	5	6	8	10
6	G					0	0	1	1	3	5	6	8	8
7	U						0	0	0	2	2	5	5	7
8	U							0	0	2	2	5	5	5
9	A		G-C = 3						0	0	0	3	3	3
10	C		A-U = 2							0	0	3	3	3
11	G		G-U = 1								0	0	3	3
12	C											0	0	0
13	A		4th condition of max{S(i,k)+S(k+1,j)}.										0	0

Figure 30 scoring and traceback in N-J Algorithm.

There can be many traceback. Each traceback is used to make the RNA secondary structure. And traceback with highest number of nucleotide coupling is selected.

Handout Chapter 04: RNA Secondary Structure Prediction

Module 024: Comparison of Algorithms

All algorithm have two main strategies:
1. Nucleotides stacking
2. Energy minimization

 ENERGY BASED ALGORITHM.

Zuker's Algorithm involves energy minimization. It is updated version and incorporate the phylogenetic information. It is improved. Overcomes the pseudoknots assumes them and accommodate them. And this algorithm helps to predict the structures of RNA based on nucleotides.

 NUCLEOTIDES STACKING ALGORITHM.

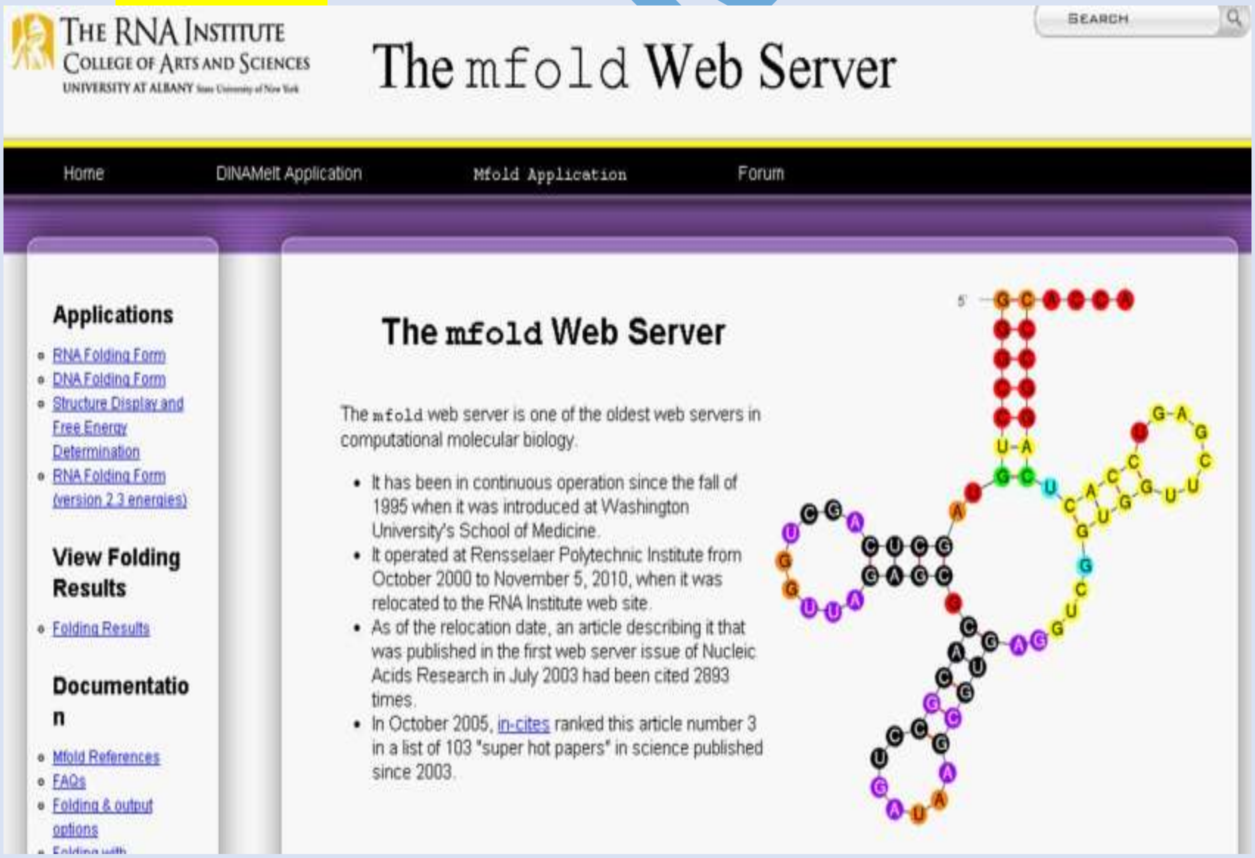
NJ's Algorithm comes under this category. It involves the maximizing the nucleotides pairing. Traceback helps to find best 2' structure.

It predict the 75% accurate 2’ structure. Because there may be more than two equal scores as it is calculated from four different positions.

For further improvements in results we take help from:
- Sequences
- Comparison
- Nucleotide
- Covariance analysis

Handout Chapter 04: RNA Secondary Structure Prediction
Module 025: WEB RESOURCES-I

For prediction of 1’ and 2’ structure of RNA we use different algorithm like Zuker’s, Martinez and N-J. Online tools also.
The mfold web server is one of the oldest web servers in computational molecular biology.
Mfold is upgraded version of Zuker’s algorithm.
MFOLD is computationally expensive and can give results for 1’ and 2’ structures that have sequences less than 8000 nucleotides.

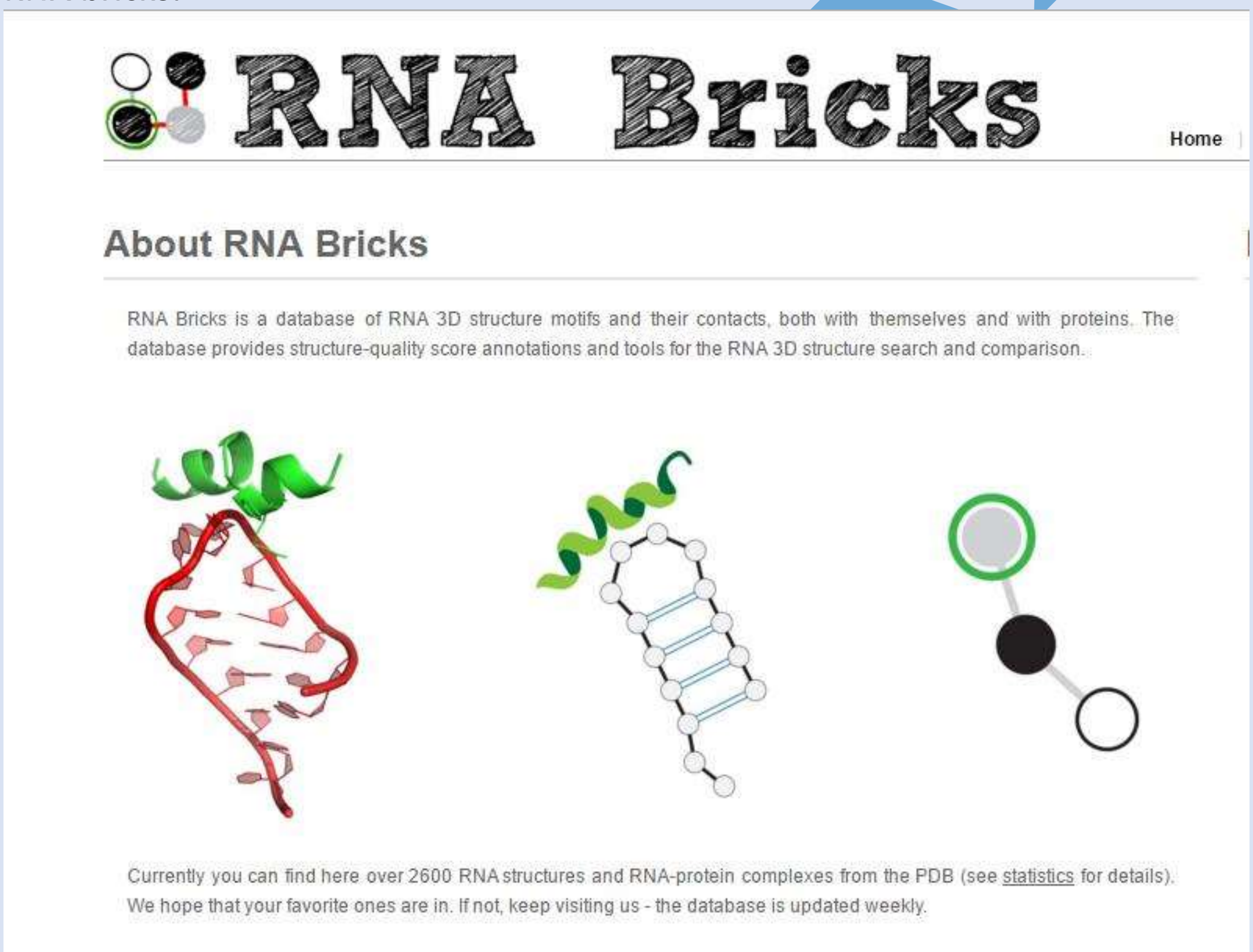


MFOLD helps fold an RNA nucleotide sequence into its possible 2’ structures. MFOLD gives out several structures along with their energetic stability!

Handout Chapter 04: RNA Secondary Structure Prediction
Module 026: WEB RESOURCES-II

RNA nucleotides folds to form 2’ structure from simple portion of 1’ nucleotides. For example CUUCGG occurs a wide variety in RNA and it mostly forms the stable hairpin loop. So we can make the list of all likely 2’ structures arising from 1’.

RNA bricks:



RNA 1’ folds and makes RNA 2’ structure and this online database is established for 2’ RNA structure and it act as dictionary for 2’ RNA structure.

Chapter 5 - Protein Sequences
Module 001: FROM DNA/RNA SEQUENCES TO PROTEIN

We are aware that DNA has four nucleotides bases (A, C, T & G). RNA contains (A, C, U & G).
And protein contain 20 different amino acids. DNA to RNA then Protein is called as central dogma. Which includes translation, transcription and protein modifications.
A set of three nucleotides called codon, codes the information for specific amino acids in protein synthesis.

Codons select the amino acids and ribosomes make the protein by polymerization process and these nucleotides coil together to form 3D structure.

Handout Chapter 05: Protein Sequences
Module 002: CODING OF AMINO ACIDS

Nucleotides (A, G, C, and T) make set of three called codons for amino acid selection in protein synthesis. More than one codon can code for same amino acids as there are 20 amino acids involved in protein synthesis.

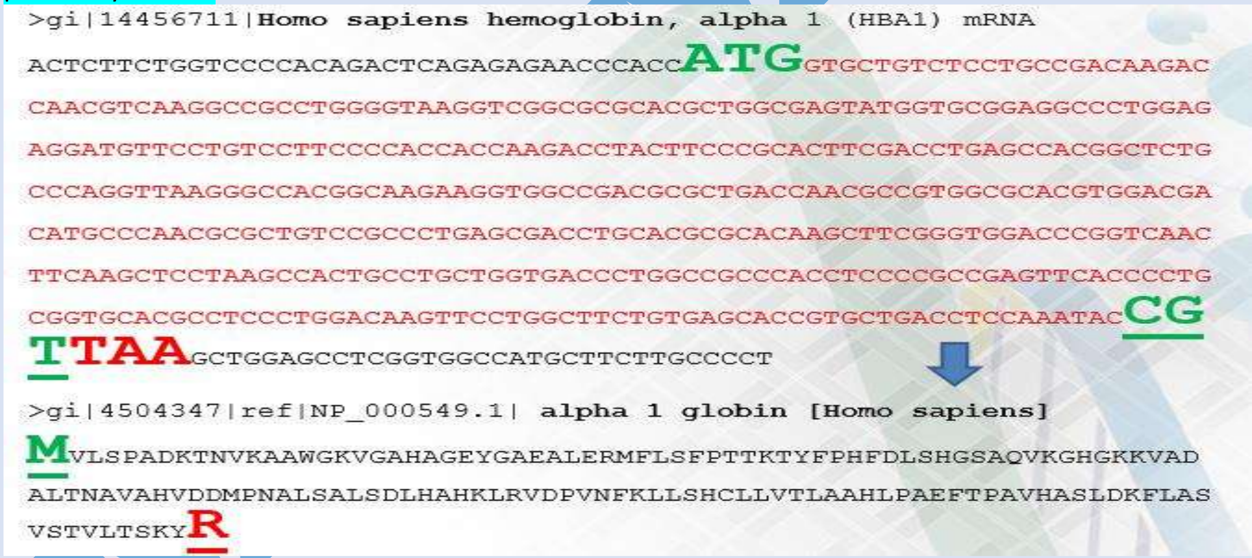


Figure 4 Start Codon ATG and Stop Codon TAG, TGA or TAA

Handout Chapter 05: Protein Sequences
Module 003: OPEN READING FRAMES

Codons codes information for amino acid and there are three stop codons and one start codon.
For the valid open reading frame it must have longest sequence.

In molecular genetics, an open reading frame (ORF) is the part of a reading frame that has the potential to code for a protein or peptide. An ORF is a continuous stretch of codons that do not contain a stop codon (usually UAA, UAG or UGA).

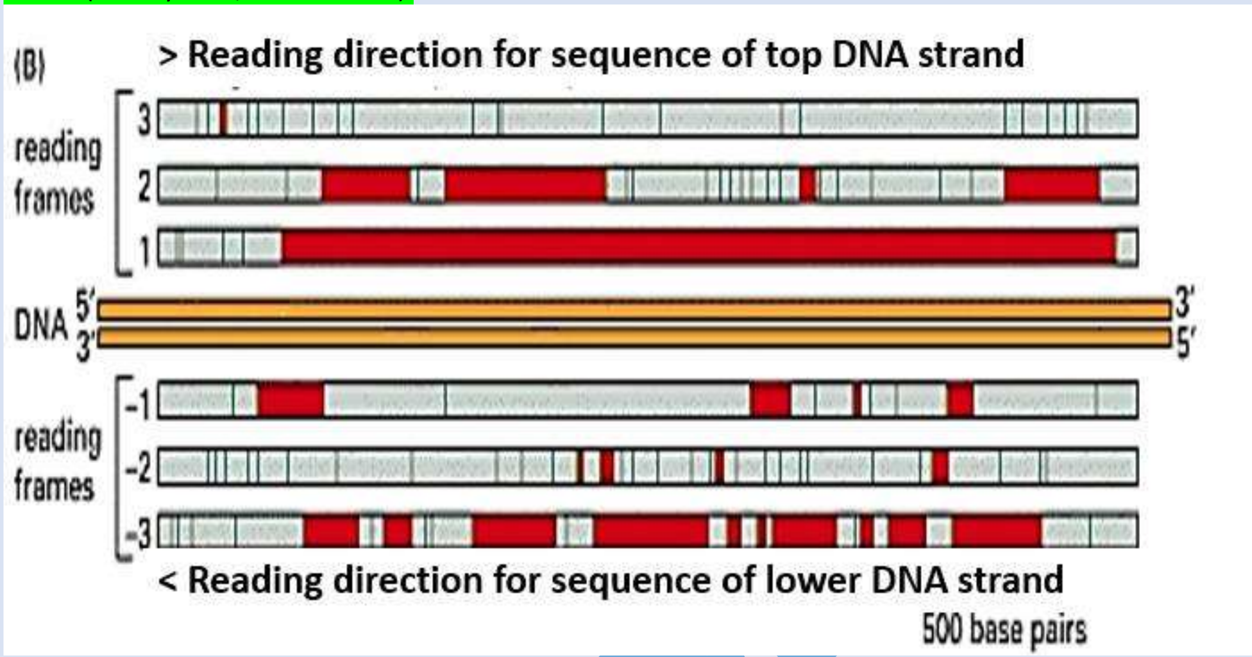


Figure 5 ORF 1 is valid, as it is the longest

What are online tools from which we can find ORF in any sequence?
NCBI, ORF Finder
Six ORF exist in any DNA sequence and longest one is marked and first stop codon will be marked end of the protein.

Handout Chapter 05: Protein Sequences
Module 004: ORF Extraction – A Flowchart

Codons of 3 nucleotides code for each Amino Acid. There are 1 start and 3 stop codons.
Selection of ORF is based on its length if it the longest one from others than it would be suitable for protein synthesis reaction.

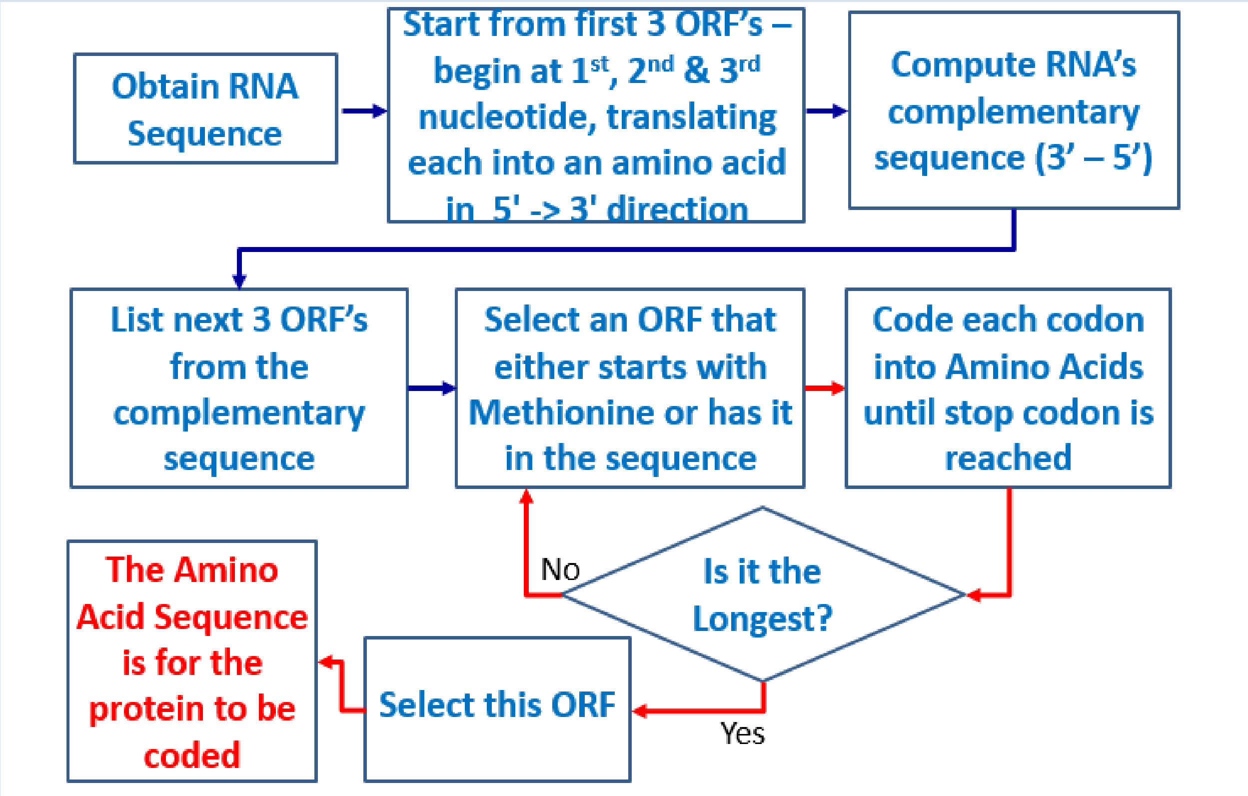


Figure 7 ORF extraction flowchart

Both reverse and forward RNA sequences are considered which may have many ORF and selection is based upon longest protein sequences having.

Handout Chapter 05: Protein Sequences

Module 005: SEQUENCING PROTEINS

Given the DNA/RNA sequence, ORFs can be extracted and protein sequence can be determined. But there are chances that protein may be unknown, that's why we use Adam degradation method in protein sequencing.

Edman degradation, developed by Pehr Edman, is a method of sequencing amino acids in a peptide.
In this method, the amino-terminal residue is labeled and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues.it starting from the N-terminal and removing one amino acid at a time.

Cyclic degradation of peptides by Phenyl-iso-thio-cyanate (PhNCS). **PhNCS attaches to the free amino group at N-terminal residue. 1 amino acid is removed as a PhNCS derivative.**

 DRAWBACKS
 It is restricted to chain of 60 residues.
 It is very time consuming process 40-50 amino acids per day.
Modern techniques for this is Tandem mass spectrometry.

Handout Chapter 05: Protein Sequences
Module 006: Application of Mass Spectrometry in Protein Sequencing

Edam Degradation methods helps us to sequence the protein which is unknown. But it is restricted to 60 amino acids only.
Protein can be charged with electrons or protons and if moving charges are placed in between the magnetic field they get deflected. And their deflection is proportional to their momentum.

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \text{(Lorentz force law)}$$

$$\mathbf{F} = m\mathbf{a} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \quad \text{(Newton's second law of motion)}$$

$$\left(\frac{m}{Q}\right)\mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B}.$$

Where:
- F is the force applied to the ion
- m is the mass of the particle,
- a is the acceleration
- Q is the electric charge,
- E is the electric field
- v × B is the cross product of the ion's velocity and the magnetic flux density.

- COMPUNENTS
- Sample Injection
- Ionization Source
- Mass Analyzer
- Ion Detector
- Spectra search using computational tools

Module 007: Techniques for MS Proteomics

MS proteomics works on the principle of protein ionization which are placed in very high magnetic field. Each protein deflect to its proportion which is equal to its molecular weight in this way molecular mass is measured. The protein mass of unknown protein is compared with the masses of proteins in database and matching one is selected.
Example for protein sequences database is uniProt, swissprot etc.
Proteins are measured and sequenced if are unknown than matched with existing database if matched than are shortlisted.

Handout Chapter 05: Protein Sequences
Module 008: Types of MS-based proteomics

Proteins can be sequenced by Edam's degradation and Mass spectrometry. MS based proteomics helps us to sequence the larger and bigger proteins more quickly.
Following steps are involved in MS:
- Separation
- Ionization
- Mass analysis
- Detection

Two methodologies are involved
1. Bottom up proteomics
2. Top down proteomics

Bottom up proteomics measures the peptide masses produced after protein enzymatic digestion. And Top down proteomics measures the intact proteins followed by peptides after fragmentation.

- BOTTOM UP PROTEOMICS

In this methodology the protein complex is treated with site specific enzymes which cleaves them into amino acid residue and resultant peptides are measured for their masses. One peptide is selected at one time for processing and when all are processed than protein search engine is used for matches.

- TOP DOWN PROTEOMICS

In this methodology proteins are ionized and measured for their masses and one protein is mass selected at a time for fragmentation. And resultant peptide fragments are measured for mass.
We can say that bottom up proteomics deals with peptides while top down proteomics can handle the whole protein.

Handout Chapter 05: Protein Sequences
Module 009: BOTTOM UP PROTEOMICS

There are two types of proteomics protocol that are usually employed.
1. Bottom up proteomics
2. Top down proteomics

(Take concept)
PROTOCOL
1. Sample containing the mixture of protein from cells and tissues is obtained.
2. Enzymes such as trypsin is use to cleave the proteins.
3. Enzyme cleaves the amino acids at specific sites of amino acid.
4. Several peptides are formed when protein is cleaved.
5. Number of peptide depends upon the number of sites where enzymes cleaved the protein. For example trypsin cleaves the protein at lysine (k)
6. Mass of each peptide is measured.
7. One peptide is selected at a time.
8. Different enzyme is use to cleave the protein at different site.
9. This process keep going until the possible number of peptides are formed or searched.

10. Peptides are searched in data base and matched.

Handout Chapter 05: Protein Sequences
Module 010: Two Approaches for Bottom up Proteomics (BUP)

There are two approaches for bottom up proteomics.
1. Peptide Mass Fingerprinting.
2. Shotgun Proteomics

Figure 0.33 Peptide mass fingerprinting

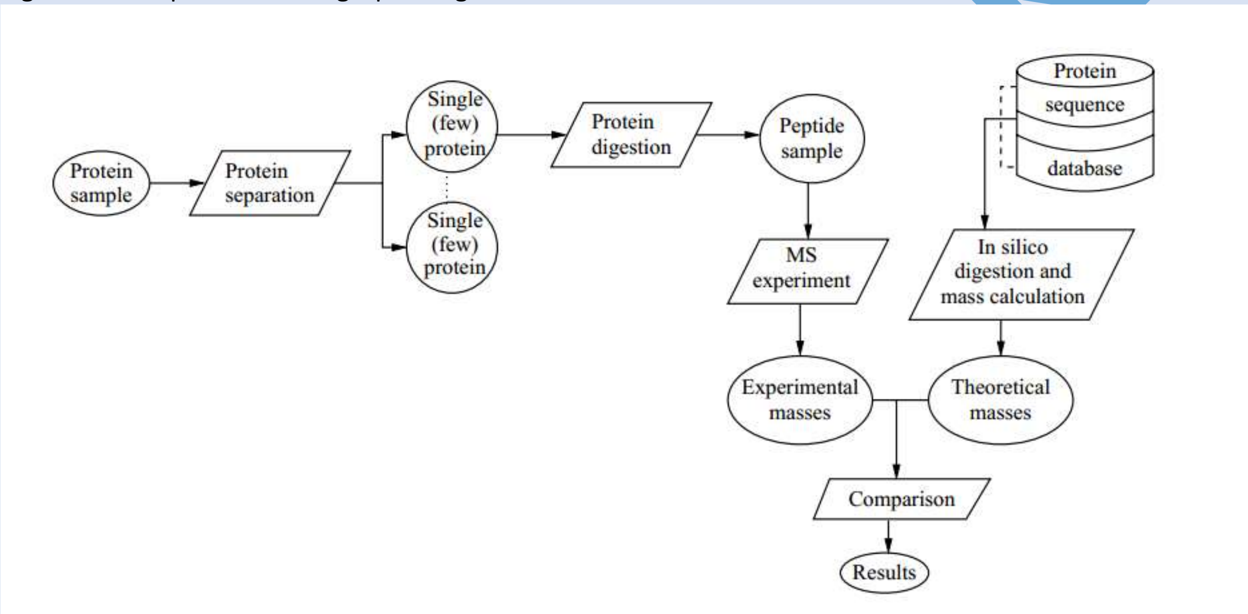
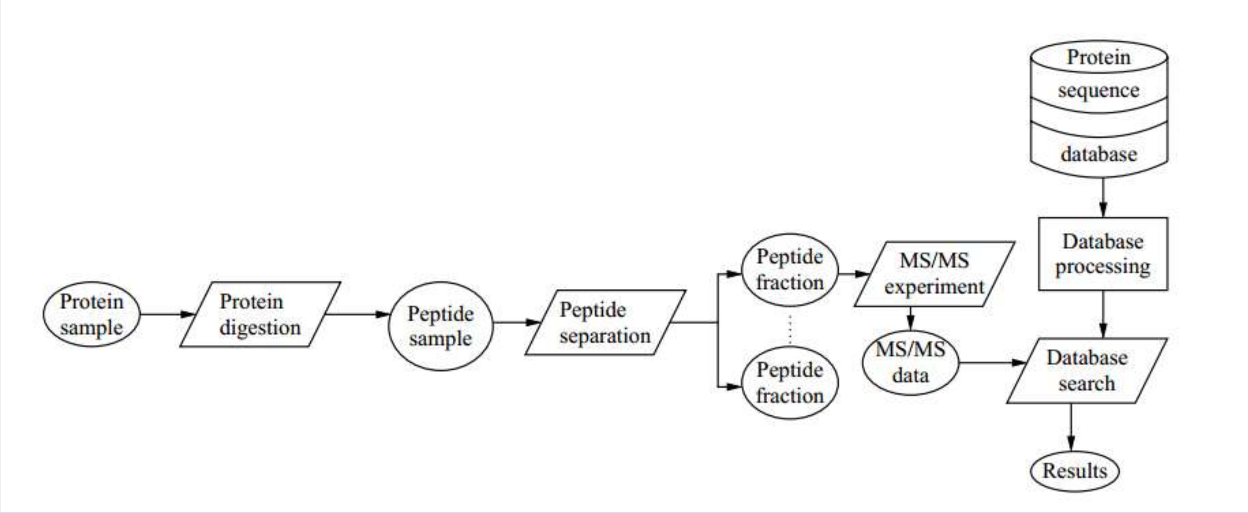


Figure 14 Shotgun Proteomics



Shotgun Proteomics digest the whole protein and mix first and compared with database. And peptide mass finger printing involves in protein separation followed by single protein’s peptide analysis.

Handout Chapter 05: Protein Sequences
Module 011: TOP DOWN PROTEOMICS

Bottom up proteomics identifies the proteins by cleaving them into segments at specific sites and was not suitable to measure the direct protein masse.

(Take concept)

□ PROTOCOL

1. Sample containing the protein mixture from cells and tissues is obtained
2. The entire protein is mixed and analyzed for masses.
3. The list of masses is obtained.
4. TDP Measures all post translational masses of protein.
5. After MS1 one protein is selected at a time and fragmented to obtain its peptides.
6. The process is repeated many times.

Comparison is done from protein database uniProt and swissProt.

TDP also measure the masses of intact proteins and masses of post transcriptional changes.

Handout Chapter 05: Protein Sequences
Module 012: PROTEIN SEQUENCE IDENTIFICATION

Mass spectrometry helps us to measure the molecular weight of proteins and peptides, but several proteins can have same masses to identify them we follow the flow chart of following techniques.

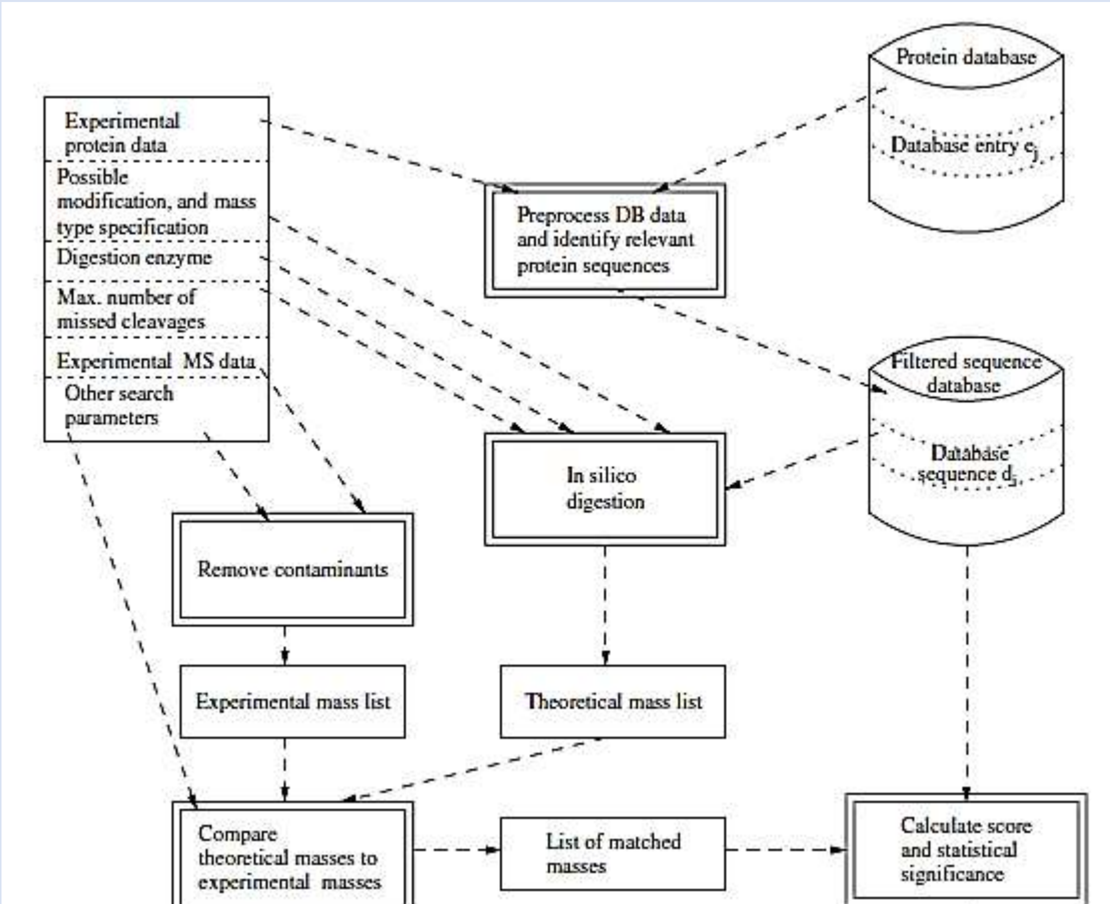


Figure 0.45 protein sequence identification flowchart

Handout Chapter 05: Protein Sequences Module 013: PROTEIN IONIZATION TECHNIQUES (Important lecture)

Protein ionization is used in Mass spectrometry based on proteomics protocols. Ionization involves loading of proton in protein or removal of protein. Ionizations can increase or decrease the mass of protein or peptide.

SALIENT IONIZATION(short question)

Is the technique which include Matrix Assisted Laser Desorption Ionization MALDI) & Electro Spray Ionization (ESI)

For example:

$$[M+H]^+, [M], [M-H]^-$$

monoisotopic or average masses

MALDI

In this technique one proton is added to protein or peptide and the molecular weight is increases by one and Mass spectrometry reports the molecule at +1.

- ESI

ESI adds many protons to protein or peptides and molecular weight is increased by the number of protons added. But it is difficult in ESI to find the molecule with +1.

- EXAMPLE

Example Suppose we have a peptide with mass 2000.0 Da, and that the ionization yields peptide ions of charge +1, +2, *and* +3, by the attraction of one, two, or three protons, respectively. The peptide ions will then be detected at

$$\text{ions with charge } +1\text{: } m/z = (2000 + 1)/1 = 2001$$

$$\text{ions with charge } +2\text{: } m/z = (2000 + 2)/2 = 1001$$

$$\text{ions with charge } +3\text{: } m/z = (2000 + 3)/3 = 666.7$$

MS data from MALDI ionization is easier to handle as the product ions masses are mostly at "1+mass". ESI is difficult to use as it does not easily give away the +1 charged ion

Handout Chapter 05: Protein Sequences
Module 014: MS1 & INTACT PROTEIN MASS

When we ionize the protein, it can be deflected by a magnetic field in proportion to its mass and the mass of protein can be measured by spectrometry.

Mass/charge helps us to calculate the mass of protein, "Mass Select" can help to select specific MS1 for further analysis.

MS1 results the intact masses of the peptides.

Handout Chapter 05: Protein Sequences
Module 015: SCORING INTACT PROTEIN MASS

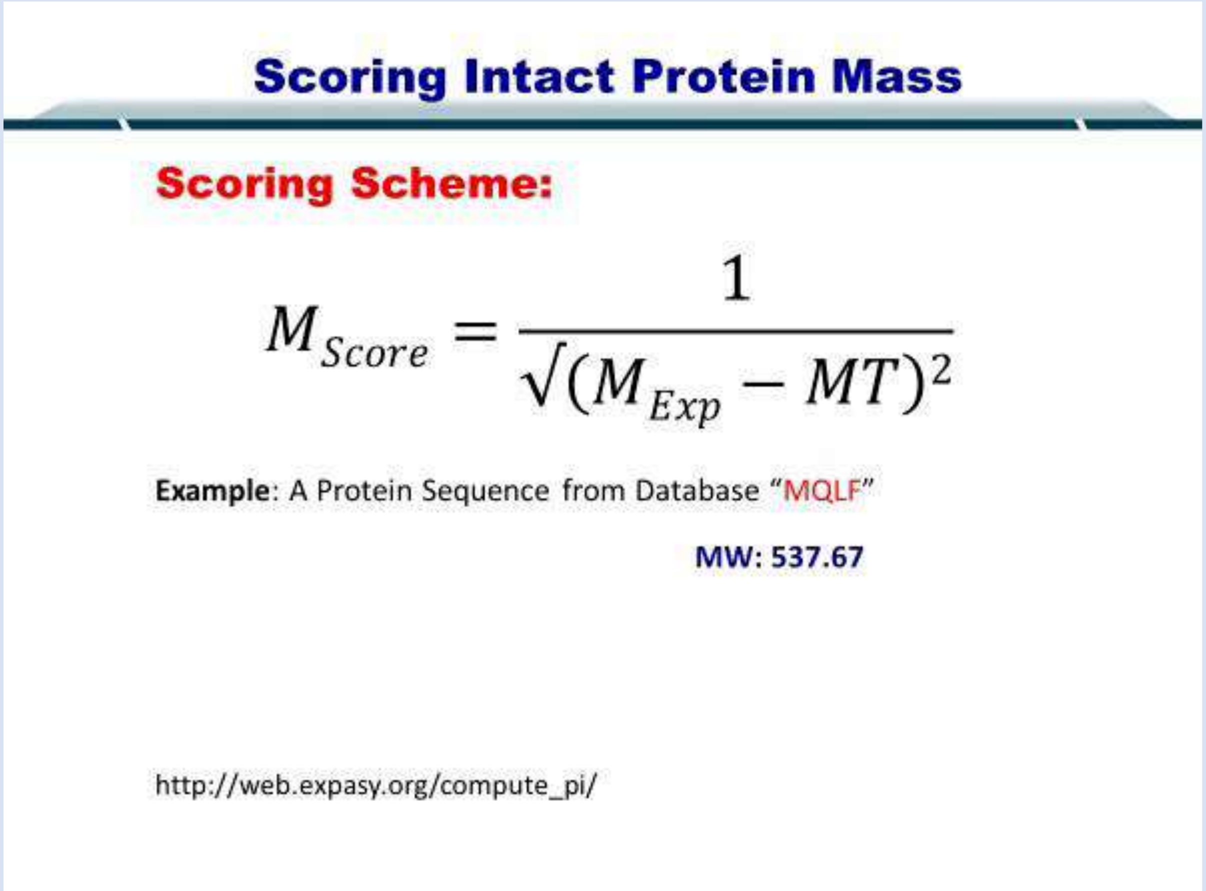
MS1 helps us to obtain the intact masses of precursor molecules which depend upon the proteomics and protocol applied. Protein masses reported by MS1 are matched with protein database, but before match the masses are converted into +1 of all molecule.

- SCORING

We can score each protein in the way that it get maximum score and low quality matches should get low scores.

After filtering the multiple charges we get the only the peaks having charge 1. And after this filter we compare it with protein data base.

- SCORING SCHEME



Example: A Protein Sequence from Database "MQLF"

Handout Chapter 05: Protein Sequences Module 016: PROTEIN FRAGMENTATION TECHNIQUES

We compare the experimental mass with theoretical data base mass of protein and on base of closeness we rank or score it.

If several proteins have same score than selection is done by using another technique protein fragmentation. We fragment the protein or peptide and ionize it, it helps us to measure the fragment masses as the same ways as their precursor.

There are different techniques for protein fragmentation.

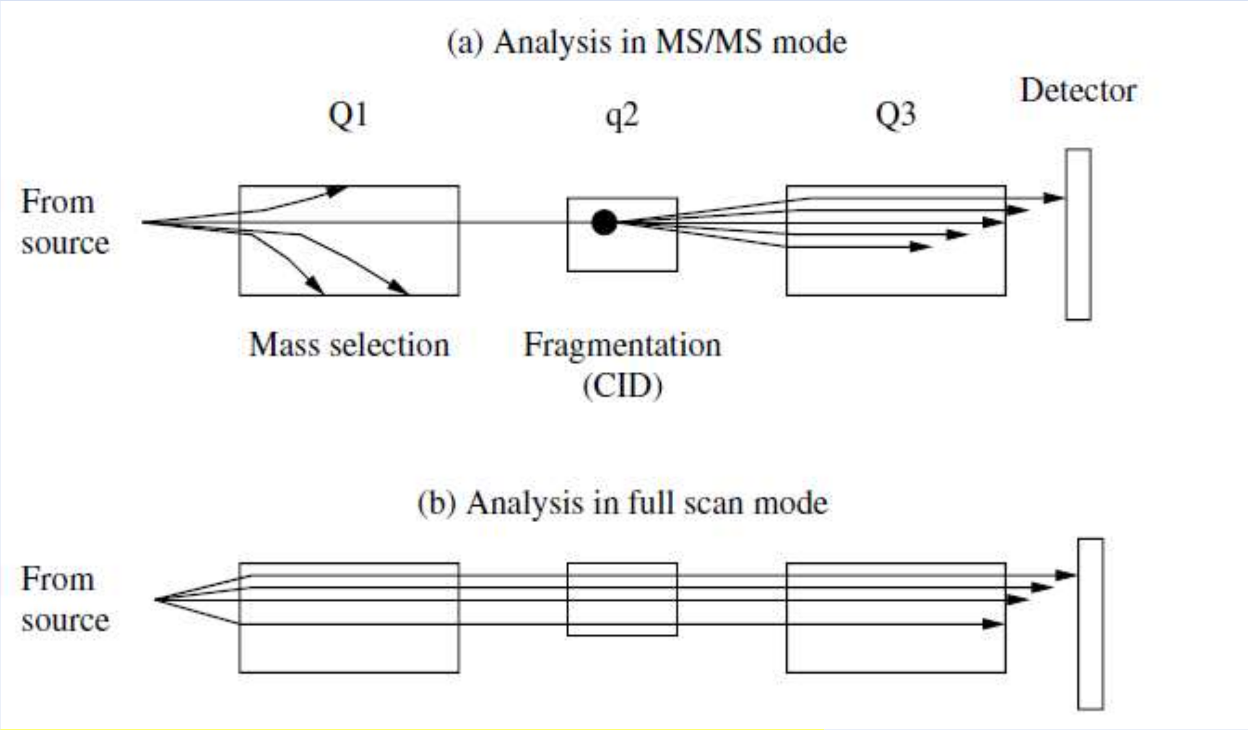
- Electron Capture Dissociation (ECD)
- Electron Transfer Dissociation (ETD)
- Collision Induced Dissociated (CID)

Each fragmentation technique gives result of specific type of fragments.

ECD gives out 'C' and 'Z' ions. CID gives out 'B' and 'Y' ions, etc.

Handout Chapter 05: Protein Sequences
Module 017: TANDEM MS

Intact masses can measure the intact proteins or peptides. And this can be followed up by their fragmentation in MS chamber.



Tandem MS can be extended to the fragments of the intact fragment. All you need is the MS instrument capability to,

(i) Select fragment's mass range.
(ii) Fragment the precursor fragment.

Handout Chapter 05: Protein Sequences
Module 018: MEASURING EXPERIMENTAL FRAGMENT'S MASS

In MS1, the molecular weight of intact sample molecule is measure and then intact molecule is fragmented in two afterward, these two fragments are measured by MS or MS2

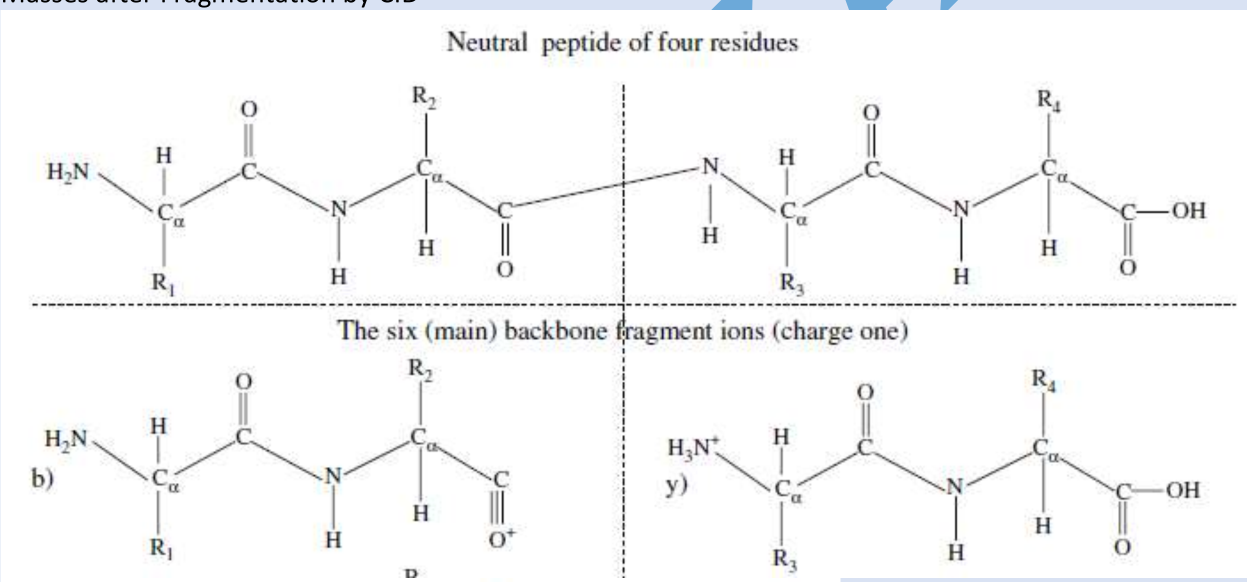
FRAGMENTATION TECHNIQUES AND MOLECULAR WEIGHT
Fragmentation techniques include ECD, CID etc. intact molecule fragmentation splits the molecule into two parts.

FRAGMENT MASS
Mass of fragment is produced by MS2 deepening upon the technique because each techniques splits the protein or peptide at different location.

Experimental mass reported from MS2 is matched with theoretical peptides of candidate proteins (from DB). Score is awarded on the basis of the closeness between experimental and theoretical masses.

Handout Chapter 05: Protein Sequences
Module 019: Calculating Theoretical Fragment's Mass

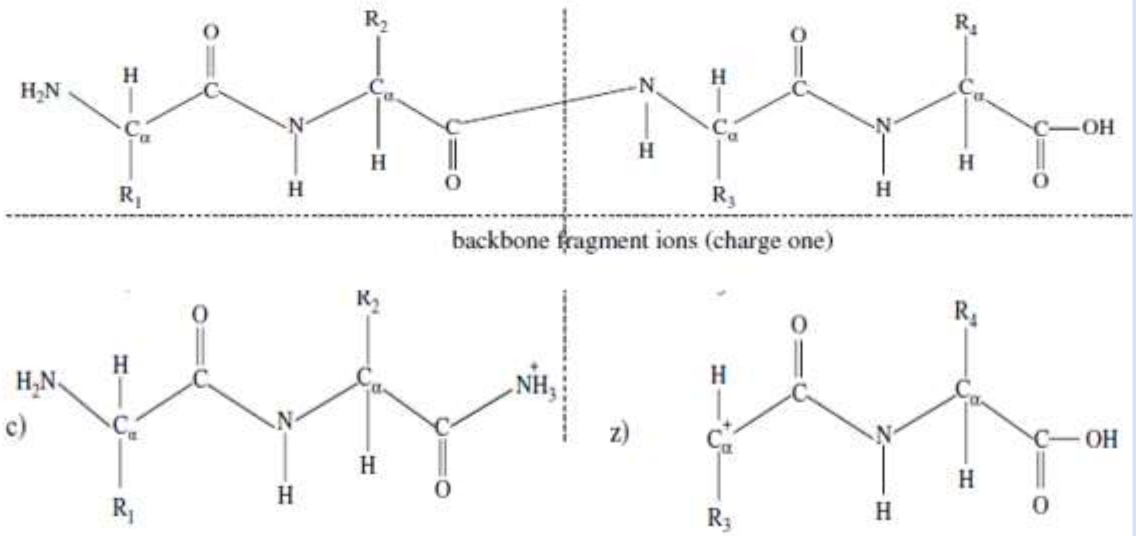
After measuring the mass of intact molecule from MS2 we compare that mass with theoretical mass of database proteins.
Masses after Fragmentation by CID



$$M_b = R_F + \langle N \rangle \qquad M_y = R_F + \langle C \rangle + 2\mathrm{H}$$

Masses after Fragmentation by ECD

Neutral peptide of four residues



$$M_c = R_F + \langle N \rangle + \mathrm{N} + 3\mathrm{H} \qquad M_z = R_F + \langle C \rangle - \mathrm{NH}$$

$$M_b = R_F + \langle N \rangle \qquad M_y = R_F + \langle C \rangle + 2\mathrm{H}$$
$$M_a = R_F + \langle N \rangle - \mathrm{CO} \qquad M_x = R_F + \langle C \rangle + \mathrm{CO}$$

$$M_c = R_F + \langle N \rangle + \mathrm{N} + 3\mathrm{H} \qquad M_z = R_F + \langle C \rangle - \mathrm{NH}$$

$$M_b + M_y = R_P + \langle N \rangle + \langle C \rangle + 2\mathrm{H} \; (= M_P + 2\mathrm{H})$$

$$M_a + M_x = R_P + \langle N \rangle - \mathrm{CO} + \langle C \rangle + \mathrm{CO} \; (= M_P)$$

$$M_c + M_z = R_P + \langle \mathrm{N} \rangle + \mathrm{N} + 3\mathrm{H} + \langle C \rangle - \mathrm{N} - \mathrm{H} \; (= M_P + 2\mathrm{H})$$

Do Hard Work your OWN and becomes a Chooser

REMEMBER ME IN YOUR PRECIOUS PRAYERS

M ZAMAN

M ZAMAN